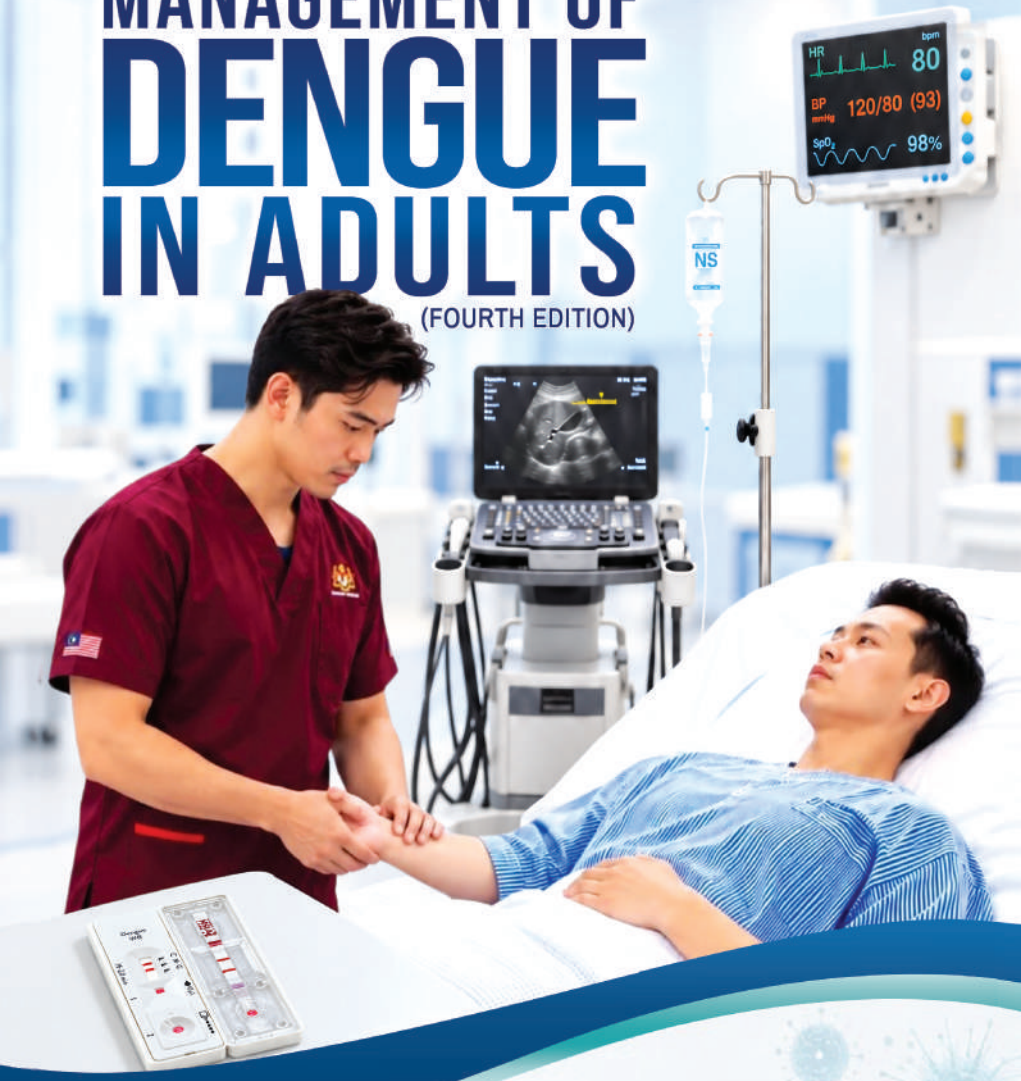
2026 MOH/P/PAK/629.26(GU) - e

CLINICAL PRACTICE GUIDELINE

MANAGEMENT OF DENGUE IN ADULTS

(FOURTH EDITION)



KEMENTERIAN KESIHATAN MALAYSIA
ACADEMY OF MEDICINE OF MALAYSIA
MALAYSIAN SOCIETY OF INFECTION CONTROL & INFECTIOUS DISEASES
PERSATUAN PAKAR-PAKAR PERUBATAN KELUARGA MALAYSIA

Published by:
Malaysian Health Technology Assessment Section (MaHTAS)
Medical Development Division, Ministry of Health Malaysia
Level 4, Block E1, Precinct 1
Federal Government Administrative Centre 62590
Putrajaya, Malaysia

Copyright
The copyright owner of this publication is MaHTAS. Content may be reproduced in any number of copies and in any format or medium provided that a copyright acknowledgement to MaHTAS is included and the content is not changed, not sold, nor used to promote or endorse any product or service, and not used in an inappropriate or misleading context.

ISBN:

Available on the following websites:
http://www.moh.gov.my
http://www.acadmed.org.my
https://mymahtas.moh.gov.my

STATEMENT OF INTENT

This clinical practice guidelines (CPG) are meant to be guides for clinical practice, based on the best available evidence at the time of development. The guidelines should not override the responsibility of the practitioners to make decisions appropriate to the circumstances of the individual. This should be done in consultation with the patients and their families or guardians, taking into account the management options available locally.

TABLE OF CONTENTS

No.	Title	Page
	Key Recommendations	i
	Guidelines Development and Objectives	iv
	Levels of Evidence and Formulation of Recommendations	vii
	Development Group	x
	Review Committee	xii
	External Reviewers	xiii
	Algorithm 1: Management of Dengue Patients with Warning Signs (Group B2)	xiv
	Algorithm 2: Fluid Management in Compensated Shock (Group C)	xv
	Algorithm 3: Fluid Management in Decompensated Shock (Group C)	xvi
	Algorithm 4: Management of Refractory Shock (Group C)	xvii
1.	**INTRODUCTION**	**1**
2.	**EPIDEMIOLOGY**	**2**
	2.1 Epidemiology	2
	2.2. Dengue Virus & Serology	3
3.	**PATHOPHYSIOLOGY**	**5**
	3.1. Pathophysiology of Plasma Leakage in Severe Dengue Infection	5
	3.2. Pathophysiology of Bleeding	7
	3.3. Pathophysiology of Shock	7
4.	**CLINICAL MANIFESTATION AND COURSE OF DENGUE INFECTION**	**9**
	4.1. Overview of the Spectrum of Dengue Infection	9
	4.2. Clinical Course of Dengue Infection	9
	4.3. Severe Dengue	11
	4.4. Differential Diagnosis	13
5.	**DISEASE CLASSIFICATION**	**14**
	5.1. Classification of Dengue Severity	14
	5.2. Expanded Dengue Syndrome	15
6.	**SURVEILLANCE/DISEASE NOTIFICATION**	**16**
7.	**LABORATORY INVESTIGATIONS**	**17**
	7.1. Disease Monitoring Tests	17
	7.2. Diagnostic Tests	17
	7.3. Post-vaccination Diagnostic Considerations	21
	7.4. Dengue Post-mortem Investigations	21

TABLE OF CONTENTS

No.	Title	Page
8.	**DENGUE ASSESSMENT**	**23**
	8.1. Predictors of Severe Dengue	23
9.	**MANAGEMENT OF DENGUE INFECTION**	**27**
	9.1. Outpatient Management of Dengue Infection	29
	9.2. Management of Dengue in the Emergency Department	34
	9.3. Inpatient Management of Dengue Infection	37
	9.4. Management of Complications	53
	9.4.1. Metabolic Acidosis in Severe Dengue	53
	9.4.2. Bleeding in Dengue Infection	54
	9.4.3. Organ Involvement/Expanded Dengue Syndrome	59
	a. Dengue Related Liver Involvement	60
	b. Cardiac Complications	62
	c. Hemophagocytic lymphohistiocytosis	62
	d. Neurological complications	63
	9.5. Dengue Management in ICU	64
10.	**SPECIAL POPULATIONS**	**70**
	10.1 Dengue in Pregnancy	70
	10.2 Dengue in Patient with Heart Failure	73
	10.3 Dengue in Patients with End-Stage Kidney Failure/Chronic Kidney Disease	74
	10.4 Dengue in Patients with Liver Cirrhosis	75
	10.5 Dengue in Patients with Diabetes Mellitus	75
	10.6 Dengue in Patients with Hypertension	76
	10.7 Dengue in Geriatric Population	77
	10.8 Dengue in Obese Patients	78
	10.9 Dengue in Patients with Haemoglobinopathies	79
11.	**COMPLEMENTARY TREATMENT IN DENGUE**	**80**
12.	**DENGUE PREVENTION**	**81**
	12.1. Role of Dengue Vaccination in Dengue Prevention	81
13.	**DISCHARGE CRITERIA**	**84**
14.	**IMPLEMENTING THE GUIDELINES**	**85**
	14.1 Facilitating and Limiting Factors	85
	14.2 Potential Resource Implications	86

TABLE OF CONTENTS

No.	Title	Page
	REFERENCES	**88**
	Appendix 1: Clinical Questions	97
	Appendix 2: Example of Search Strategy	99
	Appendix 3: Sample Collection for Dengue Tests	100
	Appendix 4: Dengue Clerking Sheet in Primary Care	101
	Appendix 5: A. Dengue Monitoring Record	104
	B. Home Care Advice	105
	Appendix 6: Capillary Refill Time Assessment	107
	Appendix 7: Systematic Triage of Dengue in Adults	108
	Appendix 8: PoCUS Guided Dengue Management	109
	Appendix 9: Inpatient Dengue Assessment and Clerking Sheet	110
	Appendix 10: Dengue Monitoring Chart	113
	Appendix 11: Fluid Quota for Dengue in Critical Phase Without Shock	114
	Appendix 12: Passive Leg Raise Manoeuvres	115
	Appendix 13: ROX Index and HACOR Score	117
	List of Abbreviations	118
	Acknowledgements	120
	Disclosure Statement	120
	Source of Funding	120

KEY RECOMMENDATIONS

The following recommendations are highlighted by the CPG DG as the key recommendations that answer the main questions addressed in the CPG and should be prioritised for implementation.

Diagnosis

- Rapid Diagnostic Test (RDT) with combined non-structural protein 1 (NS1) Antigen/ IgM/IgG markers should be used as the first-line test to diagnose dengue in adults.
- In dengue vaccinated individuals within 30 days post-vaccination:
  - dengue polymerase chain reaction (PCR) or NS1 antigen may be used to diagnose dengue.
  - dengue IgM and IgG should not be used as confirmatory tests.

  However, healthcare providers need to rely on clinical evaluation and judgement to diagnose and treat patients accordingly.

Outpatient Management

- Suspected and confirmed dengue patients should be prescribed oral fluid therapy of 200 ml per hour while awake. Patients requiring fluid restriction should follow an individualised fluid regimen.
- Urine output should be monitored and documented in all dengue patients. The frequency of urination should be at least 4 to 6 times a day.
- Non-steroidal anti-inflammatory drugs (NSAIDs) and corticosteroids should not be used in dengue patients. In patients already receiving these medications, continuation should be guided by clinical judgement and individual clinical indications.

Emergency Department Management

- Point-of-care ultrasound should be considered as an adjunctive bedside tool in the emergency and inpatient management of dengue to support early detection of plasma leakage and haemodynamic assessment, particularly in patients at risk of deterioration.
  - Its use should be guided by clinical indication, ultrasound availability and operator competency.

Inpatient Management

- For dengue patients in the febrile phase the following measures should be done:
  - Encourage adequate oral fluid intake.
  - Commence Intravenous (IV) fluids only if the patient is unable to tolerate oral fluids or dehydrated.
  - Calculate the total fluid requirement (oral and IV) using the Holliday-Segar formula.
- In patients with dengue in the critical phase without shock, the total fluid quota should be administered over 48 hours.
- IV fluid adjustment should be done frequently based on clinical response guided by hemodynamic status (including skin colour, capillary refill time, temperature, pulse volume, and pulse rate - CCTVR), vital signs, urine output, laboratory markers (haematocrit), and evidence of ongoing plasma leakage.
- Crystalloids should be used as the initial and preferred fluid for resuscitation in patients with dengue shock.
- Colloids may be considered in patients with persistent shock; however, colloids should not be used for maintenance therapy.
- Fluid administration in patients with dengue shock should be guided by dynamic assessment of fluid responsiveness using appropriate bedside parameters.
- In refractory dengue shock, other causes of shock should be aggressively identified and treated accordingly.

Management of Complications

- Occult bleeding should be suspected in dengue patients with persistent haemodynamic instability despite adequate fluid resuscitation.
- Blood and blood products transfusion should be started promptly in patients with clinically significant bleeding without waiting for a drop in haematocrit.
- Packed red blood cells should be used for transfusion in patients with significant bleeding.
- Routine prophylactic transfusion of platelets or plasma should not be performed in dengue patients without active bleeding.

ICU Management

- Patients with severe dengue should be promptly identified and referred to intensive care unit.
- High-flow nasal cannula and non-invasive ventilation may be used in selected dengue patients with respiratory failure.

- In severe dengue patients, the need for mechanical ventilation should be based on the same established criteria used for other critically ill patients.
- Vasopressor may be used in dengue patient with persistent shock despite adequate fluid resuscitation. Noradrenaline is the preferred vasopressor.

Dengue in Special Population

- All pregnant patients with confirmed dengue should be hospitalised and managed by a multidisciplinary team.
- In dengue patients with heart failure, end-stage kidney disease and liver cirrhosis, fluid management should be individualised with close monitoring.
- In dengue patients with diabetic ketoacidosis (DKA)/hyperosmolar hyperglycaemic state (HHS), fluid requirement should follow DKA/HHS regimen whereas in plasma leakage without DKA/HHS, fluid therapy should be guided by clinical assessment.
- In dengue patients with hypertension, blood pressure should be interpreted carefully relative to baseline.
- In acutely unwell elderly patients with leucopenia, dengue fever should be considered, even in the absence of classical dengue features.
- In obese dengue patients in shock, fluid management should be done with caution by applying adjusted body weight-based fluid dosing with frequent monitoring.

GUIDELINES DEVELOPMENT AND OBJECTIVES

OBJECTIVES

The objective of this CPG on the Management of Dengue in Adults (Fourth Edition) is to provide evidence-based recommendations and guidance to healthcare professionals in the holistic management of dengue in adults on the following aspects:

- prevention
- diagnosis
- assessment/prognosis
- complications
- treatment

POPULATION, USERS AND SETTINGS

TARGET POPULATION

Inclusion Criteria

- Adults at risk of dengue infection
- Adults with suspected and confirmed dengue

Exclusion Criteria

- Nil

TARGET USERS

This document is intended to guide healthcare professionals and relevant stakeholders in primary and secondary/tertiary care of both public and private sector in the management of dengue in adults including:

- doctors
- allied health professionals
- trainees and medical students
- patients and their advocates
- professional societies

HEALTHCARE SETTINGS

Primary, secondary and tertiary care

SPONSORSHIP & FUNDING

The development of the CPG on Management of Dengue in Adults (Fourth Edition) was financially supported in its entirety by the MoH.

GROUPS AND ROLES

Development Group (DG): A multidisciplinary team that used a standard methodology based on systematic evidence review to plan, critically appraise and formulate evidence-based recommendations.

Review Committee (RC): A senior multidisciplinary group actively involved throughout the development process, providing technical oversight.

External Reviewers (ER): Local and international subject matter experts who provided an independent assessment of the draft.

Disclosure Statements/ Conflicts of interest (CoI): All panel members both DG and RC declared financial and non-financial interests. None held shares in pharmaceutical firms or acts as consultants to such firms. Details are available upon request from the CPG Secretariat.

Meetings: The DG members met 30 times throughout the development of these guidelines

CLINICAL QUESTIONS AND SCOPE (*Refer* **Appendix 1**)

A total of 13 main **clinical questions (CQs)** were developed under specific sections with patient-important outcomes prioritised. DG members were assigned individual CQs within these sections.

EVIDENCE RETRIEVAL (*Refer* **Appendix 2**: *Example Search Strategy*)

Databases & sources: Medline (Ovid) and Cochrane Database of Systematic Reviews; supplementary searches in PubMed and others.
Limits: literature published in the last 10 years from 2015 to 2025 for all CQs on Humans and in English; In addition, the reference lists of all retrieved literature and guidelines were searched to further identify relevant studies. Experts in the field were also contacted for studies related to the issues addressed
Search period: All searches were conducted from 12 January 2025 to 25 January 2025. The literature search was repeated for all CQs at the end of the CPG development process, allowing any relevant papers published before 25 January 2026 to be included. Future CPG updates will consider evidence published after this cut-off date. Full details of the search strategy can be obtained from the CPG Secretariat.

EVIDENCE APPRAISAL AND FORMULATION OF RECOMMENDATIONS

All retrieved literature was appraised by at least two DG members using the Critical Appraisal Skills Programme (CASP) and Risk of Bias (RoB) checklists presented in evidence tables and discussed at DG meetings. Statements and recommendations were agreed upon by both the DG and RC. Where evidence was insufficient, the recommendations were made by consensus of the two groups. The CPG is largely based on systematic reviews, meta-analyses and clinical trials, with local practice taken into consideration.

Levels/ Grades: Individual studies were graded using the U.S. Preventive Services Task Force (2015) levels of evidence; recommendation strength followed GRADE principles. The writing of the CPG adheres to AGREE II requirements.

LEVELS OF EVIDENCE

Level	Study design
I	Properly powered and conducted randomised controlled trial; well-conducted systematic review or meta-analysis of homogeneous randomised controlled trials
II-1	Well-designed controlled trial without randomisation
II-2	Well-designed cohort or case-control analysis study
II-3	Multiple time series, with or without the intervention; results from uncontrolled studies that yield results of large magnitude
III	Opinions of respected authorities, based on clinical experience; descriptive studies or case reports; reports of expert committees

SOURCE: *United States (U.S.) Preventive Services Task Force. U.S. Preventive Services Task Force Procedure Manual. Rockville, MD: USPSTF; 2015.*

FORMULATION OF RECOMMENDATION

- In line with the new development in CPG methodology, the CPG Unit of MaHTAS is adapting **Grading Recommendations, Assessment, Development and Evaluation (GRADE)** in its work process. The quality of body of evidence and related effect size are carefully assessed/reviewed by the CPG DG.
- Recommendations are formulated based on **certainty of evidence** and the wording used denotes the **strength of recommendations**. This takes into account:
  - quality and level of the evidence
  - balance of benefits and harms of the options
  - patient's preference and values
  - resource implications
  - relevancy and applicability to the local target population
- The more criteria being fulfilled, the more certain is the evidence leading to strong recommendations using the word "**should**" being considered. Otherwise, weak recommendations use the word "**may**" in proposing an action to be made.
- In the CPG, a yellow box highlights important message(s) in the management while a blue box contains evidence-based recommendation(s) for the particular condition.

USE OF EXISTING GUIDELINES AND QUALITY APPRAISAL

References were also made to other guidelines on dengue in adults as listed below:

- World Health Organization (WHO) guidelines for clinical management of arboviral diseases: dengue, chikungunya, Zika and yellow fever Geneva: WHO; 2025.
- World Health Organization Regional Office for South-East Asia (WHO-SEARO). Dengue case management for primary health care and home-based care: Booklet for health professionals and health-care workers. WHO-SEARO; 2025.
- Pan-American Health Organization (PAHO). Guidelines for the Clinical Diagnosis and Treatment of Dengue, Chikungunya, and Zika. PAHO; 2022.

The evidence-based CPGs were evaluated using the Appraisal of Guidelines for Research and Evaluation (AGREE) II prior to them being used as references.

EXTERNAL REVIEW, PUBLIC CONSULTATION & APPROVAL

The draft on completion underwent external peer review and was posted on the MOH Malaysia website for public consultation (≥ 30 days). The draft was then presented to the Technical Advisory Committee for CPG and the HTA & CPG Council, MOH Malaysia for review and approval. Further details on the CPG development methodology by MaHTAS are available in the Manual on Development and Implementation of Evidence-based Clinical Practice Guidelines.
(available at: https://www.moh.gov.my/moh/resources/CPG_MANUAL_MAHTAS.pdf)

IMPLEMENTATION, AUDIT & MONITORING (*Detailed out in* Chapter 10)

Priority recommendations include **audit indicators** (numerator/ denominator, target, data source, suggested frequency) to enable quality improvement and monitoring.

UPDATING POLICY

These guidelines were issued in 2026 and will be reviewed in a minimum period of four years (2030) or sooner if new evidence warrants earlier revision. When it is due for updating, the Chairperson of the CPG or National Advisor of the related specialty will be informed. A discussion will be conducted on the need for a revision including the scope of the revised CPG. A multidisciplinary team will be formed and the latest systematic review methodology used by MaHTAS will be employed. Every care has been taken to ensure that this publication is correct in

every detail at the time of publication. However, in the event of errors or omissions, corrections will be published in the web version of this document, which will be the definitive version at all times. This version can be found on the websites mentioned above.

DEVELOPMENT GROUP

Chairperson

Dato' Dr. Chow Ting Soo
Senior Consultant in Infectious Disease
Hospital Pulau Pinang

Members (alphabetical order)

Dr. Abinash Kaur A/P Karam Singh
Consultant in Acute Internal Medicine
Hospital Shah Alam, Selangor

Dr. Andrew Chang Kean Wei
Infectious Disease Specialist
Hospital Umum Sarawak, Sarawak

Dr. Amelia Mohamed Ain
Consultant Intensivist & Anaesthesiologist
Hospital Sultan Abdul Halim, Kedah

Dr. Azureen Azmel
Consultant in Infectious Disease
Hospital Tengku Ampuan Rahimah
Klang, Selangor

Dr. Fazlina Mohamed Yusoff
Consultant Family Medicine Specialist
Klinik Kesihatan Anika, Selangor

Dr. Guan Han Lin
Infectious Disease Specialist
Hospital Seberang Jaya, Pulau Pinang

Dr. Hidayah Shafie
Consultant Emergency Critical Care Physician
Hospital Kuala Lumpur, Kuala Lumpur

Dr. Karen Sharmini Sandanasamy
Public Health Physician & Certified Guideline Methodologist
Head of Clinical Practice Guidelines Unit
Malaysian Health Technology Assessment Section (MaHTAS), Putrajaya

Dr. Noor Ayuni Bazura Muhamad
Senior Principal Assistant Director
MaHTAS, Putrajaya

Dr. Noorlina binti Nordin
Infectious Disease Specialist
Hospital Tengku Ampuan Afzan, Pahang

Dr. Noor Sheereen Adzaludin
Transfusion Medicine Specialist
Pusat Darah Negara, Kuala Lumpur

Dr. Norita binti Zainol
Pathologist (Microbiology)
Hospital Kuala Lumpur, Kuala Lumpur

Dr. Nurul Shaliza Shamsuddin
Emergency Critical Care Physician
Hospital Sultan Idris Shah, Serdang, Selangor

Dr. Nur Zaidah Mohd Tahir
Family Medicine Specialist
Klinik Kesihatan Ulu Tiram, Johor

Dr. Rathimalar a/p Kerisnan
Obstetrician & Gynaecologist
Specialist/ Maternal Fetal Medicine
Hospital Tunku Azizah, Kuala Lumpur

Dr. Sharifah Baizura Syed Alwi
Consultant in Infectious Disease
Hospital Sultanah Bahiyah, Kedah

Dr. Mohd Noor Azreey Mukri
Infectious Disease Specialist
Hospital Sultanah Aminah, Johor

Assoc. Prof Dr. Sharifah Faridah Syed Omar
Consultant in Infectious Disease
Pusat Perubatan Universiti Malaya,
Kuala Lumpur

Dr. Mohd Suhaidin Bin Che Ngah
Consultant Obstetrician & Gynaecologist
Hospital Sultan Ismail Petra, Kelantan

Dr. Sharifah Khairul Atikah Syed
Kamaruddin
Pathologist (Chemical Pathology)
Hospital Sg. Buloh, Selangor

Dr. Muhamad Hafizzi bin Mohd
Intensivist
Hospital Melaka, Melaka

Dr. Wan Ming Keong
Public Health Medicine Specialist
Cheras Health Office, Kuala Lumpur

REVIEW COMMITTEE

The draft guidelines were reviewed by a panel of experts from both public and private sectors. They were asked to comment primarily on the comprehensiveness and accuracy of the interpretation of evidence supporting the recommendations in the guidelines.

Chairperson
Dato' Dr. Suresh Kumar a/l Chidambaram
Pakar Perunding Kanan Penyakit Berjangkit
Hospital Sungai Buloh

Members (alphabetical order)

Dr. Asmah binti Zainudin
Senior Consultant Intensivist
Hospital Sultanah Bahiyah, Kedah

Dr Buvanes Chelliah
Consultant Fetomaternal Medicine
Hospital Tunku Azizah, Kuala Lumpur

Dr Fatahul Laham Mohamed
Senior Consultant Emergency Medicine
Hospital Sultanah Bahiyah, Kedah

Dr. Ho Bee Kiau
Consultant Family Medicine Specialist
KK Seksyen 7, Shah Alam

Dato' Dr. Mahiran Mustafa
Infectious Disease Consultant
Lecture,
International Medical School/
Management & Science University,
Kelantan

Dr. Nor Hafizah Ahmad
Transfusion Medicine Consultant
Pusat Darah Negara, Kuala Lumpur

Prof. Madya Dr. Petrick @ Ramesh K. Periyasamy
Infectious Disease Consultant
Hospital Canselor Tuanku Muhriz
Universiti Kebangsaan Malaysia,
Kuala Lumpur

Dr. Redzuan Ab Hamid
Patient Advocate
Hospital Sultanah Aminah, Johor

Dr. Syaqirah Akmal
Public Health Physician & Deputy Director
Malaysian Health Technology Assessment Section (MaHTAS), MoH, Putrajaya

Dr. Sahlawati Mustakim
Consultant Pathologist (Microbiology)
Hospital Sungai Buloh, Selangor

Dr. Shanty Ratnam
Consultant Intensivist
Hospital Thompson, Kuala Lumpur

Dr. Yoon Chee Kin
Head of Acute Internal Medicine Subspecialty Service & Acute Internal Medicine Consultant
Hospital Pulau Pinang, Pulau Pinang

Dr Zailiza Suli
Consultant Public Health Medicine Specialist & Head of Sector
Vector-borne Disease Sector
Disease Control Division, MoH, Putrajaya

EXTERNAL REVIEWERS

The following external reviewers provided feedback on the draft:

Dato' Dr. Asri Ranga Bin Abdullah Ramaiah
Senior Consultant Cardiologist
Hospital Sultan Idris Shah Serdang

Dr Azlinsuraya Mohamad Yusof
Consultant Obstetrician & Gynaecologist
Hospital Tengku Ampuan Rahimah

Dr. Faheem Ahmed Khan
Senior Consultant in Intensive Care Medicine & Emergency Medicine,
Chairman Chapter of Intensivist (AMS) Singapore,Ng Teng Fong General Hospital, Singapore

Dr. Timothy Williams
Infectious Disease Consultant
Hospital Seberang Jaya

Dr Koh Kah Chai
General Practitioner
Poliklinik Kepong Baru

Prof Nasikarn Angkasekwinai
Infectious Disease Consultant
Department of Medicine
Mahidol University, Thailand

Dr. Nurain Mohd Noor
Consultant Endocrinologist
Hospital Putrajaya

Dr. Nurul Zaynah Nordin
Consultant Nephrologist
Hospital Kuala Lumpur

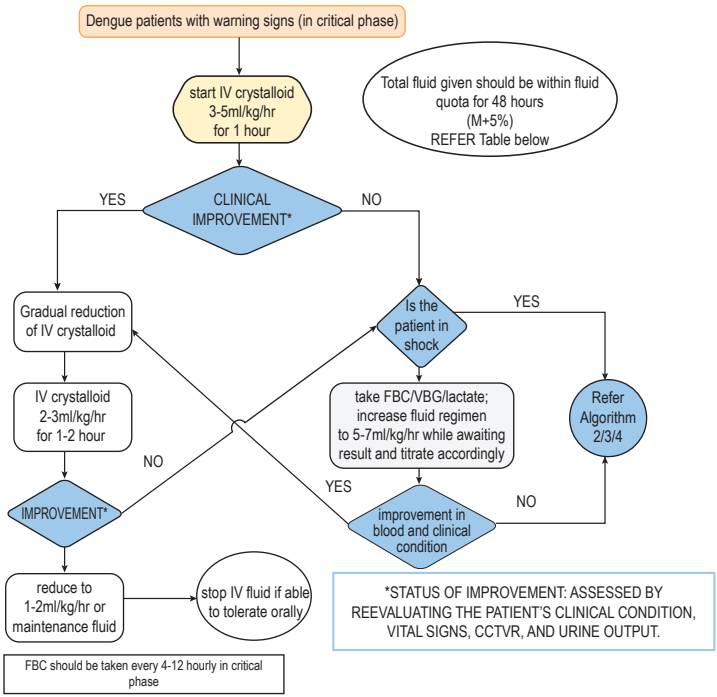
Dr Tan Lian Huat
Infectious Disease Consultant
Sunway Medical Centre

Dato' Dr. Tunku Muzafar Shah bin Tunku Jaafar
Senior Consultant Geriatrician
Hospital Selayang

Dr. Zalwani Zainuddin
Consultant Gastroenterologist & Hepatologist
Hospital Sultanah Bahiyah

Dr Adi Osman,
Senior Consultant Emergency Critical Care Physician,
Hospital Raja Permaisuri Bainun, Ipoh.

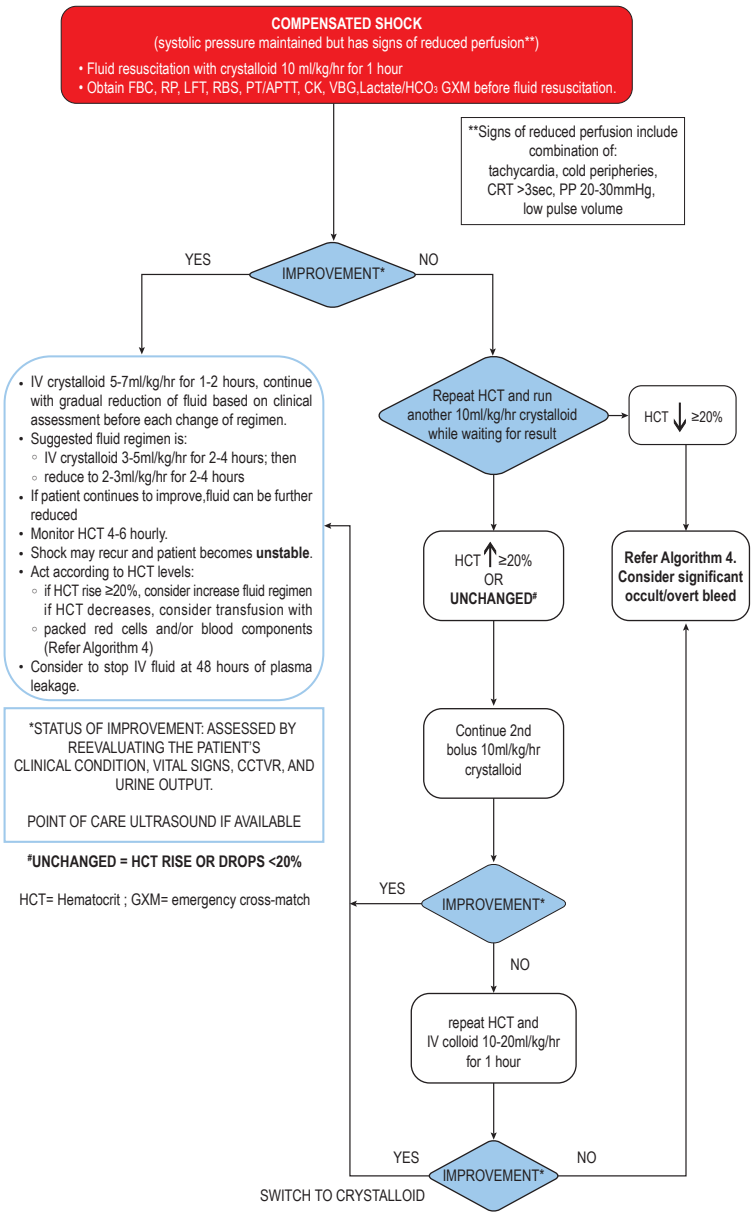
ALGORITHM 1: MANAGEMENT OF DENGUE PATIENTS WITH WARNING SIGNS (GROUP B2)



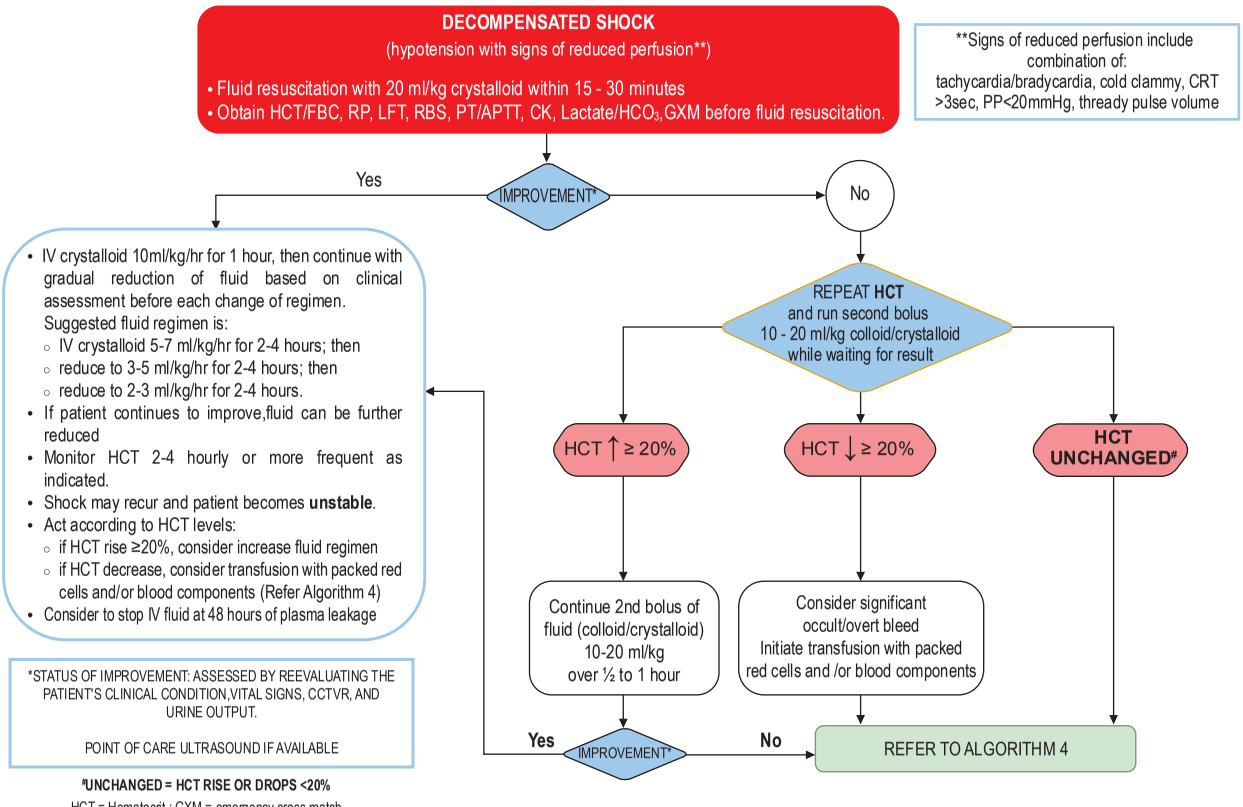
Body weight (kg)*	Fluid quota M+5% (Oral and IV) over 48 hours (ml)	Rate of IV fluid regime (ml/hr)				
		R1 (1ml/kg/h)	R2 (2ml/kg/h)	R3 (3ml/kg/h)	R5 (5ml/kg/h)	R7 (7ml/kg/h)
30	3200	30	60	90	150	210
35	3600	35	70	105	175	245
40	3900	40	80	120	200	280
45	4300	45	90	135	225	315
50	4700	50	100	150	250	350
55	5000	55	110	165	275	385
60	5400	60	120	180	300	420
65	5700	65	130	195	325	455
≥ 70	6000	70	140	210	350	490

*weight limit at 70kg

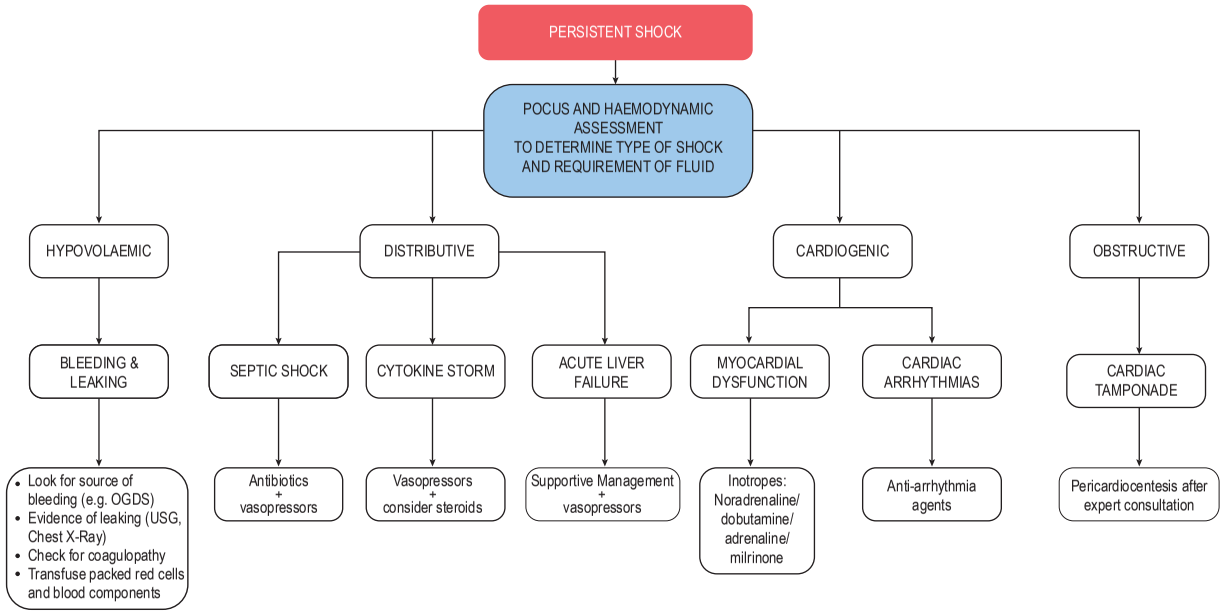
ALGORITHM 2: FLUID MANAGEMENT IN COMPENSATED SHOCK (GROUP C)



ALGORITHM 3: FLUID MANAGEMENT IN DECOMPENSATED SHOCK (GROUP C)



ALGORITHM 4: MANAGEMENT OF REFRACTORY SHOCK (GROUP C)



Patient may have more than one type of shock

OGDS= oesophagogastroduodenoscopy; USG= ultrasonography; POCUS = point of care ultrasound

1. INTRODUCTION

Dengue infection remains a major public health challenge in Malaysia, with periodic outbreaks contributing to significant morbidity, mortality, and strain on healthcare services. In adults, dengue follows a highly dynamic and unpredictable clinical course, ranging from a self-limiting febrile illness to severe dengue with plasma leakage, shock, severe bleeding, and organ dysfunction. As deterioration may occur abruptly, clinical outcomes depend largely on sustained vigilance, continuous monitoring, and serial clinical assessments, rather than initial presentation alone.[1]

Since the publication of the 3rd Edition Clinical Practice Guidelines (CPG) for the Management of Dengue in Adults in 2015, new evidence has emerged that warrants updating the existing recommendations. These include advances in risk stratification and prognostic markers, improvements in clinical assessment and disease monitoring using point-of-care ultrasound (POCUS), refined approaches to fluid management, and evolving evidence on the role of dengue vaccination as a part of a comprehensive dengue control strategy.

This 4th Edition of CPG incorporates current best evidence to support clinicians in early identification of patients at risk of disease progression, informed decision-making on admission and escalation of care, and safe, individualised fluid therapy across all phases of illness. Special considerations for vulnerable populations and complex clinical scenarios are also emphasised.

This guideline is intended for use by healthcare professionals across all levels of care, including primary care, emergency departments, medical wards, and intensive care units (ICU). It aims to provide clear, practical, and evidence-based recommendations to standardise care, optimise clinical outcomes, and reduce preventable complications of dengue in adults.

In addition to updated clinical recommendations, this guideline introduces practical implementation tools to promote consistent, high-quality dengue care across healthcare settings. These include standardised clinical audit tools for both primary care and inpatient management, as well as revised dengue monitoring records and clerking sheets. Collectively, these tools are designed to support systematic clinical assessment, accurate documentation, and timely identification of disease progression.

2. EPIDEMIOLOGY

2.1. Epidemiology

Dengue is a viral disease caused by infection with four distinct serotypes of the Flaviviridae family and is transmitted through bites of female Aedes aegypti and Aedes albopictus mosquitoes. Dengue has expanded from affecting fewer than 10 endemic countries to more than 130 countries over the past 5 decades. In 2024, Grade 3 public health emergency (the highest level of emergency requiring intense organization-wide support) was declared by World Health Organization in response to more than 11 thousand deaths and 14 million dengue cases reported globally, with Latin America (~13 million cases), South East Asia (~800,000 cases) and parts of Africa (~150,000 cases) having among the highest burdens.[2]

Dengue is an endemic disease in Malaysia with fluctuating incidence rates reported in recent years. A cyclical trend of dengue is observed in Malaysia every 4 to 5 years, with previous epidemics reported in 2015, 2019 and 2023. Mirroring global trends, Malaysia's dengue burden remained high throughout 2024 before the incidence dropped by more than half in 2025.2 A total of 117 dengue deaths were reported in 2024 with case fatality rate (CFR) of 0.096% which was below the national target of 0.2%.3 The trend for dengue cases and incidence is shown in **Figure 1**, and the trend for dengue deaths and case fatality rate is shown in **Figure 2** below. In light of Malaysia's current dengue burden (low to moderate transmission intensity), the Ministry of Health aligns with the WHO's May 2024 position paper and does not consider the use of TAK-003 (Qdenga) vaccine in the National Immunization Program at present.[4; 5]

Few chronic medical conditions are commonly reported among dengue mortality cases, for example hypertension 27%, diabetes mellitus 23%, obesity 23%, hyperlipidemia 8%, heart disease 7% and chronic renal disease 6%.[6] Nevertheless, 30% of the deceased in 2023 were without any prior co-morbidity, which indicates that any healthy individual can succumb to dengue complications.

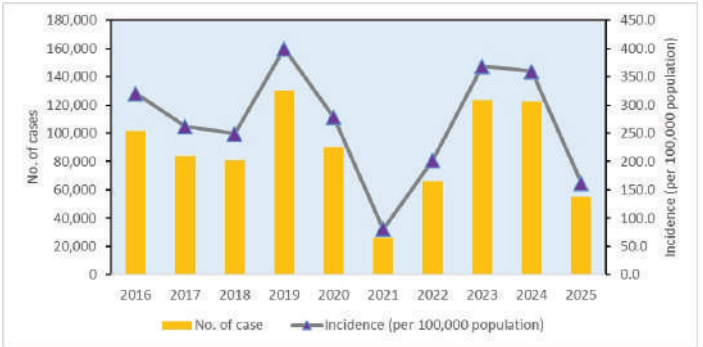


Figure 1: Ten-year trend (2016-2025) for dengue cases and incidence in Malaysia

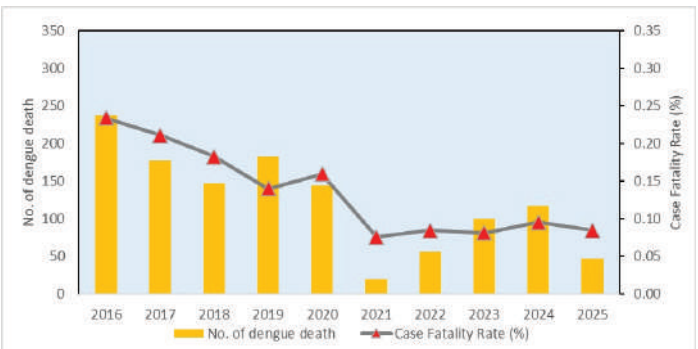


Figure 2: Ten-year trend (2016-2025) for dengue death and case fatality rate in Malaysia

Source: Vector-borne Disease Sections, Disease Control Division, Ministry of Health

2.2. Dengue Virus & Serotypes

Flavivirus is a genus of viruses within Flaviviridae family that encompasses over 70 species including dengue virus. Dengue virus is categorised into four serotypes, namely DENV-1, DENV-2, DENV-3 and DENV-4 based on the antigenicity of envelope protein E. Although an initial infection provides lifelong immunity to that specific serotype, it only offers transient and partial immunity against the other serotypes. Subsequent infection with a different serotype (secondary infection) significantly increases the risk of severe clinical outcomes, a phenomenon known as antibody-dependent enhancement (ADE).[7]

Dengue virus in Malaysia demonstrated heterogeneity across space and time, with constant fluctuation in the proportion of serotypes and

lineages within genotypes. (Refer to **Figure 3** showing a ten-year trend for circulating dengue serotype in Malaysia) Nonetheless, there is no significant association between virus lineages and clinical severity.[8]

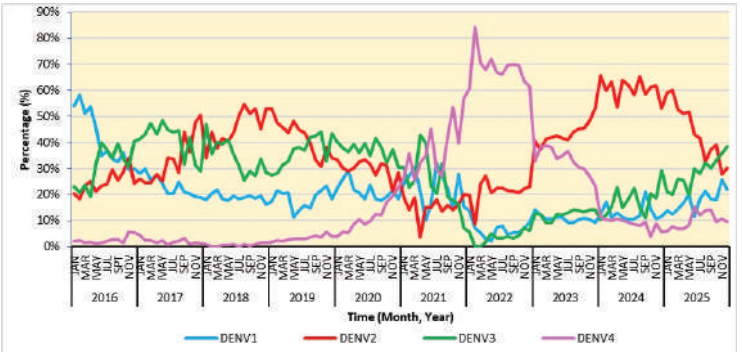


Figure 3. Ten-year trend (2016-2025) for circulating dengue virus serotype in Malaysia

3. PATHOPHYSIOLOGY

3.1. Pathophysiology of Plasma Leakage in Severe Dengue Infection

A key pathophysiologic feature of dengue is a transient but marked increase in microvascular permeability, leading to plasma leakage. Evidence suggests that leakage arises from a combination of viral–host immune interactions and endothelial injury, particularly involving the endothelial glycocalyx layer (EGL), which lines the inner surface of blood vessels covering the endothelial cells. It is composed of proteoglycans and glycosaminoglycans. In healthy adults this layer plays a crucial role in regulating trans-vascular fluid and protein movement, thus preventing excessive plasma leakage and protein into interstitial space.[9; 10; 11] Refer to **Figure 4**, a diagram illustrating normal endothelial glycocalyx.

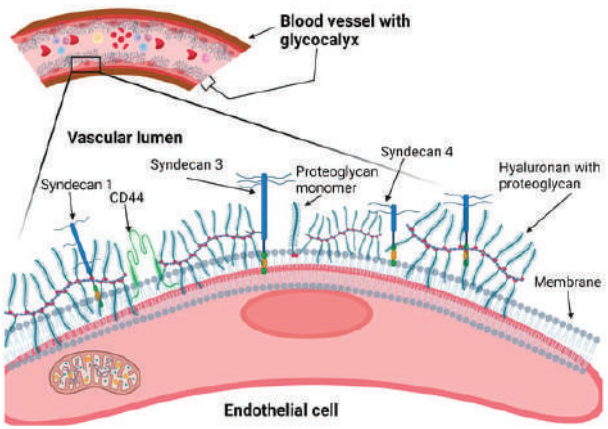


Figure 4: Normal endothelial glycocalyx

Source: Sembajwe LF, Ssekandi AM, Namaganda A, et al. Glycocalyx-Sodium Interaction in Vascular Endothelium. Nutrients. 2023 Jun 25;15(13):2873.

3.1.1. Pathogenesis Leading to Endothelial Injury

- Dengue viral proteins, especially NS1, directly injure the endothelium and EGL through enzymatic activation (heparinase, sialidases, cathepsin L).[10]
- The host immune response amplifies endothelial damage via cytokine release (e.g. IL-6, IL-8, TNF-α, IFN-γ), complement activation, mast cell degranulation, and antibody-dependent enhancement (ADE) in secondary infection.[9; 11]
- Recent cohort data show that hyperinflammation and endothelial activation (elevated IL-6, ferritin, Ang-2, syndecan-1) correlate strongly with plasma leak severity and organ dysfunction.[12]

Refer **Figure 5** a diagram illustrating the Pathophysiology of plasma leakage.

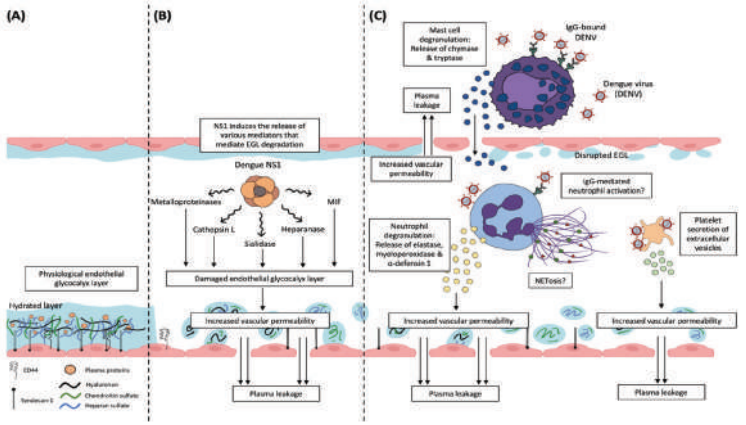


Figure 5: Pathophysiology of plasma leakage

Note (A) Healthy blood vessel, (B) NS1-induced plasma leakage, (C) Other mechanism such as cytokine-release-mediated plasma leakage

Source: Teo A, Chua CLL, Chia PY, et al. Insights into potential causes of vascular hyperpermeability in dengue. PLoS Pathog 17(12):2021.

3.1.2 Microvascular Permeability and Revised Starling Law

Disruption of the EGL and endothelial junctions alters microvascular permeability, allowing plasma to escape into the third space i.e. interstitial, pleural and peritoneal spaces. Leakage typically begins at the end of febrile phase, peaks during the critical phase, and resolves during recovery as endothelial integrity is restored.[9; 11]

The classical Starling model does not fully explain the plasma leakage seen in dengue, where aggressive fluid administration was traditionally assumed to maintain intravascular volume.

The revised Starling model identifies the EGL as the key regulator of microvascular fluid exchange.With EGL injury, as in dengue, the barrier is lost and fluid shifts into the interstitial space.In this setting, both crystalloids and colloids will leak into extravascular compartments and contribute to fluid accumulation.[12; 13; 14; 15]

Venous reabsorption is minimal and filtered fluid returns to the circulation via lymphatics. If large intravenous volumes are administered during the leaking phase, the combined lymphatic return and residual fluids may lead to intravascular overload during recovery phase.[12; 13; 14]

Figure 6 below illustrates the difference between the classical and revised Starling Model

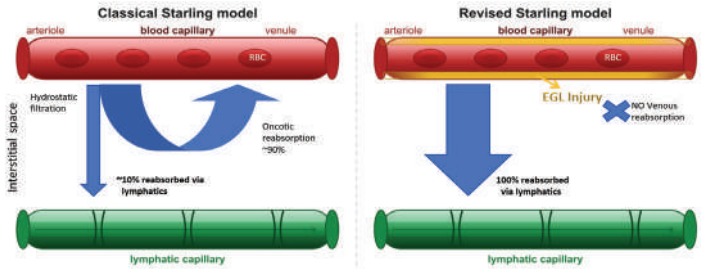


Figure 6: Classical vs Revised Starling Model of Microvascular Fluid Exchange

Adapted from: Tactile Medical. The Revised Starling Principle [Internet]. Available from:https://tactilemedical.com/lymphedema-resources/er-the-revised-starling-principle/ (Accessed 13 Jan 2026)

3.1.3 Clinical Implication of Fluid Therapy

Fluid therapy should therefore be phase-specific and individualised, guided by the degree of leakage and timing of endothelial recovery, to avoid both inadequate perfusion and fluid-related complications.[13; 14] The degree of leakage is inferred clinically from haematocrit trends, haemodynamic parameters, urine output, and evidence of third-space fluid loss, with PoCUS as a useful adjunct.

3.2. Pathophysiology of Bleeding

Severe bleeding in dengue reflects abnormalities of haemostasis encompassing thrombocytopenia, qualitative platelet dysfunction, coagulopathy, enhanced fibrinolysis, endothelial injury and hepatic impairment.[9; 11] Bleeding can manifest as mucosal, gastrointestinal or intracranial haemorrhage and may occur independently of shock. Prolonged hypoperfusion and acidosis in dengue shock can precipitate disseminated intravascular coagulation (DIC) and massive haemorrhage.[9; 11, 12]

3.3. Pathophysiology of shock

Plasma leakage can lead to intravascular volume depletion and hypoperfusion. Persistent hypoperfusion results in anaerobic metabolism and elevated lactate. Inadequate perfusion will lead to increased anaerobic glycolysis and lactic acidosis. If hypovolaemia is not corrected promptly, tissue hypoperfusion may progress to decompensated shock, which may become refractory to volume

resuscitation or vasopressor drugs. The resultant lactic acidosis further depresses the myocardium and worsens the hypotension. Complications of prolonged shock include DIC and multiorgan failure.[13]

4. CLINICAL MANIFESTATION AND COURSE OF DENGUE INFECTION

4.1. Overview of the Spectrum of Dengue Infection

Dengue infection presents with a wide spectrum of clinical manifestations, ranging from asymptomatic infection to severe, life-threatening disease. Many individuals may remain asymptomatic or develop a mild, undifferentiated febrile illness. At the severe end of the spectrum, dengue may be complicated by plasma leakage, severe bleeding, or organ involvement. Epidemiological data indicates that only 0.5 - 5.0% of dengue will progress to severe dengue.[16, level III]

The World Health Organization (WHO) South-East Asia Regional Office (SEARO) 2025 framework recognises Expanded Dengue Syndrome (EDS) as a separate entity, referring to atypical or uncommon manifestations of dengue that do not fit neatly into the classic categories. These manifestations may occur with or without plasma leakage.[17]

Symptomatic dengue infection is a systemic and dynamic disease, with clinical, haematological, and serological parameters changing from day to day. Understanding the evolving pathophysiology of dengue infection across its phases allows for a rational and safe approach to clinical management.

4.2. Clinical Course of Dengue Infection

The incubation period of dengue infection is 4 to 7 days. After the incubation period, the illness will be followed by two possible clinical courses. A systematic review of 43 studies globally found that 36.8% of symptomatic hospitalised dengue patients developed plasma leakage.[18, level III] This is illustrated in **Figure 7** below

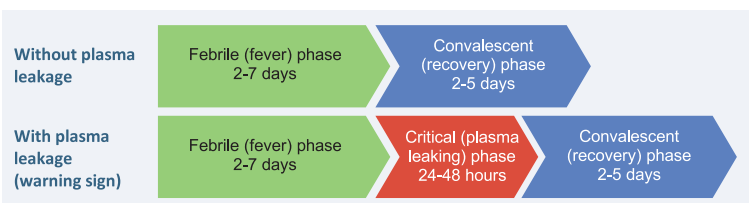


Figure 7: Clinical Course of Dengue Infection

Adapted from: World Health Organization Regional Office for South-East Asia. Dengue case management for primary health care and home-based care: Booklet for health professionals and health-care workers. World Health Organisation Regional Office for South-East Asia, 2025.

4.2.1. Febrile Phase

The febrile phase usually lasts 2 - 7 days, during which patients are generally haemodynamically stable. The majority of symptoms are due to viraemia.

Clinical features during this period include:[17; 19]

- sudden onset of high-grade fever
- headache
- retro-orbital pain
- myalgia and arthralgia
- gastrointestinal symptoms: nausea, vomiting and diarrhoea
- facial flushing or transient erythematous rash
- mild bleeding manifestation:
  - petechiae
  - positive tourniquet test
  - mild mucosal bleeding (nose bleeding, gum bleeding)

Common laboratory abnormalities during this phase include:

- leucopenia
- mild thrombocytopenia

4.2.2. Critical Phase

The critical phase occurs when fever subsides, usually between days 3 and 7 of illness. A subset of patients may develop severe plasma leakage resulting in:[1]

- hypovolaemia
- shock
- multiorgan failure

Warning signs in critical phase are as follows:[17; 19; 20;]

- Lack of clinical improvement or worsening symptoms during the period when fever subsides
- Intense abdominal pain or tenderness
- Persistent vomiting (>three times per day)
- Lethargy, restlessness, irritability or confusion
- Fluid accumulation
- Postural hypotension
- Liver enlargement >2 cm
- Reduced or absent urine output for 4 - 6 hours
- Bleeding manifestation
- Laboratory findings: rising haematocrit (HCT) with rapidly falling platelet count

4.2.3. Recovery/reabsorption phase

The recovery phase is characterised by:[19]

- improving general conditions and appetite

- convalescent rash – characterised as confluent petechial rash with scattering white halo on extremities, itching, especially on palms and soles
- diuresis
- increasing platelet count
- stabilisation of haematocrit
- normalisation of organ dysfunction

The following figure shows the course of dengue infection with plasma leakage:

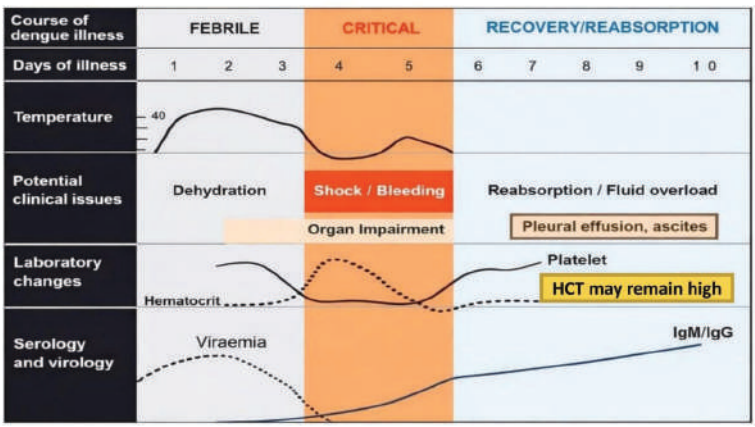


Figure 8: Course of Dengue Infection with Plasma Leakage

Adapted from:

1. World Health Organisation. Regional Office for South-East Asia. Dengue case management for primary health care and home-based care: Booklet for health professionals and health-care workers. World Health Organisation Regional Office for South-East Asia, 2025.
2. Ministry of Health Malaysia. CPG Management of Dengue Infection in Adults Third Edition. MoH; Putrajaya; 2015

4.3. Severe Dengue

Dengue complications arise primarily from plasma leakage, severe bleeding, and organ involvement. These complications may occur during the critical phase (plasma leakage) or at any stage of illness and require prompt recognition and management.[19]

4.3.1. Severe Plasma Leakage

Severe plasma leakage typically occurs during the critical phase and results from acute increases in vascular permeability, leading to extravasation of plasma into the extravascular compartment. This causes:[19]

- haemoconcentration
- hypovolaemia
- dengue shock

Hypovolemia may lead to vasoconstriction with the following manifestations:[19]

- Skin: coolness, pallor and delayed capillary refill time (CRT)
- Cardiovascular: tachycardia, increase in diastolic pressure, narrow pulse pressure.
- Renal: reduce urine output
- Gastrointestinal system: persistent vomiting and abdominal pain.
- Central nervous system: lethargy, restlessness, apprehension, reduced level of consciousness
- Respiratory system: tachypnoea (RR >20/min)

The following table shows a summary of the continuum of the various changes in a patient who progresses from a normal circulatory state to hypovolemic shock.

Table 1: Features comparing Compensated and Decompensated Shock

Normal Circulation	Compensated Shock	Decompensated/ Hypotensive Shock
Clear consciousness	Clear consciousness	Lethargy, restlessness, irritability or confusion
CRT≤3sec	CRT>3 sec	CRT>3 sec
Warm peripheries	Cool peripheries	Cool peripheries
Heart rate (HR) normal	Tachycardia	Severe tachycardia or bradycardia in late shock
Blood pressure (BP) normal	Normal systolic pressure with raised diastolic pressure	Hypotension
Normal pulse pressure	Narrowing pulse pressure (20 - 30 mmHg)	Narrowed pulse pressure (<20 mmHg)
Normal respiratory rate	Tachypnoea	Tachypnoea or Kussmaul's breathing
Normal urine output	Reduced urine output	Oliguria or anuria

Adapted from: Ministry of Health Malaysia. CPG Management of Dengue Infection in Adults Third Edition. MoH; Putrajaya; 2015

4.3.2. Severe Bleeding

Severe bleeding is bleeding which is significant enough to require medical intervention. It may occur at any phase, and patients can present with:[1]

- gastrointestinal haemorrhage
- intracranial bleeding
- massive occult bleeding (e.g. retroperitoneal haematoma)

4.3.3. Severe Organ Involvement

Severe organ involvement may occur independent of dengue shock which include:[1]

- Liver: severe hepatitis, acute liver failure
- Central nervous system: encephalopathy, encephalitis, seizures
- Heart: myocarditis, arrhythmias
- Kidneys: acute kidney injury

4.4. Differential Diagnosis

Clinical features of dengue are rather non-specific and may mimic other diseases. Dengue fever should be suspected in any patient presenting with acute febrile illness, myalgia, headache, retroorbital pain and rash.[20]

Differentiation of dengue should be guided by epidemiological risk, clinical progression, laboratory trends (haematocrit, platelet count and liver enzymes) and appropriate investigations in addition to dengue confirmation tests (NS1 antigen, PCR, serology). **Table 2** shows common mimics of dengue fever.

Table 2: Differential Diagnosis of Acute Febrile Illness and Severe Dengue

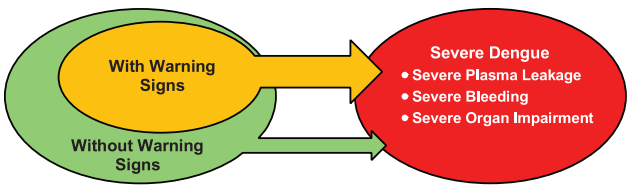
Categories	Febrile Phase	Severe Dengue
Viral infections	• Influenza • Measles • Rubella • Chikungunya • Zika • Covid-19 • Viral haemorrhagic fevers • Infectious Mononucleosis • Acute HIV seroconversion • Viral gastroenteritis • Hepatitis	
Bacterial infections	• Leptospirosis • Typhoid • Rickettsial infections • Meningococcal Infection, Scarlet Fever • Syphilis • Bacterial gastroenteritis	• Sepsis • Severe leptospirosis
Parasitic infections	• Malaria	• Severe Malaria
Others	• Systemic Lupus Erythematosus	• Disseminated intravascular coagulopathy • Other haematological disorders • Acute abdomen

5. DISEASE CLASSIFICATION

5.1. Classification of Dengue Severity

This guideline adopts the dengue classification proposed by the Pan American Health Organisation (PAHO), which stratifies patients according to clinical severity and level of care[20] similar to WHO 2009 classification.[1]

Table 3: Classification of Dengue Severity



Dengue without warning signs (DNWS)	Dengue with warning signs (DWWS)	Severe Dengue (SD)
Person who lives or has travelled to areas with dengue transmission in the last 14 days and presents with fever, usually of 2 to 7 days duration, and at least 2 of the following criteria: • Nausea / vomiting • Rash • Headache/ retro-orbital pain • Myalgia / arthralgia • Petechiae • Leukopenia	Dengue case near or in **critical phase**, that presents with one or more of the following signs: • Lack of clinical improvement or worsening symptoms during the period when fever subsides • Intense abdominal pain or tenderness • Persistent vomiting (>three times per day) • Lethargy, restlessness, irritability or confusion • Fluid accumulation • Postural hypotension • Liver enlargement >2 cm • Reduced or absent urine output for 4–6 hours • Bleeding manifestation • Laboratory findings: ○ Rising haematocrit with rapidly falling platelet count	Dengue case that has one or more of the following manifestations: • Severe plasma leakage leading to shock or respiratory distress • Severe bleeding which is significant enough to require medical intervention • Severe organ involvement (liver impairment, myocarditis, encephalitis, etc.)
	Requires strict monitoring and immediate medical intervention	

Adapted from: Pan American Health Organization (PAHO). Algorithms for the Clinical Management of Dengue Patients. Washington DC: PAHO; 2020. (Available from: https://www.paho.org)

Interpreting Warning Signs

- Warning signs are most predictive for severe dengue during the critical phase.
- Symptom such as persistent vomiting during febrile phase should not be interpreted as warning signs.
- Interpretation should be integrated with:
  - haemodynamic state
  - haematocrit trend
  - urine output
  - perfusion markers (e.g. CRT >3 sec)
- Phase-based interpretation of symptoms and signs reduces over-triaging.

5.2. Expanded Dengue Syndrome

In some guidelines, expanded dengue syndromes (EDS) refers to uncommon or atypical manifestations of dengue beyond plasma leakage, which include conditions mentioned in severe organ involvement (WHO 2009 Classification) and other conditions as illustrated in **Table 15**. EDS is more frequently observed in older adults and patients with underlying comorbidities.[21, level III] For further reading, refer to **Chapter 9.4.3** on **Organ Involvement/Expanded Dengue Syndrome**.

6. SURVEILLANCE / DISEASE NOTIFICATION

Dengue Fever and Dengue Haemorrhagic Fever have been listed as part of the notifiable diseases under First Schedule of the Prevention and Control of Infectious Diseases Act 1988 (Act 342) that mandates medical practitioners to notify suspected dengue to the nearest health office.[22] Failure to notify any suspected case by the medical practitioners can lead to charges against the individual involved with maximum compounds up to RM10,000 or imprisonment not exceeding two years or both.[23] All dengue cases has to be notified by medical practitioners within 24 hours of diagnosis being made, and compliance with this requirement is monitored as an indicator under the National Dengue Prevention and Control Program.[24]

In 1997, the WHO classified dengue clinical conditions into three distinct categories, namely (i) Dengue Fever, (ii) Dengue Haemorrhagic Fever and (iii) Dengue Shock Syndrome. This classification was later revised by WHO in 2009 into (i) Dengue with warning signs, (ii) Dengue without warning signs and (iii) Severe dengue. For billing and epidemiological reporting purpose, the Ministry of Health Malaysia is currently adopting the International Statistical Classification of Diseases, 11th Revision (ICD-11) for dengue which are reflected in **Table 4** below:

Table 4: ICD-11 Code for Dengue Diagnosis

ICD-11 Code	Diagnosis
1D20	dengue without warning signs
1D21	dengue with warning signs
1D22	severe dengue
1D2Z	dengue fever, unspecified

Various surveillance mechanisms have been established to monitor the dengue epidemiology in Malaysia.

- Sentinel-based dengue virus serotype surveillance in Malaysia has been started since 2012 to track circulating dengue virus serotypes and identify potential serotype shift in the country. Blood samples from dengue patients in 53 sentinel sites are sent to five Public Health Laboratories for weekly serotype and genotype testing.[25]
- Case-based surveillance utilises an integrated online database to track initial dengue notification and subsequent field investigations.
- Laboratory-based surveillance includes continuous testing of all dengue-related mortality and periodic dengue seroprevalence monitoring.

7. LABORATORY INVESTIGATIONS

7.1. Disease Monitoring Tests

Tests for disease monitoring in dengue include:

- **White cell count (WCC) and Platelet count**
  - In the early febrile phase WCC and platelet count are usually normal but will decrease rapidly as the disease progresses.
  - The decrease in WCC is accompanied by platelet reduction.
  - In the recovery phase, the WCC normalises, followed by platelets.
  - Leukocytosis and thrombocytopenia are predictive of severe disease.
  - There is no correlation between platelet count and prediction of bleeding.

- **Haematocrit (HCT)**

A rising HCT (≥20% from baseline) is a marker of plasma leakage in dengue infection. The median values of normal HCT level among Malaysian populations are:

- male ≤60 years – 46%
- male > 60 years – 42%
- female (all age groups) – 40%

HCT should be interpreted in conjunction with the patient's haemodynamic status, disease phase and certain medical conditions.

Other important blood tests in disease monitoring include liver function test (LFT/AST), renal profile (RP), coagulation profile, lactate and blood gases. Requests for specialised investigations, such as troponin and creatine kinase (CK), should be discussed with the treating physician.

7.2. Diagnostic Tests

The choice and interpretation of diagnostic tests depend on the phase of illness, host immune status, and the local epidemiology of dengue virus circulation. Available diagnostic modalities include direct viral detection methods; dengue non-structural protein 1 (NS1) antigen detection, dengue virus reverse transcriptase–polymerase chain reaction (RT-PCR) and viral isolation, as well as serological assays detecting dengue-specific immunoglobulin M (IgM) and immunoglobulin G (IgG).[19]

A meta-analysis of rapid immunochromatographic tests (ICT) for suspected dengue compared with PCR, ELISA, or virus isolation as reference tests concluded that among all test, the NS1/IgM/IgG combination yielded the highest sensitivity (91.0%, 95% CI 84.0 to 95.0) and specificity (96.0%, 95% CI 91.0 to 98.0). Quality assessment of the primary papers included using QUADAS-2 was mixed.[26, level III]

While a systematic review showed RT-PCR and NS1 ELISA having strong accuracy among those in early days of illness (0 – 4 days post-symptom onset):[27, level III]

- RT-PCR showed a sensitivity of 95% (95% CI 77 to 99) and specificity of 89% (95% CI 60 to 98)
- NS1 ELISA showed a sensitivity of 90% (95% CI 68 to 98) and specificity of 93% (95% CI 71 to 99)

In contrast, IgM ELISA alone is less reliable during the initial phase, reinforcing the value of multi-marker or molecular approaches for optimal diagnostic performance.[27, level III]

For rapid dengue detection, a diagnostic study on patients with acute febrile illness suspected of dengue evaluated the performance of fluorescent immunoassay (FIA) and immunochromatography (IC) against the definitive diagnostic method by RT-qPCR and ELISA, showed:[28, level III]

- For NS1 detection, both tests performed well, but FIA had slightly better agreement and predictive values
  - FIA: Sensitivity 79.1%, Specificity 92.3%
  - IC: Sensitivity 76.6%, Specificity 92.3%
- For IgM and IgG, FIA was more specific but less sensitive than IC. Both were inferior to ELISA in overall accuracy
  - FIA IgM/IgG combined: Sensitivity 40.5%, Specificity 89.8%
  - IC IgM: Sensitivity 53.8%, specificity 80.9%
  - IC IgG: Poor performance across all metrics

A new biosensor-based rapid diagnostic test was assessed in a local diagnostic study for the point-of-care diagnosis of previous and recent dengue infections showed that ViroTrack Dengue Serostate vs Panbio Dengue IgG, demonstrated high diagnostic accuracy:[29, level III]

- Previous dengue - sensitivity 91.1%, specificity 91.1%, positive predictive value (PPV) 97.5%, negative predictive value (NPV) 73.0%
- Recent dengue - sensitivity 90.2%, specificity 93.2%, PPV 55.2%, NPV 99.0%

However, ViroTrack Dengue Serostate is currently not available in Malaysia.

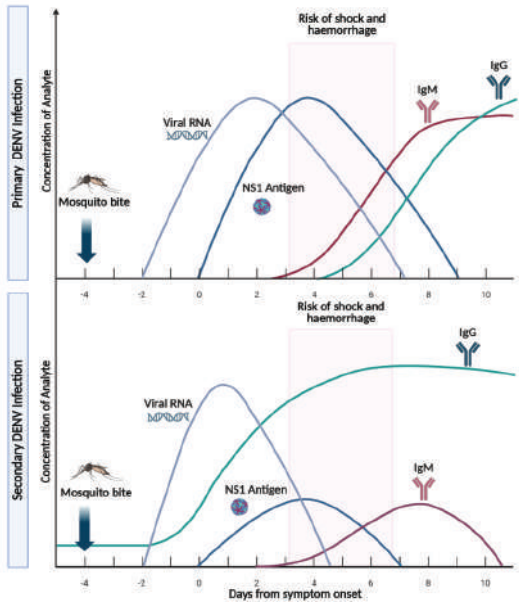


Figure 9: Timeline of dengue biomarker appearance in primary and secondary infection.

Source: World Health Organization. Laboratory Testing for Dengue Virus: Interim Guidance, April 2025. Geneva: WHO; 2025

NS1 is a highly conserved glycoprotein detectable in serum for up to 9-10 days after symptom onset in primary infections, whereas in secondary infections, detection declines by day 6 - 7. RDTs or laboratory-based ELISA are used for NS1 detection. NS1 ELISA is highly sensitive (100%) and specific (99.2%) for DENV and has been reported to be as specific as conventional molecular diagnostic tests.[30]

In the convalescent phase, IgM antibodies typically become detectable by day 5 of illness (indicating recent infection), peaking by two weeks and typically waning by three months, whereas IgG antibody response starts around day 8 and lasts a lifetime (suggesting past infection).

In primary and early febrile phase infections, viral RNA and NS1 antigens are detectable early, with RNA persisting for up to seven days and NS1 up to nine days, or longer in some cases. IgM antibodies emerge around days 3 - 5, while IgG appears later.

In secondary infections, IgG levels rise rapidly within 1-2 days of symptoms onset, NS1 and viral RNA are cleared more quickly, and IgM

is either absent or present at lower levels. In secondary infections, the sensitivity of NS1 declines (67.1 - 77.3%) compared to 94.7 - 98.3% for primary infections, reported for ELISA and RDT tests, due to pre-existing IgG antibodies forming antigen–antibody complexes.[30]

It is important to note that usage of point-of-care rapid diagnostic tests (RDTs) that combine NS1 and IgM/IgG provides rapid results, and the positive predictive value can be higher when both analytes are considered together than when relying on a single analyte (NS1 or IgM alone).

Single-specimen IgG testing is not recommended for diagnosing acute dengue due to potential cross-reactivity with past infections; however, a change from negative to positive IgG in paired samples can signify a current infection.[30]

Rapid IgG tests detect only the presence of IgG and cannot quantify antibody level. Therefore, they should not be used to diagnose secondary dengue. However, capture enzyme-linked immunosorbent assays (ELISA), such as the Panbio Dengue IgG capture ELISA, can detect high-titre dengue-specific IgG, which is more consistent with secondary dengue infection when interpreted in the appropriate clinical and temporal context.[30; 31 - 32, level III] Refer **Table 5** on **Interpretation of Dengue Diagnostic Tests**.

Table 5: Interpretation of Rapid Diagnostic Tests for Dengue

Clinical History	Test	Result			Interpretation
		NS1	IgM	IgG	
History of fever less than 5 days	Rapid Diagnostic Test (RDT) for Dengue	⊕	⊖	⊖	Acute dengue infection
		⊕	⊕	⊖	Acute dengue infection
		⊕	⊖	⊕	Acute dengue infection or secondary infection
		⊖	⊖	⊖	Not dengue, but if history suggestive of dengue, repeat RDT
History of fever more than 5 days	RDT for Dengue	⊕	⊕	⊖	Acute dengue infection. High risk for severe dengue
		⊖	⊕	⊖	Acute or recent dengue infection
		⊖	⊕	⊕	Acute, recent or secondary dengue
		⊖	⊖	⊕	Unable to rule out secondary dengue, proceed with dengue IgG high-titre
		⊖	⊖	⊖	Not dengue infection

For details on sample collection for dengue tests refer to **Appendix 3** on **Sample Collection for Dengue Tests**.

Negative test results require differential diagnosis to confirm cause of infection. Interpretation of results requires consideration of clinical presentation, epidemiological context and local dengue transmission patterns.

7.3. Post-Vaccination Diagnostic Considerations

In patients presenting with suspected dengue within 30 days after receipt of TAK-003/Qdenga vaccine, laboratory confirmation should preferably be performed using RT-qPCR on serum collected within 5 days of symptom onset. A positive RT-qPCR result may reflect wild-type dengue virus, vaccine-strain viraemia, or both. Therefore, genomic sequencing should be considered where available, particularly in severe cases, fatal cases, clusters, or cases requiring vaccine safety investigation. Dengue IgM and IgG should not be used as standalone confirmatory tests in this period, as vaccine-induced antibodies may resemble those seen in natural dengue infection. NS1 antigen testing may be considered when RT-qPCR is unavailable, but a positive result does not differentiate wild-type from vaccine-derived virus, and a negative result does not fully exclude dengue or vaccine-strain viraemia.[33, level III; 34] Healthcare providers need to rely on clinical evaluation and judgement to diagnose and treat patients accordingly.

Recommendation 1

- Rapid Diagnostic Test (RDT) with combined non-structural protein 1 (NS1) Antigen/ IgM/IgG markers should be used as the first-line test to diagnose dengue in adults.
- In dengue vaccinated individuals within 30 days post-vaccination:
  - dengue polymerase chain reaction (PCR) or NS1 antigen may be used to diagnose dengue.
  - dengue IgM and IgG should not be used as confirmatory tests

  However, healthcare providers need to rely on clinical evaluation and judgement to diagnose and treat patients accordingly

7.4. Dengue Post-Mortem Investigations

Post-mortem evaluation supports confirmation of dengue-related mortality and identification of complications. Post-mortem investigations in dengue-related mortality cases include:[19]

- RT-PCR for cerebrospinal fluid (CSF) or tissue (liver, spleen and kidney) samples for confirmation
- NS1 antigen testing on serum samples for confirmation

- Assessment for plasma leakage, haemorrhage, organ congestion, and secondary infections. Repeat diagnostic test, such as dengue serology, if warranted
- Blood culture if bacteraemia is suspected

8. DENGUE ASSESSMENT

8.1. Predictors of Severe Dengue

Patients with suspected or confirmed dengue should be systematically assessed for predictors of severe dengue using clinical evaluation, serial laboratory monitoring, and focused imaging when available. The World Health Organization 2009 classification framework reflects disease progression, identifies high-risk patients, and guides appropriate intervention and escalation of care.[1]

8.1.1. Clinical/ Warning Signs

A large body of observational evidence indicates that multiple clinical and epidemiological factors are associated with the progression to severe dengue and an increased need for hospitalisation.

Older age was independently associated with increased mortality among patients with dengue (OR=1.037, 95% CI 1.009 to 1.066).[35, level II-2] Overweight or obesity (BMI >25 kg/m²) has also been identified as a predictor of severe dengue (OR=1.784, 95% CI 1.040 to 3.057).[36, level II-2]

Pre-existing comorbidities further increase the risk of severe disease. Poorly controlled Type- 2 Diabetes Mellitus was strongly associated with severe dengue (OR=6.207, 95% CI 2.464 to 15.636).[37, level II-2] In addition, the presence of two or more comorbidities significantly increases the likelihood of severe dengue (OR=10.15, 95% CI 3.53 to 29.23) and substantially increased mortality risk.[38, level III]

Patients presenting with warning signs are at increased risk of progressing to severe dengue and require close monitoring with timely escalation of care. While individual warning signs may have limited sensitivity, the presence of multiple warning signs provides improved predictive value in detecting dengue progression.[1] Altered mental status (OR=8.591, 95% CI 2.914 to 25.330) and gastrointestinal (GI) bleeding (OR=1.939, 95% CI 1.037 to 3.626) were independently associated with mortality in dengue patients.[35, level II-2]

The development of in-hospital complications, including acute kidney injury (OR=2.627, 95% CI 1.373 to 5.028) and pneumonia (OR=2.250, 95% CI 1.054 to 4.802), was also independently associated with increased mortality.[35, level II-2]

8.1.2. Laboratory Test

Key laboratory parameters have been consistently identified as predictors of severe dengue and adverse outcomes. During the first 72 hours of illness, patients with severe dengue had significantly lower platelet counts (SMD= -0.65, 95% CI -0.97 to -0.32) and higher AST

levels (SMD=0.87, 95% CI 0.36 to 1.38), highlighting the prognostic value of early thrombocytopenia and transaminase elevation during the febrile phase.[39, level II-2]

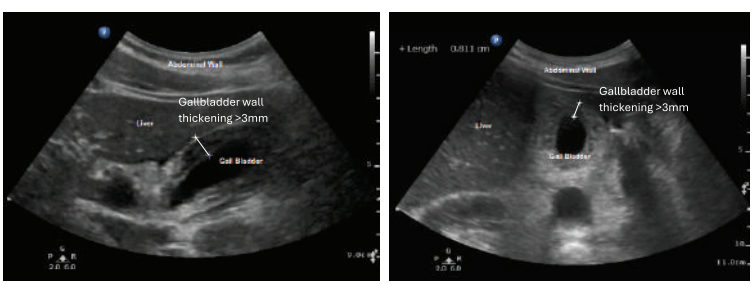
Marked thrombocytopenia (<50 × 10^9/L) has been strongly associated with mortality (OR=15.70, 95% CI 5.65 to 43.53)[38, level III] while leukocytosis independently predicted both severe disease (OR=1.52, 95%CI 1.21 to 1.90) and mortality (OR=2.504, 95% CI 1.124 to 5.578), warranting close monitoring on admission.[35, level II-2; 40, level II-2]

Severe hepatitis (ALT >1000 U/L) during hospitalisation was also predictive of dengue mortality (OR=5.324, 95% CI 2.199 to 12.889). Furthermore, prolonged prothrombin time (PT) (OR=1.745, 95% CI 1.26 to 2.40) was independently associated with severe outcomes in dengue.[40, level II-2]

8.1.3. Point-of-care Ultrasound (PoCUS)

Point-of-care ultrasound (PoCUS) has emerged as a valuable tool for early detection of plasma leakage in dengue, often preceding laboratory abnormalities and haemodynamic compromise.

Gallbladder wall oedema or thickening (>3 mm) independently predicted severe dengue (OR=2.74, 95% CI 1.71 to 4.41).[41, level II-2] Additionally, wall thickness also increased with dengue severity (p<0.05),[42, level III] thus supporting its value as an early sonographic marker of severe dengue. (**Figure 10** shows gallbladder wall thickening on PoCUS) Similarly, presence of ascites (OR=2.04, 95% CI 1.20 to 3.43) and pleural effusion or organomegaly (OR=2.62, 95% CI 1.63 to 4.20) was associated with increased risk of severe disease.[41, level II-2]



Note: Ultrasound appearance of Gall Bladder Wall Thickening >3 mm and striated pattern (multiple linear hypoechoic layers separated by parallel echogenic layers)

Figure 10: Ultrasound appearance of Gallbladder wall thickening in dengue fever with warning signs

Source: Emergency and Trauma Department, Hospital Kuala Lumpur

8.1.4 Dengue Severity Scoring Tools

The Aspartate Aminotransferase-to-Platelet Ratio Index (APRI), calculated from AST levels and platelet counts, reflects hepatic involvement and thrombocytopenia in dengue. Elevated APRI scores (range of ≥3.2 to 9.04) have been independently associated with an increased risk of severe dengue.[40, level II-2; 43, level II-2; 44, level III] When combined with white blood cell count (WCC) and prolonged prothrombin time (PT), the predictive accuracy of APRI improved substantially (AUC=0.877).[40, level II-2] However, optimal APRI cut-off values vary across studies, and external validation in diverse populations remains limited.

Plasma Leak Scores, which integrate parameters such as BMI >25 kg/m², platelet count <100 × 10^9/L, and AST or ALT ≥100 U/L on days 3 - 4, have also been shown to correlate with dengue severity and identify patients at higher risk of plasma leakage. Each incremental increase in the score (range 1 to 3) was independently more predictive of severe dengue [specificity increased from 29.8% (score ≥1) to 77.7% (95% CI 69.2 to 84.8) and 93.4% (95% CI 87.4 to 97.1)] for a score of ≥2 and 3.[36, level II-2] However, these scoring systems currently lack standardisation and broad external validation.

At present, neither tool is endorsed by the WHO. The WHO 2009 classification remains the globally accepted framework for dengue severity assessment.[1] These scores should be considered adjunctive tools that complement clinical, laboratory, and imaging assessments. Their role as a predictive tool remains tentative until additional validation studies confirm their effectiveness.

Predictors of Severe Dengue

- Patient's factor:
  - Older age
  - Overweight/obesity (BMI >25 kg/m²)
  - Co-morbid conditions: Type-2 DM with poor glycaemic control, presence of one or more co-morbidities
- Clinical Signs:
  - Presence of multiple warning signs
  - Gastrointestinal bleeding
  - Clinical features suggestive of plasma leakage
  - Altered mental status
  - In-hospital complications e.g., acute kidney injury or pneumonia
- Laboratory Markers:
  - Hematologic: Early decline in platelet count in febrile phase, severe thrombocytopenia (platelet count <50 × 10^9/L), leucocytosis or rising haematocrit with rapid drop in platelet

- Biochemical: Early elevation of AST/ALT in febrile phase or severe hepatitis >1000 U/L
- Coagulation: Prolonged PT

• Imaging: Gallbladder wall oedema/ increased gallbladder wall thickness, pleural effusion, ascites and/or organomegaly

Severe dengue is predicted by a combination of patient vulnerability and early clinical deterioration, laboratory abnormalities, underlying comorbidities, and ultrasound evidence of plasma leakage with outbreak locality contributing to mortality risk.

9. MANAGEMENT OF DENGUE INFECTION

Management of dengue patients require a thorough clinical assessment by integrating dengue severity classification with the PAHO 2020 recommended level of care, ensuring appropriate disposition to outpatient management, hospital observation, emergency treatment or admission according to disease severity. Refer **Table 6** below showing Dengue Severity Classification and Management.

Table 6: Dengue Management Based on Severity

Category	Group A	Group B1	Group B2	Group C
Severity classification	**Dengue without warning signs (DNWS)**	**Dengue without warning signs (DNWS)**	**Dengue with warning signs (DWWS)**	**Severe dengue (SD)**
Group criteria	• Stable general well-being (able to drink adequately, good urine output and able to ambulate) • Absence of warning signs in critical phase • Haemodynamically stable • Stable blood investigations • Absence of unstable comorbidities or social factors	Presence of **comorbidities:*** • Hypertension • Diabetes mellitus • Asthma/ COPD • Chronic Kidney disease • Haemolytic diseases • Chronic liver disease • Peptic ulcer disease or gastritis of any aetiology • Being treated with anticoagulants/ antiplatelet • others **High risk group:** • Pregnancy • ≥65 years old • Obesity • Patients with special needs **Social risk:** • The patient lives alone or far from health facilities • Does not have transportation *Unstable comorbidities must be referred for assessment	Dengue case near or in **critical phase**, that presents with one or more of the following signs: 1. Lack of clinical improvement or worsening of symptoms when fever subsides 2. Intense abdominal pain or tenderness 3. Persistent vomiting (>three times per day) 4. Lethargy, restlessness, irritability or confusion 5. Fluid accumulation 6. Postural hypotension (lipothymia) 7. Liver enlargement >2 cm 8. Progressive increase in haematocrit with drop in platelets 9. Bleeding manifestation 10. Reduced or absence of urine output 4-6 hours	Every dengue case that has one or more of the following manifestations: • Severe plasma leakage leading to shock or respiratory distress • Severe bleeding which is significant enough to require medical intervention • Severe organ involvement (liver impairment, myocarditis, etc.)
Management of care	**Outpatient monitoring**	**Possible referral to hospital** May require close observation and treatment of their associated condition.	**Admit to hospital**	**Intensive Care Unit** Requires emergency treatment

Adapted from: 1. Pan American Health Organization (PAHO). Algorithms for the Clinical Management of Dengue Patients. Washington DC: PAHO; 2020. (Available from: https://www.paho.org)

2. Ministry of Health, Malaysia. Emergency Medicine and Trauma Services Policy. Putrajaya: MoH; 2012. (Available from: https://www.moh.gov.my/moh/resources/Polisi/EMTS_Book.pdf.)

The decision for referral and admission must not be based on a single clinical parameter instead should depend on the total assessment of the patient.

9.1. Outpatient Management of Dengue Infection

The management of dengue infection is symptomatic and supportive. An approach to outpatient evaluation is as suggested in **Table 7**.[19] Management issues vary according to the three phases of the clinical course (Refer **Figure 8** illustrating the Course of Dengue Infection with Plasma Leakage). It is crucial to recognise plasma leakage, early shock and severe organ dysfunction. This can be achieved by serial clinical and laboratory monitoring using the Dengue Clerking Sheet for Adults in Primary Care (Refer **Appendix 4**). Dengue patients managed in the outpatient setting should be provided with an outpatient dengue monitoring record (Refer **Appendix 5**) to ensure that relevant information is available for continuity of care by healthcare providers.

Table 7: Dengue Clinical Assessment in Primary Care

Steps	Key Components
History Taking	● Day of illness and date of fever onset. ● Symptoms suggestive of dengue (e.g. fever, nausea, vomiting, rash, body aches and pain such as headache, retro-orbital pain, myalgia, arthralgia) and living in/travelling to a dengue endemic area ● Presence of dengue warning signs such as intense abdominal pain, lethargy, restlessness, irritability or confusion, reduced or absent urine output for 4-6 hours, bleeding manifestations or persistent vomiting (>3 times over 24 hours) ● Daily function such as daily activity, oral fluid intake and urine output. ● Comorbidities such as obesity, heart failure, end-stage kidney disease (ESKD), chronic lung disease, diabetes mellitus, hypertension, ischemic heart disease, chronic obstructive pulmonary disease ● Special group populations such as pregnancy, elderly > 65 years old and infant. ● Social factors affecting follow-up and access to care ● Dengue vaccination status and date of vaccination.
Clinical Examination	● General condition, mental status and Glasgow Coma Scale, hydration status ● Haemodynamic status such as blood pressure, pulse pressure, pulse rate, respiratory rate and temperature. ● Anthropometry such as weight, height and BMI ● Assessment of CCTVR (Colour, Capillary Refill Time (CRT), Temperature, Pulse Volume, Pulse Rate). **Refer to Appendix 6 on Capillary Refill Time Assessment.** ● Assessment of shock. ● Respiratory status (tachypnoea, pleural effusion) ● Abdominal examination (tenderness, hepatomegaly, ascites) ● Assess bleeding tendency (skin, mucosa, injection sites)
Investigations	● Point-of-care test for dengue (Rapid Combo Test or NS1 antigen). If unable to perform, patients should be referred to the nearest healthcare facility where it can be done. ● Full Blood Count (FBC) ● AST (if available) ● Ultrasound (PoCUS - if trained and available)

Diagnosis	● **Suspected dengue:** Fever + ≥2 symptoms +/- supportive lab findings
	● **Confirmed dengue:** laboratory evidence of dengue infection
	● A complete diagnosis should include the following: ○ Day of dengue fever, ○ Dengue phase: febrile, critical or recovery ○ Classification of dengue: ▪ DNWS ▪ DWWS (during critical phase) ▪ Severe dengue ○ Presence of shock: not in shock, compensated shock or decompensated shock. ○ Comorbidities or special populations
Management Decision, Patient Education & Safety Netting	● Notify the district health office ● Determine suitability for outpatient management (**Group A**), referral (**Group B1**) or admission (**Group B2** and **C**). Refer to **Table 6 on Dengue Severity and Management**. ● If managed at an outpatient department (**Group A - DNWS)** ○ Arrange daily follow-up and monitoring until patient is afebrile for **24 - 48 hours without antipyretics** ○ Provide **Outpatient Dengue Monitoring Record** (**Appendix 5**) ○ Educate the patient on DOs and DON'Ts, warning signs and fluid intake. ○ Advise immediate return if the patient has warning signs.

Note:
Suspected dengue: A case compatible with clinical description
Confirmed dengue: A case compatible with the clinical description and laboratory confirmed

Sources:
1. Ministry of Health Malaysia. Management of Dengue Infection in Adults (Third Edition). Putrajaya: MoH; 2015.
2. Ministry of Health Malaysia. Case definitions for infectious diseases in Malaysia (4th Edition). Putrajaya: MoH; 2026.

Prerequisites for transfer to the hospital
- All efforts should be taken to stabilise the patient's condition before and during transfer.
- The Medical Department and/or Emergency Department of the receiving hospital must be informed before transfer.
- Adequate and essential information must be sent together with the patient, which includes dengue clerking sheet, fluid chart, monitoring chart and investigations

9.1.1. Outpatient Monitoring

All dengue patients who can be monitored as outpatients need to have daily follow-up, particularly when they are in the febrile and critical phase.[19] All fields in the dengue clerking sheet are required to be completed and dengue monitoring record must be updated at each follow up visit.

Given the dynamic nature of dengue infection, the diagnosis should be re-evaluated and updated in accordance with the patient's current clinical findings and laboratory results at every follow-up visit. For patients who remain suitable for outpatient care, reinforce on the home care advice and education, and to seek immediate medical attention if warning signs develop or symptoms worsen.[19]

Recommendation 2
- Dengue clerking sheet and dengue monitoring record must be completed and updated at each follow-up visit for all suspected/ confirmed dengue patients.
- The diagnosis of dengue should be re-evaluated and updated in accordance with the patient's current clinical findings and laboratory results at every follow-up visit.

9.1.2. Outpatient Treatment

a. Fluid Intake

WHO recommends protocolised oral fluid therapy which is a structured administration of oral fluids. To maintain adequate hydration, adults should drink approximately 200 ml of fluid every hour while they are awake. Patients with fluid restrictions must adhere to their fluid regimen. Fluids containing electrolytes, such as fruit juice, oral rehydration salts (ORS), coconut water or soup, are recommended to maintain hydration and normal urine output. Plain water alone should be avoided because it can lead to electrolyte imbalance.[17]

- Patients with signs of moderate to severe dehydration require intravenous (IV) fluid therapy and should be referred for hospital management.[17]

b. Urine output Monitoring

Maintaining a good urine output of minimum 0.5 ml/kg/hour is essential for proper hydration. Urine output should be monitored and documented as any reduction in urine output can be promptly identified and addressed.[17] A patient should pass urine 4 to 6 times per day, with urination occurring at least once every 6 hours.

c. Treatment of fever

Fever in dengue needs to be adequately managed. WHO suggests the use of paracetamol for pain relief and/or fever in patients with suspected or confirmed non-severe dengue.[17] However, paracetamol should be used with caution as hepatic involvement is common in dengue. An RCT comparing paracetamol (median dose 2.5 g) with placebo as an antipyretic in hospitalised adult dengue patients found a higher rate

of transaminase elevation in the paracetamol group.[45, level I] Therefore, paracetamol should be used judiciously at the lowest effective dose and not to exceed 4 g/day. Alternative non-pharmacological measure for fever reduction such as warm bath may be used as supportive management.[17]

d. Use of other Medications

- **Non-steroidal anti-inflammatory drugs**

The WHO strongly advised against using non-steroidal anti-inflammatory drugs (NSAIDs) in all dengue patients, regardless of disease severity. For patients who are already taking NSAIDs for chronic conditions, the decision to continue therapy should be made on a case-by-case basis, carefully weighing the potential risks and benefits.[17]

- **Steroids**

The WHO advised against the use of corticosteroids in patients with acute suspected or confirmed non-severe dengue. For patients already receiving corticosteroids for other medical conditions, the decision to continue or withhold treatment should be based on an individual assessment of risks and benefits.[17]

- **Antiplatelet**

In dengue patients, the continuation or discontinuation of antiplatelet therapy such as aspirin should be based on clinical judgement. A retrospective cohort study of patients who had been on antiplatelet therapy for at least 180 days before dengue infection found non-significant (NS) differences in major adverse cardiac and cerebrovascular events (MACCE) or all-cause mortality between those who continued therapy and those who discontinued it. Continuation of antiplatelet therapy also was not associated with an increased need for platelet, blood transfusion or occurrence of dengue haemorrhagic fever (DHF).[46, level II-2]

- The CPG DG opines that continuation of antiplatelet in patients who have had recent stenting requires individualised assessment and consultation with a cardiologist/ neurologist/vascular surgeon. The ultimate decision should weigh the risk of in-stent thrombosis against the risk of bleeding associated with dengue.

- **Anticoagulant**

Patients with dengue who are receiving anticoagulant therapy should be closely monitored and may require referral to hospital for further assessment and management.[20]

- **Antibiotics**

There was no retrievable evidence on the use of antibiotics in the treatment of dengue. Inappropriate antibiotic use in dengue fever may lead to antimicrobial resistance.

- **Statin**

A large cohort study showed no statistically significant difference between statin users and non-users in dengue severity (including DHF/DSS and severe dengue), liver enzyme levels (ALT/AST), length of hospital stay, ICU admission, or mortality. Statin use was not associated with increased liver inflammation and did not reduce dengue severity.[47, level II-2]

- The CPG DG opines that statins may be continued during acute illness if there are no contraindications.

- **Metformin**

A retrospective cohort study that determined the association between metformin use and dengue severity among diabetic patients with dengue reported metformin use was associated with a 33% reduction in the risk of severe dengue (cRR=0.67, 95% CI 0.46 to 0.96).[48, level II-2]

However, in a recent open-label RCT among obese dengue patients, the use of higher dose of metformin (≥1 to 1.5 g/day) was associated with significantly higher adverse events (e.g. severe diarrhoea, dengue shock, increased lactate >3.0 mmol/L, hypoglycaemia and persistent vomiting) compared to control (p=0.00031) resulting in 42% of patients discontinuing therapy.[49, level I]

- The CPG DG opines that metformin may be continued in diabetic patients with dengue infection if there are no contraindications (e.g., renal impairment, dehydration, haemodynamic instability), however, there is insufficient evidence to support its use solely for the prevention of severe dengue.

- **Sodium-Glucose Transport 2 inhibitors**

Patients on Sodium-Glucose Transport 2 inhibitors (SGLT2i) must follow “sick day rules” to reduce the risk of the rare but serious complication of euglycaemic diabetic ketoacidosis. The medication should be temporarily withheld during periods of dehydration (such as significant vomiting or diarrhoea), acute illness (including fever or suspected infection), or when oral intake is reduced. It can be safely restarted in the recovery phase.[50, level III]

Recommendation 3

- Suspected and confirmed dengue patients should be prescribed oral fluid therapy of 200 ml per hour while awake. Patients requiring fluid restriction should follow an individualised fluid regimen.
- Urine output should be monitored and documented in all dengue patients. The frequency of urination should be at least 4 to 6 times a day.
- Paracetamol should be used cautiously in dengue fever at the lowest effective dose and shortest duration. Paracetamol should be avoided or discontinued if liver enzymes are deranged.
- Non-steroidal anti-inflammatory drugs (NSAIDs) and corticosteroids should not be used in dengue patients. In patients already receiving these medications, continuation should be guided by clinical judgement and individual clinical indications.
- Continuation of antiplatelet in dengue patients should be guided by clinical judgement.
- Statins and metformin can be safely continued during dengue infection in the absence of contraindications.

9.1.3. Dengue Home Care Advice

All dengue patients must be given clear verbal and written home care advice.[19] Refer to **Appendix 5** on **Dengue Monitoring Record and Home Care Advice**.

9.2. Management of Dengue in The Emergency Department

9.2.1. Triage

The objective of triage is to stratify patients according to clinical severity to ensure timely assessment and appropriate treatment.[51] Triage also aims to preserve life while reducing unnecessary dengue admissions by identifying patients at risk of complications.[1] Early recognition and understanding of clinical problems across the different phases of dengue are essential, as accurate triage is a key determinant of clinical outcome.[1]

a. Primary triage

Primary triage begins with a focused 'first ask' on history of fever and outbreak locality, accompanied by a rapid 'first look' assessment of appearance, airway, breathing, circulation, and disability (AVPU) to identify life-threatening conditions.[51] This is followed by a 'first touch' evaluation of peripheral perfusion to detect shock, including assessment of skin colour, capillary refill time, temperature, pulse volume, and pulse rate (CCTVR).[1; 52; 53]

b. Secondary triage

Patients without life-threatening features should subsequently undergo secondary triage, which includes assessment for warning signs, identification of co-morbidity, vital sign evaluation, and review of available point-of-care test results.[51; 54]

c. Integration Between Malaysian Triage Scale and Dengue Classification

Integration of the Malaysian Triage Scale (MTS) with dengue classification allows prioritisation of care based on severity and risk of deterioration directly correspond to the PAHO-recommended level of care and management setting, enabling timely resuscitation, appropriate hospital admission, or safe outpatient management according to disease severity.[20]

Table 8: Integration of MTS with Dengue Classification and Recommended Management

MTS Level	Zone	Dengue Classification	PAHO 2020 Recommended Management*
Level 1 Resuscitation	Red	Decompensated dengue shock syndrome	Immediate resuscitation, ICU admission (Group C)
Level 2 Emergency	Red	Compensated dengue shock syndrome	Emergency treatment, IV fluid therapy and admission (Group C)
Level 3 Urgent	Yellow	Dengue with warning signs or without warning sign but unstable co-morbidity	IV fluid therapy, admission, management of associated conditions (Group B2)
Level 4 Early care	Green	Dengue without warning signs with stable comorbidity	Observation or admission, management of associated conditions (Group B1)
Level 5 Routine care	Outpatient	Dengue without warning signs, comorbidity and social risk	At-home outpatient management with follow-up (Group A)

Adapted from:

1. Pan American Health Organization (PAHO). Algorithms for the Clinical Management of Dengue Patients. Washington DC: PAHO; 2020. (Available from: https://www.paho.org)
2. Ministry of Health, Malaysia. Emergency Medicine and Trauma Services Policy. Putrajaya: MoH; 2012.

Refer to **Appendix 7** on **Systematic Triage of Dengue in Adults**

d. Human Resource and Training for Triage

Triage personnel should be appropriately trained to recognise and understand the evolving clinical problems across the different phases of dengue, ensuring accurate severity assessment and timely intervention.[55] An organised educational programme should therefore be developed to support effective and consistent dengue triaging within clinical services.[1; 55]

e. Triage During Dengue Outbreak

During a dengue outbreak, a dedicated fever clinic should be established to facilitate early triage of suspected dengue cases, and adequate space and bed capacity should be designated for patient admission.[52; 53]

9.2.2. Intervention in the Emergency Department

Patients with suspected or confirmed dengue require proactive and ongoing reassessment of airway, breathing, circulation, and neurological status (ABCD), with particular attention to early signs of haemodynamic compromise or clinical deterioration. Clinical evaluation should be repeated at regular intervals and after any therapeutic intervention to detect evolving plasma leakage, shock, bleeding, or organ dysfunction, ensuring timely escalation of care and appropriate resuscitation.

a. Laboratory Tests

These tests must be done immediately in cases of suspected dengue infection:

- FBC
- Rapid diagnostic test (RDT) for dengue
- Blood glucose

In suspected severe dengue, the following investigations should be done:

- Blood gases and lactate
- LFT and AST
- RP
- PT/APTT
- GXM
- Electrocardiogram (ECG)
- Troponin and CK in suspected myocarditis and rhabdomyolysis

b. Imaging in the Emergency Department

i. Chest x-ray (CXR)

CXR is not routinely required in uncomplicated dengue but is indicated in patients with respiratory distress to detect pleural effusion due to plasma leakage and to exclude alternative diagnoses such as pneumonia or cardiogenic pulmonary oedema.

ii. Point of Care Ultrasound (PoCUS)

PoCUS provides rapid bedside assessment that complements clinical evaluation and laboratory parameters. It should be performed by trained and competent healthcare personnel where available. Its major role is in predicting severe dengue by:[41, level II-2, 42, level III]

- Detection of early plasma leakage (gall bladder wall oedema)
- Evaluation of third space fluid accumulation (pleural effusion, ascites)

PoCUS can also be used to assist in:

- Assessment of intravascular volume status
- Assessment of fluid responsiveness
- Evaluation of fluid tolerance in patients with cardiac or renal dysfunction
- Differentiation of haemodynamic profiles in dengue shock (hypovolaemic, cardiogenic, distributive, or obstructive components)
- Monitoring of plasma resorption during recovery phase, which precedes clinical improvement

While current evidence is predominantly observational and does not yet establish definitive predictive thresholds, PoCUS offers real-time physiological assessment that can support management decisions, guide fluid resuscitation, and detect complications. Refer to **Appendix 8** on **PoCUS Guided Dengue Management**.

Recommendation 4

- Point-of-care ultrasound should be considered as an adjunctive bedside tool in the emergency and inpatient management of dengue to support early detection of plasma leakage and haemodynamic assessment, particularly in patients at risk of deterioration.
  - Its use should be guided by clinical indication, ultrasound availability and operator competency.

9.3. Inpatient Management of Dengue Infection

9.3.1. Inpatient Monitoring

All admitted patients should receive a thorough clinical assessment with appropriate ongoing monitoring, guided by the phase and severity of illness. The plan of management is tailored accordingly and adjusted based on dynamic clinical and laboratory findings, and must also be documented in the inpatient dengue monitoring chart.[19] Refer to **Appendix 9** on **Inpatient Dengue Assessment and Clerking Sheet** and **Appendix 10** on **Dengue Monitoring Chart.** During the critical phase, monitoring must be intensified, and frequent adjustments to the intravenous fluid regimen may be required. Early recognition of the reabsorption phase is equally important, as intravenous fluids should be progressively reduced or discontinued to avoid fluid overload.

Table 9 and **10** below summarise the issues, parameters and frequency of monitoring according to the different phases of the illness.[19]

Table 9: Clinical Issues and Monitoring Consideration Across the Phases of Dengue Illness

Phases of Illness	Issues
Febrile	• Differentiation of dengue illness from other febrile illnesses.
Critical	• Plasma leakage occurs as the patient progresses to late febrile phase or as temperature begins to defervesce (T <38.0°C). • Clinical deterioration occurs during this phase due to plasma leakage. • Plasma leakage results in haemoconcentration and hypovolaemia/ shock. • Excessive plasma leakage, which may be exacerbated by intravenous fluid therapy, can lead to respiratory distress. • Bleeding can be precipitated by prolonged shock and shock can be perpetuated by bleeding. • Organ dysfunction may occur. • May mimic acute abdomen from other causes. • May be confused with septic shock or other forms of shock.
Reabsorption	• Cessation of plasma leakage. • Reabsorption of fluid from the extravascular compartment. • Haemodilution occurs following fluid reabsorption. • Fluid overload and pulmonary oedema if intravenous fluid therapy is continued.

Adapted from: Ministry of Health Malaysia. Management of Dengue Infection in Adults (Third Edition). Putrajaya: MoH; 2015

Table 10: Parameters and Frequency of Monitoring According to the Phases of Dengue Illness

Parameters for Monitoring	Frequency of Monitoring		
	Febrile Phase	**Critical Phase**	**Reabsorption Phase**
Clinical Parameters			
General well being/ Appetite/ oral intake/ Warning signs	Daily or more frequently towards late febrile phase	At least twice a day and more frequently as indicated	Daily or more frequently as indicated
Temperature **Hemodynamic status** - CCTVR - BP, HR - Pulse Pressure **Respiratory status** - RR - SpO_2 **Neurological status** - Conscious level - Restlessness - Seizures	4-6 hourly depending on clinical status	2-4 hourly depending on clinical status **In shock** Every 15-30 minutes till stable then 1-2 hourly	4-6 hourly
Urine output	6-8 hourly	2-4 hourly **In shock,** hourly	6-8 hourly

Laboratory Parameters			
FBC	At least daily. More frequent as indicated	4-12 hourly depending on clinical status **In shock** Repeated before and after each fluid resuscitation and as frequent as indicated	Daily
RP/Creatinine **LFT + AST**	As clinically indicated	At least daily or more frequently as indicated	As clinically indicated
Blood gas, lactate **Coagulation profile** **CRP, Troponin** **Fibrinogen,** **LDH, ferritin,** **triglyceride**	As clinically indicated	As clinically indicated **In severe dengue** Monitor blood gas, lactate and coagulation profile closely	As clinically indicated
ECG	As clinically indicated	As clinically indicated	As clinically indicated
PoCUS	As clinically indicated	As clinically indicated	As clinically indicated

Adapted from: Ministry of Health Malaysia. Management of Dengue Infection in Adults (Third Edition). Putrajaya: MoH; 2015

In hospitalised dengue patients, PoCUS may be used in conjunction with laboratory monitoring to enhance early detection of plasma leakage. A cohort study showed that serial PoCUS is more likely to identify early leakage during the initial phase of illness than haematocrit monitoring alone (OR=4.42, 95% CI 2.85 to 6.86). If accessibility to the USG is limited, platelet count (<97 x 10^9/L) and serum transaminase levels (>51 U/L) are useful to select patients for targeted US assessment as they are independent predictors of US-detected plasma leakage with OR=4.56 (95% CI 2.53 to 8.22) and OR=2.97 (95% CI 1.62 to 5.44) respectively.[56, level II-2]

Recommendation 5

- All admitted dengue patients should be clinically assessed and monitored at appropriate intervals guided by the phases and severity of illness.

9.3.2. Inpatient Treatment

Patients with dengue may be admitted at different phases of illness. Management would include fluid therapy and supportive care.

a. Fluid Management

In the management of dengue fever, meticulous fluid management constitutes the foundational, life-saving intervention. The primary clinical challenge is navigating the dynamic 24-to-48-hour critical phase

characterised by systemic plasma leakage resulting from increased capillary permeability. The goal is to achieve a precise balance, administering isotonic crystalloids meticulously to maintain adequate intravascular volume and organ perfusion, while actively preventing the serious complication of fluid overload that occurs once the plasma reabsorption phase begins.

Common Pitfalls in Fluid Therapy

1. Excessive fluid administration in stable cases:
   - Administering unnecessary fluid boluses during the febrile phase.
   - Employing excessive and prolonged fixed or graded fluid regimens in clinically stable patients.
   - Continuing intravenous fluids unnecessarily into the recovery phase.
2. Inappropriate monitoring and assessment:
   - Infrequent monitoring and subsequent lack of timely adjustment of infusion rates.
   - Giving unnecessary fluid boluses based solely on a single parameter (e.g. rising HCT, tachycardia, or cool peripheries) without a comprehensive assessment of other clinical indicators.
   - Inadequate and inappropriate fluid resuscitation in patients with dengue shock syndrome.
3. Injudicious use of colloids:
   - Unnecessary administration of colloids, such as human albumin, for maintenance fluid therapy in leaking dengue patients.
4. Failure to adjust for comorbidities:
   - Providing excessive fluid therapy to patients with pre-existing conditions like heart failure or kidney diseases.

In Malaysia, the mean adult body weight was 62.65 kg based on the Malaysian Adults Nutrition Survey (MANS) 2009.[57, level III] The prevalence of overweight and obesity is high and increasing, reaching 53.5% according to National Health Morbidity Survey (NHMS) 2023.[58, level III] Considering these two data, the CPG DG opine that the current mean body weight would be higher than 62.65kg. Therefore, the estimated current Malaysian adults mean weight would be in the range of 66 to 68 kg. To mitigate the risk of fluid overload, especially during critical phases, the CPG DG has decided to adopt a 70 kg weight limit for weight-based fluid calculations in maintenance and fluid quota.

i. Febrile phase

For patients with dengue in febrile phase, it is important to ensure:

- Vascular Access: Maintain a cannula in situ for access.
- Fluid Management:
  - Encourage adequate oral fluid intake.

- For patients presenting with inadequate oral intake, IV crystalloid solutions may be required.
- The fluid requirement (oral + IV) should be determined using the Holliday-Segar formula.[52, level III; 55] (Refer to **Table 11** for calculation of fluid maintenance)
- Additional fluid may be required in patients with acute fluid loss from vomiting, diarrhoea and dehydration.
- Reduce the rate or discontinue IV infusion promptly once oral intake has improved and dehydration corrected.

Table 11: Calculation of maintenance fluid as per Holliday Segar formula

Weight	Fluid Regime
First 10kg	4ml/kg/hr
Next 10kg	2ml/kg/hr
Subsequent body weight in kg	1ml/kg/hr
Example for 50kg male patient	
1st 10kg	40ml/hr
2nd 10kg	20ml/hr
Subsequent 30kg	30ml/hr
Total per day	90ml/hr = 2160ml/day

The following table provides a guide for an hourly maintenance fluid regimen based on body weight.

Table 12: Guide for Hourly Maintenance Fluid Regimen According to Body Weight

Weight*	Fluid requirement (Oral and IV) based on Holliday Segar formula (ml/hr)
30	70
40	80
50	90
60	100
≥70	110

*weight limit at 70kg

Recommendation 6

- For dengue patients in the febrile phase the following measures should be done:
  - Encourage adequate oral fluid intake.
  - Commence Intravenous (IV) fluids only if the patient is unable to tolerate oral fluids or dehydrated.
  - Calculate the total fluid requirement (oral and IV) using the Holliday-Segar formula.
  - Reduce the rate of fluid therapy or stop IV fluids once oral intake improves and dehydration is corrected.

ii. Critical Phase

Not all dengue patients will enter the critical phase and develop significant plasma leakage. Severe plasma leakage is the primary cause of shock, and prolonged shock can lead to bleeding, multi-organ failure, and death.

Key Indicators of critical phase:

- **Temperature change**: A rapid drop in temperature may signal entry into the critical phase.
- **Rising haematocrit level**: An increase of HCT ≥20% from baseline.[59]
- **Platelet count drop**: A platelet count below 100 × 10^9/L is the earliest and most useful indicator.[59]
- **Presence of plasma leakage by PoCUS**: PoCUS (chest and abdomen) can detect objective evidence of fluid leakage, such as pericholecystic fluid collection, thickened gallbladder wall, ascites, or pleural effusion.

Note that some patients may still have a low-grade fever during the early phase of leakage.

- **Plasma leakage and Fluid Management**

The duration of plasma leakage in dengue is variable among patients. Accurate identification of the onset and resolution of this phase is essential to guide appropriate fluid therapy. Plasma leakage typically progresses gradually, peaks at approximately 24 hours after onset, and subsequently declines, usually resolving within 48 hours (Refer to **Figure 11** illustrating plasma leakage during critical phase of dengue fever). However, this duration may be altered under certain clinical circumstances, including the following:[59]

- Patients presenting in shock may have already been in the critical phase for up to 24 hours, suggesting a possibility of another 24 hours remaining in this phase

- Patients with uncorrected dehydration during the febrile phase may only be at the early stage of the critical phase at presentation hence the remaining duration of plasma leakage may exceed 24 hours.

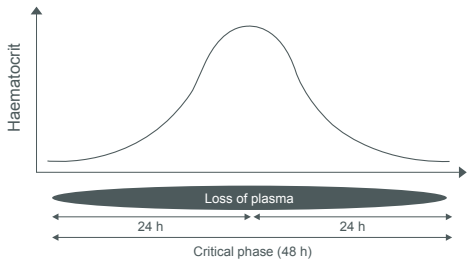


Figure 11: Plasma leakage during critical phase of dengue fever

Source: Kularatne SA, Dalugama C. Dengue infection: Global importance, immunopathology and management. Clin Med. 2022;22(1):9-13.

- **Fluid Quota Regimen in Critical Phase (Without Shock)**

The fluid quota of maintenance + 5% deficit has been used in several international dengue guidelines in the critical phase (refer **Appendix 11** on **Fluid Quota**). It may prevent unnecessary excess fluid in patients in critical phase without shock.[52; 59; 60; 61; 62] The 5% deficit represents the expected plasma volume loss during critical phase and crystalloid is the intravenous fluid of choice.

A cohort study on adults with DHF investigated the impact of fluid regimens on outcomes related to total fluid volume administered, incidence of fluid overload, and causes of fluid excess. Key findings reported that patients with DHF grades III and IV received higher fluid volumes (≥maintenance + 7.5% deficit)($p<0.0001$) as they developed dengue shock syndrome. In addition, half of patients in the ≥ maintenance + 7.5% deficit group developed fluid overload ($p<0.0001$) and the most common reasons for fluid excess were reduced urine output (52.5%), narrow pulse pressure (32.5%), rising packed cell volume (7.5%), shock (2.5%), and unidentified causes (5%). Fluid overload was not documented in patients with DHF Grades I and II who were managed with the maintenance + 5% deficit fluid quota regimen.[63, level II-2]

Hence CPG DG decided to introduce the fluid quota regimen (IV and oral) in managing dengue patients in critical phase. The fluid quota is the maximum fluid to be given over 48 hours. The full fluid quota is not necessary for all patients; many will require less because the rate, peak, and duration of plasma leakage differ among individuals.

This measure is crucial to enhance patient safety and optimise clinical outcomes, especially in patients who are susceptible to capillary leak

and third-space loss, where excessive intravenous fluids administration may be harmful.

The fluid quota is reserved solely for non-shock dengue patients during the critical phase. However, certain groups of patients may need more fluid than the calculated quota such as those in diabetic ketoacidosis (DKA) and hyperosmolar hyperglycaemic state (HHS).

Guide for fluid administration in dengue patients with warning signs (Group B2):

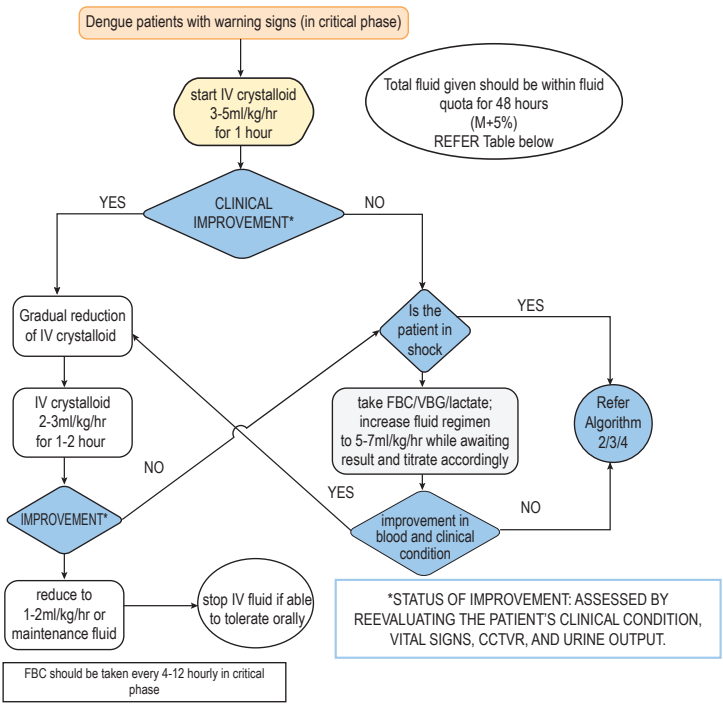
1. **Quota Calculation**:
   - The maximum body weight used for calculation is 70 kg.
   - The total fluid quota (Maintenance + 5% deficit) must be calculated at the beginning of the critical phase.
2. **Initial Management**:
   - Encourage oral fluid intake.
   - Patients entering the critical phase may require IV infusion of crystalloids.
3. **IV Fluid Therapy**:
   - Patients exhibiting warning signs and unable to tolerate oral fluids may require an initial IV fluid infusion of 3 - 5 ml/kg/hour for 1 - 2 hours.
   - The duration for this regimen depends on the patient's clinical response and hemodynamic status.
   - Following reassessment, the IV fluid rate should be adjusted.
   - The rate can be reduced to 2 - 3 ml/kg/hour for the next 1 - 2 hours, followed by a maintenance rate.
   - **If a patient can tolerate oral fluid intake equivalent to 1 - 2 ml/kg/hour, IV fluid administration is not necessary**.
   - Important Note: The total fluid administered must remain within the calculated fluid quota.
   - Frequent monitoring of CCTVR, BP, pulse pressure, HCT, and urine output will determine the subsequent IV fluid requirement.
   - Plasma leakage severity varies among patients and can fluctuate over time, even within the same patient.
   - **If a patient progresses to shock due to worsening plasma leakage, the fluid quota is no longer applicable**.
   - Refer to **Algorithm 1** on **Management of Dengue Patients with Warning Signs (Group B2)**.
4. **Monitoring Urine Output**:
   - Calculate urine output in ml/kg/hr at each void.
   - **Optimal urine output**: 0.5 – 1 ml/kg/hour indicates adequate perfusion.
5. **Urinary catheterisation**:
   - Consider early catheterisation if urine output monitoring is difficult and is essential for high-risk categories.

Algorithm 1 illustrates the approach to fluid management for dengue fever patients exhibiting warning signs in critical phase.

Recommendation 7

- In patients with dengue in the critical phase without shock, the total fluid quota should be administered over 48 hours.
- Intravenous fluid adjustment should be done frequently based on clinical response guided by hemodynamic status (including skin colour, capillary refill time, temperature, pulse volume, and pulse rate - CCTVR), vital signs, urine output, laboratory markers (haematocrit), and evidence of ongoing plasma leakage.

Algorithm 1: Management of Dengue Patients with Warning Signs (Group B2)



Body weight (kg)*	Fluid quota M+5% (Oral and IV) over 48 hours (ml)	Rate of IV fluid regime (ml/hr)				
		R1 (1ml/kg/h)	R2 (2ml/kg/h)	R3 (3ml/kg/h)	R5 (5ml/kg/h)	R7 (7ml/kg/h)
30	3200	30	60	90	150	210
35	3600	35	70	105	175	245
40	3900	40	80	120	200	280
45	4300	45	90	135	225	315
50	4700	50	100	150	250	350
55	5000	55	110	165	275	385
60	5400	60	120	180	300	420
65	5700	65	130	195	325	455
≥ 70	6000	70	140	210	350	490

*weight limit at 70kg

iii. Dengue Shock Syndrome (DSS)

Dengue shock syndrome is a medical emergency that requires immediate attention. Early recognition of compensated shock and prompt fluid resuscitation are essential for a favourable clinical outcome.[19] Failure to detect compensated shock will lead to decompensated (hypotensive) shock, resulting in bleeding and multiorgan failure. Refer to **Table 1** for detailed information on features of compensated and decompensated shock.

Patients with dengue shock must be managed in high dependency unit (HDU) or intensive care units (ICU). Fluid resuscitation is critical and must be initiated promptly; it should not be delayed while waiting for HDU/ICU admission. Recurrent episodes of shock may occur after initial resuscitation as plasma leakage can persist for 24 - 48 hours.[59; 60; 61; 52; 20; 19]

Crystalloids are the initial and preferred fluid for resuscitation in patients with dengue shock.[61; 64; 65] Choice of crystalloids includes balanced crystalloids (e.g. Hartman's, Sterofundin, etc.) and 0.9% saline.

A Cochrane systematic review comparing balanced solutions with 0.9% saline in critically ill adults and children found no statistically significant differences in outcomes such as overall in-hospital mortality, acute kidney injury, or organ systemic dysfunction between the two types of fluids. However, balanced solutions led to significantly lower chloride levels (MD= -3.02, 95% CI -5.24 to -0.80), higher bicarbonate levels (MD=2.26, 95% CI 1.25 to 3.27), and higher pH (MD=0.04, 95% CI 0.02 to 0.06). It is worth noting that only four RCTs involving children with dengue fever were included in this systematic review.[66, level I]

Although there is no evidence to suggest the superiority of colloids over crystalloids, colloid solutions, such as albumin may be considered in patients with refractory shock where haemodynamic instability persists following adequate fluid resuscitation.[19; 59] If colloids are not available, resuscitation should be continued with crystalloids and consider blood transfusion if there is suspicion of bleeding.[19, 59]

The available colloids are albumin 5% and gelatin solutions. However, several guidelines recommend against using gelatin for resuscitation in critically ill patients due to no demonstrated benefit with potential harms.[67; 68] Therefore, the CPG DG opines that, albumin 5% is the colloid of choice to be used in resuscitation for DSS. The use of colloids should be reserved primarily for resuscitation. Extended utilisation of colloids as the fluid for maintenance therapy should be avoided.[19; 64]

- **Principles of fluid resuscitation in Dengue Shock Syndrome**

The initial and subsequent volume of fluid resuscitation, which may range from 10 - 20 ml/kg adjusted body weight (ABW), depends on the degree of shock. Refer to **Table 13** on ABW calculation. The volume and rate must be carefully titrated based on the clinical response to maintain effective circulation while preventing over-replacement. Adequate and effective fluid resuscitation should lead to improvements in the specified clinical parameters.[19] (Refer to **Table 14** for indicators of effective fluid resuscitation).

Table 13: Weight Adjustment for Fluid Therapy

For obese patients (BMI >27.5 kg/m^2), maintenance fluid calculation should be based on **Adjusted Body Weight (ABW)**.

The formulas for calculating ABW and Ideal Body Weight (IBW) are:

1. **Adjusted Body Weight (ABW)**:

ABW = IBW + 0.4 (actual weight – IBW)

2. **Ideal Body Weight (IBW)**:
   - Female: 45.5 kg + 0.91(height in cm -152)
   - Male: 50.0 kg + 0.91(height in cm -152)

References:

1. Malaysian Society for the Study of Obesity. *Clinical Practice Guidelines: Management of Obesity*. Ministry of Health, Malaysia; 2023.
2. GlobalRPH 2015, calculator adjusted body weight (available at http://www.globalrph.com/ibw_calc.htm)

Table 14: Indicators of Effective Fluid Resuscitation

All these parameters should be interpreted collectively and not in isolation
A. Clinical Parameters ▪ Improvement of general well-being/mental state ▪ Warm peripheries ▪ Capillary refill time ≤ 3sec[64] ▪ Stable BP ▪ Improving pulse pressure ▪ Reducing tachycardia ▪ Improving urine output ▪ Less tachypnoea
B. Laboratory Parameters ▪ Appropriate decrease in HCT ▪ Improvement in metabolic acidosis and lactate

Further haemodynamic assessments of fluid responsiveness in dengue shock, such as passive leg raise test (PLR), end-expiratory occlusion test (EEOT), pulse pressure variation (PPV) and stroke volume variation (SVV) are explained below.

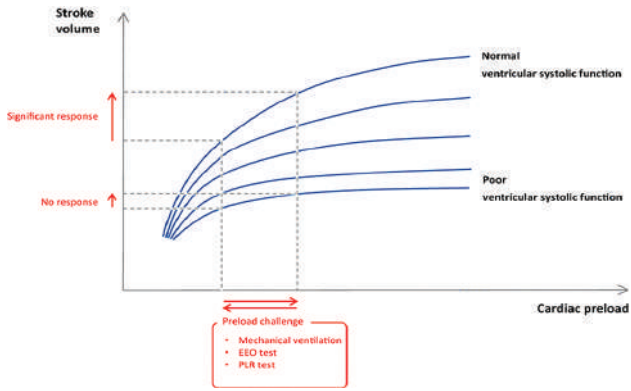
- **Assessment of fluid responsiveness**

In patients who remain hypotensive despite fluid resuscitation, further fluid administration should be guided by assessment of fluid responsiveness and fluid tolerance.

Fluid responsiveness should be assessed using dynamic rather than static indices, as dynamic tests consistently demonstrate better diagnostic performance than static filling measures.[69, level III]

Fluid responsiveness is generally defined as an increase in stroke volume or cardiac output of ≥10 - 15% following a standardized fluid challenge. In adults, this is commonly assessed using about 4 mL/kg of fluid administered over a short period, typically 5 - 15 minutes.[70, level III; 71, level III] However, repeated fluid boluses may be harmful and can lead to fluid accumulation and overload. Consequently, newer evidence-based approaches are increasingly being used to assess fluid responsiveness. **Figure 12** shows the Frank–Starling relationship, illustrating that a preload challenge increases stroke volume only when the heart is operating on the preload-responsive portion of the curve, whereas little or no increase occurs when the ventricle is already on the flat portion or has poor systolic function.

Figure 12: Frank-Starling Relationship



Adapted from: Monnet X, Marik PE, Teboul JL. Prediction of fluid responsiveness: an update. Ann Intensive Care. 2016 Dec;6(1):111)

Passive Leg Raising

Passive leg raising (PLR) with direct measurement of cardiac output or stroke volume is the most broadly applicable bedside test. This manoeuvre rapidly shifts approximately 150 – 300 ml of venous blood from the lower extremities to the central circulation. An immediate increase in cardiac output signifies preload responsiveness. This simulates a fluid challenge without actual fluid infusion, making the hemodynamic effects quickly reversible and minimizing the risk of fluid overload.[64] PLR can be used for both spontaneously breathing and mechanically ventilated patients. However, it may be less reliable in patients with intra-abdominal hypertension. The diagnostic accuracy of PLR depends on how the manoeuvre is performed (refer to **Appendix 12** on **Passive Leg Raise Manoeuvres**). The haemodynamic response occurs within 30 - 90 seconds and is transient; therefore, measurements should be obtained within this timeframe.[72, level III]

Pulse Pressure Variation and Stroke Volume variation

In mechanically-ventilated patients, pulse pressure variation (PPV) and stroke volume variation (SVV) have the strongest evidence among routine bedside indices. PPV and SVV are unreliable in patients with spontaneous breathing efforts, cardiac arrhythmias, low tidal volume ventilation, reduced lung compliance (e.g., ARDS), a high respiratory rate–to–heart rate ratio, right ventricular dysfunction, elevated intra-abdominal pressure, or open-chest conditions.[69, level III]

In mechanically ventilated patients receiving low tidal volume ventilation, baseline PPV becomes less reliable. However, its diagnostic performance can be restored by performing a tidal volume

challenge or by assessing the change in PPV induced by PLR. Both approaches demonstrate high diagnostic accuracy and are useful when lung-protective ventilation strategies are employed. Nevertheless, the limitations of these techniques are the same as for PLR and PPV.[73, level III; 74, level III]

End-Expiratory Occlusion Test

The end-expiratory occlusion test (EEOT) also performs well in mechanically ventilated ICU patients and is best considered a confirmatory test, particularly when combined with direct cardiac output monitoring techniques. However, the certainty of evidence is lower than for some other dynamic tests, and its performance appears more variable outside the ICU setting.[75, level III]

Left Ventricular Outflow Tract Velocity Time Integral Variation

Emerging echocardiographic data suggest that respiratory variation in left ventricular outflow tract (LVOT) velocity–time integral (VTI) may outperform PPV under low-tidal-volume ventilation and may be particularly useful when performed by operators trained in echocardiography. However, the current evidence is derived from relatively small cohort studies, and the overall evidence base remains less robust compared with other dynamic indices such as PLR, PPV/SVV, tidal volume challenge (TVC), or EEOT. Similar to PPV and SVV, VTI variability is unreliable in patients with arrhythmias or spontaneous breathing and is influenced by ventilatory conditions.[76, level III]

Inferior Vena Cava Measurements

Inferior vena cava (IVC)–based indices have only moderate diagnostic accuracy and should not be used as a sole test for guiding fluid therapy. Their performance is affected by ventilation mode, respiratory effort, elevated intra-abdominal pressure, and right heart dysfunction. IVC measurements may be used as an adjunct when other dynamic tests are not feasible, but they should be interpreted cautiously and within the clinical context.[77, level III; 78, level III; 79. level III]

Application of dynamic test for fluid responsiveness in dengue shock patients is illustrated in **Figure 13** below.

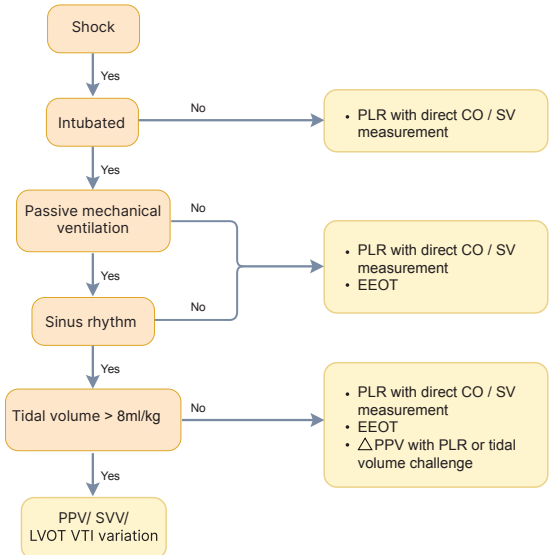


Figure 13: Application of dynamic test for fluid responsiveness in dengue shock

Diagnostic threshold for predicting fluid responsiveness are as below:[69, level III; 72 - 76, level III]

Method	Criterion of judgement	Diagnostic threshold
PLR	Increase in CO/ SV	≥10%
	Absolute reduction in PPV	1 to 2.5 points
PPV	Absolute value itself	≥12%
SVV	Absolute value itself	≥12%
Tidal Volume challenge	Absolute increase in PPV	1 to 3.5 points
EEOT	Increase in CO	≥5%
LVOT VTI variation	Absolute value itself	≥16%

- **Assessment of fluid tolerance**

Fluid tolerance refers to the capacity of a patient to receive fluid without developing organ impairment due to congestion. It identifies who may be harmed even if they are fluid-responsive.[80, level II-2]

Fluid tolerance can be assessed clinically and with PoCUS. Clinical features suggesting reduced fluid tolerance include:[80, level II-2]

- Increasing breathlessness or oxygen requirement
- Pulmonary oedema
- Elevated jugular venous pressure

PoCUS, when available, may assist in assessing fluid tolerance by identifying pulmonary congestion (e.g. diffuse B-lines) and systemic congestion [Venous Excess Ultrasound (VExUS)].

In dengue shock, the decision to give further fluid should balance the likelihood of improving tissue perfusion against the risk of fluid-related complications.

- **Non-responders to Initial Resuscitation**

Persistent Shock with High HCT: If the haemodynamic state is not stabilised and the HCT remains high after initial fluid resuscitation, colloids should be considered for subsequent fluid resuscitation. (Refer to **Algorithm 2** and **Algorithm 3** on fluid management of dengue with compensated and decompensated shock).

Persistent Shock with Dropped HCT: If the haemodynamic state is not stabilised and HCT drops after fluid resuscitation, significant (often occult) bleeding should be suspected, and prompt blood ± blood products transfusion is necessary. Refer to **Algorithm 4** on **Management of Persistent Shock**.

Other Causes of Shock: Other potential causes of refractory shock include hypovolaemic (bleeding, leaking), distributive shock (sepsis, cytokine storm, liver failure), cardiogenic shock (myocardial dysfunction, cardiac arrhythmias) and obstructive shock (cardiac tamponade). Refer to **Algorithm 4** on **Management of Persistent Shock**.

Recommendation 8
- Crystalloids should be used as the initial and preferred fluid for resuscitation in patients with dengue shock.
- Colloids may be considered in patients with persistent shock; however, colloids should not be used for maintenance therapy.
- Fluid administration in patients with dengue shock may be guided by dynamic assessment of fluid responsiveness using appropriate bedside parameters.
- In refractory dengue shock, other causes of shock should be aggressively identified and treated accordingly.

9.4. Management of Complications

9.4.1. Metabolic Acidosis in Severe Dengue

Metabolic acidosis commonly occurs in dengue shock syndrome, primarily as a consequence of severe plasma leakage and haemorrhage leading to reduced tissue perfusion and hypoxia. The most common cause of metabolic acidosis in dengue shock is lactic acidosis.

A lactate level of <2 mmol/L in a critically ill patient, including those with severe dengue, generally suggests adequate tissue perfusion. However, high lactate levels are not necessarily the result of tissue hypoxia or shock.

Although lactate reduction has been associated with improved survival, targeting lactate normalisation in a non-shock patient can precipitate excessive fluid administration. Combining lactate reduction with other perfusion markers such as capillary refill time and urine output is therefore recommended for resuscitation. Hence, lactate levels should always be interpreted with clinical findings.

If metabolic acidosis persists or worsens despite adequate fluid resuscitation, clinicians should consider:

- Bleeding that may be masked by haemoconcentration
- Worsening acute liver injury
- Worsening acute kidney failure
- Sepsis with distributive shock
- Cardiac dysfunction
- Medications related - beta 2 agonists, metformin, SGLT-2i
- Hyperchloraemic metabolic acidosis secondary to saline infusion
- Concomitant ketoacidosis – diabetic or euglycaemic ketoacidosis

Sodium bicarbonate should not be used routinely in all patients with dengue-associated AKI and metabolic acidosis. However, in patients with severe metabolic acidaemia (arterial pH ≤7.20) accompanied by moderate-to-severe AKI, intravenous sodium bicarbonate may be considered to correct severe acidaemia and reduce the need for kidney replacement therapy (KRT).[81, level I]

The decision to administer bicarbonate should be individualised, taking into consideration the patient's volume status, serum sodium concentration, calcium levels and the risk of fluid overload.

9.4.2. Bleeding in Dengue Infection

a. Haemostatic Abnormalities in Dengue Infection

Bleeding in dengue infection is multifactorial, resulting from endothelial activation, thrombocytopaenia, platelet dysfunction and coagulation activation.[82] It ranges from minor mucocutaneous manifestations to life-threatening haemorrhage.

b. Risk factors for bleeding

Risk factors for bleeding in dengue include the use of non-steroidal anti-inflammatory drugs (NSAIDs) anticoagulant therapy or antiplatelet agents, pre-existing peptic ulcer disease, severe renal

or liver failure, persistent metabolic acidosis and any form of trauma including intramuscular injection.[1; 11; 55; 83] Severe plasma leakage and prolonged shock may precipitate mucosal ischemia, coagulopathy and disseminated intravascular coagulation, thereby increasing the risk of clinically significant haemorrhage.[1;11; 55]

c. Recognising Bleeding in Dengue

Bleeding in dengue may be overt or occult and is considered clinically significant when it requires intervention or results in haemodynamic instability.[19; 59] Minor mucocutaneous bleeding such as gingival bleeding, epistaxis, petechiae or mild per vaginal bleeding is relatively common and usually self-limiting.[19; 55]

In contrast, severe overt bleeding in dengue may manifest as upper gastrointestinal haemorrhage (haematemesis or melaena), massive per vaginal bleeding, severe epistaxis requiring intervention, intracranial haemorrhage, pulmonary haemorrhage or large soft-tissue haematomas, and may lead to haemodynamic compromise requiring urgent resuscitation and blood transfusion.[1; 11]

Occult bleeding refers to bleeding that is not clinically apparent but may result in unrecognised intravascular blood loss and contribute to persistent haemodynamic instability.[19]

In the presence of shock, a haematocrit rise of ≤20% from baseline or only minimal elevation should be interpreted with caution, as it may indicate that plasma leakage alone is insufficient to explain the shock.

Suspect significant occult bleeding in the following situations[19]

- A drop in HCT without clinical improvement despite adequate fluid replacement.
- Severe metabolic acidosis and end-organ dysfunction despite adequate fluid replacement.
- HCT not as high as expected for the degree of shock to be explained by plasma leakage alone.

d. Role of Viscoelastic Testing-Rotational thromboelastrometry (ROTEM)/ Thromboelastography (TEG)

Viscoelastic testing can detect hypocoagulable states and reduced functional fibrinogen that may not be identified by conventional coagulation tests. Randomised trials in trauma and cardiac surgery demonstrate that viscoelastic-guided transfusion strategies reduce blood product use.[84, level I; 85, level I] Similar findings are reported in liver disease, where rotational thromboelastometry-guided management reduced pre-emptive transfusion from 100% to 40% in cirrhotic patients undergoing elective procedures.[86, level I]

However, evidence is limited with no randomised trials showing improved outcomes in dengue. Therefore, ROTEM/TEG may be used as an adjunct to support targeted transfusion decision in selected patients with significant or refractory bleeding and not for routine use.

e. Management of bleeding in dengue infection

Blood transfusion of blood and blood products is indicated when there is evidence of significant bleeding.[19] However, blood transfusion must be given with care because of the risk of fluid overload. Packed red blood cells are currently recommended over whole blood.[19; 55]

In rapid transfusion, blood warmers should be used to avoid hypothermia.[87] The amount of blood and blood products transfused must be included in the cumulative fluid volume.

Transfusion should not be delayed while waiting for a marked decline in haematocrit. In dengue, severe bleeding often occurs after a period of prolonged shock associated with plasma leakage, during which haematocrit levels may initially rise due to haemoconcentration. When bleeding subsequently occurs, the haematocrit may fall but may not appear markedly low because of the preceding haemoconcentration.[55] In such situations, administration of colloids may help unmask the occult bleeding by causing haemodilution and revealing a disproportionate fall in haematocrit.[19] At the same time, the source of bleeding should be identified promptly, and appropriate measures instituted to achieve haemostasis. For example, endoscopy may be required to identify and manage persistent gastrointestinal bleeding.[88]

Transfusion of blood in patients with significant bleeding[19]

- Transfuse blood (5–10 ml/kg of packed red cells) and observe the clinical response. Consider blood components if required.
- Consider repeating the blood transfusion if there is further blood loss or no appropriate rise in HCT after blood transfusion.

In patients with severe or clinically significant bleeding, coagulation parameters (PT, aPTT and fibrinogen) should be assessed. If fibrinogen is <100 mg/dL, cryoprecipitate should be administered. If fibrinogen is ≥100 mg/dL but PT or aPTT exceeds 1.5 times the normal reference range, fresh frozen plasma may be considered.[55] Correction of coagulation parameters should not be undertaken solely on the basis of laboratory abnormalities in the absence of clinically significant bleeding.

The severity of thrombocytopenia does not reliably predict clinical bleeding, and routine correction of isolated thrombocytopaenia in the absence of significant bleeding is not supported by current evidence as it does not reduce bleeding risk and may be associated with increased

mortality, longer hospital stay, and adverse events.[83, level I] Platelet transfusion is indicated in patients with significant bleeding and may be considered when undergoing procedure with high risk of bleeding.[55]

The CPG DG opines that platelet transfusion should be considered in haemodynamically stable patients with significant bleeding from gastrointestinal, genitourinary, respiratory tract, or in body cavity, muscle and soft tissue.

For neurosurgical and ophthalmic procedure, platelet transfusion may be indicated if platelet count ≤100 × 10^9/L.[55] For obstetric procedures, refer **Chapter 10.1** on **Dengue in Pregnancy**

Platelet transfusion is not recommended for oropharyngeal bleeding or epistaxis which is self-resolving. Local compression and adjunct methods may be used to mitigate bleeding.[89]

Recommendation 9

- Occult bleeding should be suspected in dengue patients with persistent haemodynamic instability despite adequate fluid resuscitation.
- Blood and blood products transfusion should be started promptly in patients with clinically significant bleeding without waiting for a drop in haematocrit.
- Packed red blood cells should be used for transfusion in patients with significant bleeding.
- Routine prophylactic transfusion of platelets or plasma should not be performed in dengue patients without active bleeding.
- Platelet transfusion should be considered for patients with significant bleeding or those requiring invasive procedures with a risk of bleeding.

f. Adjunctive therapy in dengue patients with bleeding

Pharmacological platelet-raising therapies, including IL-11, anti-D immunoglobulin and eltrombopag may increase platelet counts but have not been shown to improve clinical bleeding outcomes.[83, level I; 90, level I]

Intravenous immunoglobulin (IVIG) has not been shown to improve platelet count or clinical bleeding outcome.[83, level I]

Recombinant activated factor VII does not significantly improve the control of bleeding, transfusion requirements or mortality in dengue haemorrhagic fever.[83, level I]

Tranexamic acid has not been shown to significantly improve bleeding outcomes in dengue related bleeding and may cause adverse effects

such as abdominal pain and pruritus.[91, level II-2] However, some guidelines still recommend it for menstrual bleeding.[59]

There is no retrievable evidence supporting Prothrombin complex concentrate (PCC) in dengue associated bleeding.

The CPG DG opines that adjunctive therapy such as IL-11, anti-D immunoglobulin, eltrombopag, IVIG and Recombinant activated factor VII should not be routinely used in dengue patients with bleeding.

g. Severe bleeding in Dengue

Severe bleeding in dengue may manifest as gastrointestinal (GI) bleeding, intracranial haemorrhage, pulmonary haemorrhage, severe vaginal bleeding or bleeding into other organs. The pathogenesis is multi-factorial and, in some patients may be associated with prolonged shock and tissue hypoperfusion.[64]

- **Upper gastrointestinal bleeding**

Gastrointestinal bleeding is one of the most common haemorrhagic manifestations in dengue infection and it is often associated with high mortality. The common causes are haemorrhagic (and/or erosive) gastritis, peptic ulcer and oesophageal ulcer.[19] It may present as hematemesis, melaena, haematochezia or occult bleeding and is often associated with increased mortality. Bleeding from the upper gastrointestinal tract may not be apparent for many hours, until the first black stool or melena is passed.[55]

Management of GI bleeding in dengue include:[19]

- Haemodynamic stabilisation with appropriate fluid resuscitation.
- Blood transfusion (packed red cell) and correction of thrombocytopenia or coagulopathy.
- Administration of proton pump inhibitor.
- Role of endoscopy:
  - Endoscopy should be considered in patient with significant or persistent GI bleed despite optimal medical therapy.
  - There is no evidence to support endoscopic injection therapy in GI bleeding secondary to severe dengue.
  - In general, endoscopic homeostatic therapy is recommended in patients with history of peptic ulcers with recent haemorrhage including active arterial bleeding, nonbleeding visible vessels, nonbleeding adherent clots or persistent oozing.

- **Intracranial haemorrhage**

Intracranial haemorrhage in dengue is rare but a life-threatening complication with high mortality. The prognosis is worse with severe thrombocytopenia, coagulopathy, low GCS, large hematoma, shock or

multiorgan involvement. The management is usually supportive and includes the following:[92 - 93, level III]

- Early recognition of neurological symptoms and prompt brain imaging
- Haemodynamic stabilisation and correction of thrombocytopenia and coagulopathy.
- Selective neurosurgical intervention (e.g. decompression craniectomy) after stabilisation may be considered in selected cases after discussion with neurosurgical team:
  - Significant intracranial haemorrhage
  - Progressive neurological deterioration after correction of coagulopathy
- Multidisciplinary care management should involve critical care, neurology and neurosurgery team.

9.4.3. Organ Involvement/Expanded Dengue Syndrome

Expanded dengue syndrome refer to dengue infection with severe involvement of one or more organs (such as brain, liver, heart, or kidneys), which cannot be explained by usual plasma leakage or bleeding. These conditions are uncommon and need to be managed according to the involved organ. Organ failure secondary to prolonged shock in dengue infection is not classified under expanded dengue syndrome though overlap may occur. The manifestations of expanded dengue are shown in table below.[62]

Table 15: Expanded dengue syndrome (Unusual or atypical manifestations of dengue)

System	Unusual or atypical manifestations
Neurological	Febrile seizures in young children. Encephalopathy. Encephalitis/aseptic meningitis. Intracranial haemorrhages/thrombosis. Subdural effusions. Mononeuropathies/polyneuropathies/Guillane-Barre syndrome. Transverse myelitis.
Gastrointestinal/ hepatic	Hepatitis/fulminant hepatic failure. Acalculous cholecystitis. Acute pancreatitis. Hyperplasia of Peyer's patches. Acute parotitis.
Renal	Acute renal failure. Haemolytic uremic syndrome.
Cardiac	Conduction abnormalities. Myocarditis. Pericarditis.
Respiratory	Acute respiratory distress syndrome. Pulmonary haemorrhage.
Musculoskeletal	Myositis with raised creatine phosphokinase (CPK). Rhabdomyolysis.
Lymphoreticular / bone marrow	Infection-associated haemophagocytic syndrome. IAHS or Haemophagocytic lymphohistiocytosis (HLH), idiopathic thrombocytopenic purpura (ITP). Spontaneous splenic rupture. Lymph node infarction.
Eye	Macular haemorrhage. Impaired visual acuity. Optic neuritis.
Others	Post-infectious fatigue syndrome, depression, hallucinations, psychosis, alopecia.

Source: World Health Organization, Regional Office for South-East Asia. Comprehensive guidelines for prevention and control of dengue and dengue haemorrhagic fever. Revised and expanded edition. WHO-SEARO; India: 2011.

a. Dengue related liver involvement

Dengue-related liver involvement ranges from asymptomatic transaminitis to severe hepatitis and acute liver failure (ALF). Patients may present with abdominal pain, tender hepatomegaly, nausea, vomiting and anorexia.[94, level III]

A meta-analysis of liver enzyme abnormalities in patients with dengue found that 80% of DHF patients and 75% of dengue fever (DF) patients had AST abnormalities while 54% of DHF patients and 52% of DF patients had alanine transaminase (ALT) abnormalities.[95, level I] AST levels are typically higher than ALT levels in dengue infection, reflecting extra-hepatic sources and the systemic inflammatory nature of the disease. Transaminases generally peak within the first week of illness and normalise within 2 to 3 weeks in uncomplicated cases.[96, level I] Dengue induced severe hepatitis (DISH) is associated with severe forms of

DENV infection and is defined by a more than ten-fold rise in serum transaminase levels.[97, level II-2]

The pathogenesis of liver injury in dengue is multifactorial, involving direct viral cytopathic effects, immune-mediated hepatocyte injury and secondary insults including hypoxia from plasma leakage and shock.[98, Level III]

Severe hepatic involvement in dengue is defined by marked transaminase elevation (AST or ALT ≥1000 IU/L).[1] In addition, a subset of patients may develop acute liver failure (ALF), defined by coagulopathy (INR ≥1.5) with hepatic encephalopathy in the absence of pre-existing cirrhosis.[99, level III] Although dengue-associated ALF is relatively uncommon, occurring in approximately 1–2% of cases, it carries a high mortality rate of 40–50%, particularly in patients with dengue shock syndrome, prolonged hypotension or multi-organ dysfunction.[96, level I]

- **Risk scoring systems and additional workups**

Risk scoring systems may assist in early identification of dengue patients at risk of acute liver failure. In patients with DISH, MELD score ≥15 or an EZ-ALBI score ≥ -30 identifies patients at high risk of developing ALF. The development of ALF and elevated baseline INR are found to be independent predictors of mortality.[97, level II-2; 100, level II-2;]

In patients with atypical liver involvement (e.g. ALT predominance, disproportionate hyperbilirubinaemia or worsening function beyond day 5 - 7), an alternative diagnosis may be present. Therefore, targeted assessment and investigations have to be considered.

- **Management of acute liver failure**

Management of ALF in dengue patients is primarily supportive with avoidance of hepatotoxic medications and best managed in ICU.

The role of N-acetylcysteine (NAC) remains uncertain as current evidence is limited to observational studies and case series. Routine use is not currently recommended for severe transaminitis without ALF. However, NAC may be considered in patients with ALF. Reported intravenous regimens include an initial dose of 150 mg/kg infusion over 15 - 60 minutes followed by 12.5 mg/kg/hour for 4 hours and 6.25 mg/kg/hour thereafter, or continuous infusion at 100 mg/kg/day for up to 5 days.[101, level II-3] Low volume of diluent can be considered to avoid fluid overload. NAC may induce anaphylactoid reactions in very small percentage of patients and may cause treatment interruption.[64]

Advanced therapies such as liver transplantation and extracorporeal liver support systems including molecular adsorbent recirculating

system (MARS), therapeutic plasma exchange and continuous renal replacement therapy have been reported only in limited case reports and small series, and there is currently insufficient evidence to support their routine use.[102, level III]

b. Cardiac complications

Cardiac dysfunction in dengue can manifest as arrhythmias, functional myocardial impairment and myocarditis. Possible mechanisms include regional vulnerability to coronary hypoperfusion, cytokine storm, direct myocardial inflammation and altered calcium homeostasis. Cardiac complications of dengue should be suspected in those with fluid refractory shock or haemodynamic compromise disproportionate to capillary leakage/HCT increase.[19]

In such patients, it is recommended to do:[19; 103, level III]

- ECG: ECG alterations reported in dengue infection are often transient. ECG changes might be the only sign of cardiac involvement with normal biomarker levels and echocardiograms. Common ECG changes reported are:
  - Sinus bradycardia (commonest finding)
  - Atrioventricular block
  - Atrial fibrillation
  - T-wave and ST-segment abnormalities
- Cardiac biomarkers: Cardiac biomarkers (CK/CKMB or Troponin) have limited sensitivity and could be normal.
- Echocardiography - Measurement of ejection fraction is difficult to interpret as it is influenced by changes in preload and afterload. In severe dengue infection, there may be evidence of systolic and diastolic dysfunction with possible segmental wall abnormalities of the septum as well as right ventricular wall.

- Management of dengue infection in such patients should primarily be focused on cautious fluid resuscitation, aiming to give just sufficient IV fluid therapy to maintain adequate tissue perfusion.

A Cochrane systematic review found that intravenous immunoglobulin (IVIG) did not confer a survival benefit in adults with acute myocarditis when compared with placebo.[104, level I] There was also no retrievable evidence to support the use of antiviral and immunomodulatory treatments such as beta-interferon and corticosteroids for dengue myocarditis.[19]

c. Hemophagocytic lymphohistiocytosis

Hemophagocytic lymphohistiocytosis (HLH) is a potentially fatal syndrome of pathologic extreme immune activation leading to cytokine

storm. There is increased IL-2, TNF-α, IL-6, IL-8, interferon γ generated by uncontrolled activation of histiocytes and T-cells. The clinical course is generally severe and may mimic sepsis. It commonly presents with an unexplained persistent fever or resurgence of high-grade fever after an initial defervescence.[19]

The HScore, is a diagnostic tool used to estimate the probability of reactive hemophagocytic syndrome (secondary HLH) in adults. Since HLH is a progressive systemic disease, the lack of adequate criteria at one juncture does not exclude the diagnosis. Bone marrow examination is not necessary when enough criteria have been fulfilled, although bone marrow aspiration seems to be safe in many reports involving adults and older children with dengue-associated HLH.[105, level III]

For the treatment of dengue-associated HLH, no randomized trials are available, which makes treatment selection both challenging and uncertain. Published experience from contemporary cases mainly relies on supportive plus short courses of immunosuppressive therapy. Dexamethasone 10 mg/m² daily for 3 - 4 days is commonly used. IVIG should be considered especially if steroids alone are insufficient or if there is concern about secondary infections. Etoposide is suggested as a rescue therapy.[105, level III]

d. Neurological complications

Recent evidence suggest that neurological complications are seen in 4.2 - 13% of all hospitalised dengue patients. The neurological manifestations associated with dengue virus were categorised into:[106, level III]

1. Direct neurotropic complications leading to encephalitis and myositis.
2. Indirect effects due to metabolic complications resulting in encephalopathy, and cerebrovascular complications due to thrombocytopenia as well as platelet dysfunction and hypokalaemic paralysis.
3. Immune-mediated neurological complications (IMNC), which include acute disseminated encephalomyelitis (ADEM), Guillain–Barre syndrome (GBS), acute transverse myelitis and opsoclonus myoclonus syndrome (OMS).

Dengue encephalitis classically presents with fever, headache, seizures, altered consciousness and focal neurological signs after 5 - 7 days of onset of fever. MRI findings may be normal or reveal non-specific cerebral oedema or classic double doughnut sign.[107, level III] If there are no contraindications, lumbar puncture should be done. The CSF should be analysed for dengue virus-specific RNA. However, the sensitivity of RT-PCR is low. CSF analysis can be normal or they can reveal pleocytosis and high protein. There is no specific treatment for dengue encephalitis.[19]

Myositis associated with dengue infection could range from subclinical elevation of CK, self-limiting myalgia and weakness, and rhabdomyolysis, to life-threatening severe fulminant myositis resulting in severe quadriparesis and respiratory failure.[107, level III]

Dengue encephalopathy is the most commonly reported neurological complication in dengue infection. Encephalopathy in dengue infection may be the result of prolonged shock, hypoxia, cerebral oedema, acute liver failure, acute renal failure, hyponatraemia and cerebral haemorrhage.[19]

Acute transverse myelitis in dengue infection can occur within the first week of illness due to direct virus invasion or 1-2 weeks later as an immune mediated phenomenon.[19]

A cross-sectional study showed IVIG and plasmapheresis as the standard treatment modality for GBS can also be used in dengue infection with GBS. In addition, most immune-mediated neurological complications of dengue patients have a favourable recovery with appropriate immunomodulatory therapy.[106, level III]

9.5. Dengue Management in ICU

Early identification of clinical deterioration, coupled with timely admission to the ICU, is critical for improving patient outcomes. Patients with severe dengue require close monitoring, advanced organ support, and rapid therapeutic intervention capabilities that are often beyond the scope of care in general wards.

The following clinical presentations warrant referral to the ICU in patients with dengue:

1	Dengue shock syndrome
2	Dengue with significant bleeding
3	Dengue with significant co-morbidity needing closed monitoring (e.g. end-stage kidney disease or severe heart failure)
4	Dengue with severe organ impairment - Acute respiratory failure - Severe transaminitis with or without encephalopathy - Altered mental status, seizures, or new onset focal neurological deficit - Cardiac arrhythmias, myocarditis, or acute heart failure - Acute kidney injury with metabolic acidosis, hyperkalaemia, or fluid overload

9.5.1. Respiratory Support

Respiratory failure in severe dengue may arise from several mechanisms, including plasma leakage resulting in pleural effusion, pulmonary oedema secondary to fluid overload, and the development of acute respiratory distress syndrome (ARDS).

The main objective of respiratory support is to ensure adequate gas exchange while minimising the work of breathing and preventing further lung injury.

Evidence regarding the use of non-invasive ventilation (NIV) in dengue is limited. Available data are derived mainly from small studies in paediatric population which suggest that NIV may improve oxygenation and reduce the need for invasive mechanical ventilation in dengue-associated respiratory failure, particularly when the underlying cause is fluid overload–related pulmonary oedema or pleural effusion[19; 108, level II-3] Contraindications to NIV include cardiac or respiratory arrest, inability to protect the airway, ongoing vomiting or high aspiration risk, and haemodynamic instability.

High-flow nasal cannula (HFNC) is increasingly used for acute respiratory failure. However, there is no evidence for its use in dengue. Systematic review data in acute hypoxaemic respiratory failure demonstrated that HFNC improves oxygenation and reduces the need for NIV and tracheal intubation compared with conventional oxygen therapy, without a clear mortality benefit.[109, level I] Contraindications to HFNC include inability to protect the airway, severe haemodynamic instability, and upper airway obstruction.

The decision to intubate dengue patients requires careful and thorough assessment by the treating physician. Indications for mechanical ventilation are similar to other critically ill patients. These include:

- Impaired consciousness or inability to protect the airway
- Increased work of breathing
- Persistent shock
- Worsening respiratory failure
- Failure of HFNC or NIV - ROX index and HACOR score may help predict failure risk and guide decision to escalate.[110 - 111, level II-2] (Refer to **Appendix 13** on **ROX index and HACOR score**)

Mechanically ventilated patients, especially those with ARDS, should be ventilated using a lung-protective strategy with low tidal volumes of approximately 4 - 8 mL/kg predicted body weight, plateau pressure <30 cmH_2O, and appropriate positive end-expiratory pressure (PEEP) to minimise ventilator-induced lung injury.[67; 112; 113]

9.5.2. Haemodynamic Support

Severe dengue may lead to circulatory shock due to plasma leakage, intravascular volume depletion, and myocardial dysfunction. Although fluid resuscitation remains the cornerstone of management, some patients develop persistent hypotension despite adequate fluid therapy, necessitating vasopressor support.

The evidence specifically evaluating vasopressors and inotropes in dengue remains limited and recommendations are partly extrapolated from the broader critical care literature on distributive shock.

Among available vasopressors, noradrenaline has the most favourable haemodynamic profile, as it increases systemic vascular resistance while preserving organ perfusion.[61; 114, level I]

In contrast, dopamine has been associated with increased mortality (scored lowest on SUCRA) and a high incidence of tachyarrhythmias (1.53%) and therefore is generally not recommended as the first-line vasopressor when alternatives are available.[114, level I]

If hypotension persists or noradrenaline requirements continue to escalate, a second vasoactive agent should be added rather than further increasing noradrenaline alone. Clinical assessment, laboratory parameters and point-of-care ultrasound (PoCUS) may be used to assess haemodynamic phenotype and guide the selection of additional vasoactive therapy. Vasopressin should be considered when noradrenaline requirements exceed approximately 0.2 µg/kg/min, particularly when the clinical picture is consistent with vasoplegic shock.[67; 115, level III] In patients with dengue and evidence of myocardial dysfunction or reduced cardiac output, the addition of an inotropic agent such as dobutamine or adrenaline may be beneficial to improve cardiac performance and tissue perfusion.[19] Milrinone may be considered in patients on chronic Beta-blockers, significant RV dysfunction or pulmonary hypertension.[116, level I]

9.5.3. Kidney Support

Acute kidney injury (AKI) is one of the complications of dengue infection that is associated with high mortality and prolonged hospital stay. The complexity of the disease often necessitates intensive intervention.

Pathophysiology of AKI in dengue infection is still not completely clear. Possible causes are acute tubular necrosis due to prolonged shock, direct kidney injury by the virus, indirect immune response kidney injury, and rhabdomyolysis.[117 - 118, level III]

Management is primarily supportive and involves:[117, level III]

- volume optimization while strictly avoiding fluid overload which can exacerbate intravascular fluid extravasation
- monitoring and management of electrolytes abnormalities
- avoidance of the use of nephrotoxic medications
- kidney replacement therapy (KRT)

There is no evidence on timing of initiation of kidney replacement therapy in dengue infection. However, in non-dengue critically ill patient,

starting KRT early offers no survival advantage over waiting until specific clinical triggers are met.[119, level I]

Early KRT was associated with more KRT-related hypophosphataemia (RR=1.94, 95% CI 1.24 to 3.02) and Central Venous Catheter (CVC)-related bloodstream infection (RR=1.66, 95% CI 1.04 to 2.66). KRT should only be initiated when urgent indications are present, rather than based on the stage of AKI or the level of creatinine alone.[119, level I]

Urgent indications of kidney replacement therapy include:[68]

- Refractory metabolic acidosis, pH of less than 7.20 despite resuscitation
- Life-threatening hyperkalaemia
- Refractory volume overload
- Symptomatic uraemia

In hemodynamically unstable severe dengue patients requiring KRT, continuous kidney replacement therapy is recommended where available.

9.5.4. Electrolytes Management

Routine monitoring of serum electrolytes (potassium, sodium, calcium, magnesium, phosphate, chloride) is critical for hospitalised dengue patients.[120 - 122, level III]

In small cross-sectional and prospective observational study, electrolyte imbalances like hyponatraemia, hypophosphataemia, and hypocalcaemia were associated with the severity of dengue.[123, level III]

In critically ill patients, hypocalcaemia reduces the myocardial and vascular response to vasopressors, complicating the management of shock. Severe hypocalcaemia (ionised Calcium, iCa<0.8mmol/L) in the critically ill is independently associated with higher mortality rates.[124, level III]

Electrolyte panels should be performed at regular intervals for all patients admitted to the ICU. Any imbalances must be corrected promptly according to local ICU protocols to maintain physiological homeostasis and optimize the efficacy of supportive therapies.[120 - 122, level III]

9.5.5. Invasive Procedures in ICU

Invasive procedures in ICU may be necessary for the monitoring and management of patients with severe dengue.

a. Central Venous Line Insertion

Indications of central venous line insertion in severe dengue infection include:

- Prolonged infusion of vasopressors
- Inadequate peripheral venous access

A well-functioning peripheral venous cannula may be used for the administration of vasopressors for short durations (typically <2 hours). However, it requires vigorous monitoring for signs of extravasation and local tissue injury.

For rapid volume expansion, large-bore peripheral access (14G or 16G) is superior to a standard central venous line due to lower resistance and higher flow rates.

Although thrombocytopenia and coagulation abnormalities are a relative contraindication for central venous line insertion, the risk of bleeding following insertion in these patients is low.[125, level III] Routine prophylactic transfusion of platelets and FFP is not recommended in patients undergoing a central venous line.[126] The Consensus Statement on Patient Blood Management 2024 suggested prophylactic platelet and FFP transfusion should be considered if platelet count is less than 20 x 10^9/L or INR of more than 2. Hence, the decision for platelet and FFP transfusion should be individualised. If prophylactic platelet and FFP are given, the central line should be inserted during or within 30 - 60 minutes of transfusion.[87]

Central venous line insertion should be done by trained clinicians guided by real-time ultrasound where available. A subclavian cannulation should be avoided as subclavian vein and artery are not accessible for direct compression.[127, level I]

b. Arterial Catheter Insertion

Arterial catheter insertion may be indicated for the following:

- continuous intra-arterial pressure monitoring
- repeated arterial blood gas sampling
- haemodynamic monitoring (e.g. PPV analysis)

Severe thrombocytopenia and severe coagulopathy are relative contraindications for arterial catheter insertion. However, the rate of bleeding complications is low. Routine prophylactic transfusion of platelet and FFP is not recommended in patients undergoing arterial line insertion, thus decision for platelet and FFP transfusion should be individualised.[19]

c. Gastric Tube Insertion

If a gastric tube is required, the nasogastric route should be avoided. An orogastric tube is preferred, as it is generally less traumatic in patients with dengue.[19]

Recommendation 10

- Patients with severe dengue should be promptly identified and referred to intensive care unit (ICU).
- High-flow nasal cannula and non-invasive ventilation may be used in selected dengue patients with respiratory failure.
- In severe dengue patients, the need for mechanical ventilation should be based on the same established criteria used for other critically ill patients.
- Vasopressor may be used in dengue patient with persistent shock despite adequate fluid resuscitation. Noradrenaline is the preferred vasopressor.
- Kidney replacement therapy in severe dengue should be initiated according to standard clinical indications.
- Prophylactic platelet and FFP transfusion prior to invasive procedure in ICU should be individualised.

10. SPECIAL POPULATION

10.1. Dengue in Pregnancy

Dengue in pregnancy is high-risk because maternal physiological adaptations, like increased plasma volume and gestational thrombocytopenia, mask critical warning signs of plasma leakage. Healthcare providers must prioritise trend-based monitoring over isolated values to ensure early recognition and prevent adverse outcomes.

The physiological changes in pregnancy are depicted in **Figure 14**.

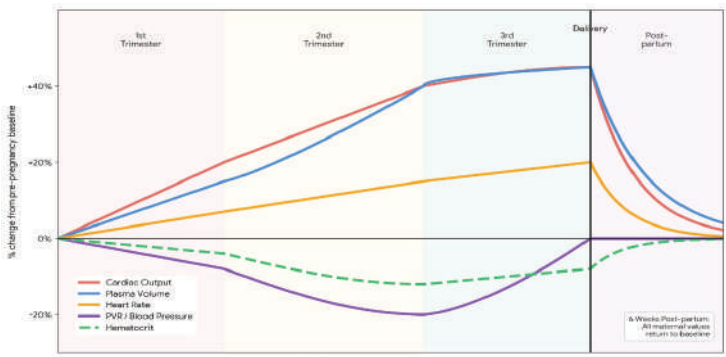


Figure 14: Physiological Changes in Pregnancy

Adapted from: De Backer J, Haugaa KH, Hasselberg NE, et al. 2025 ESC Guidelines for the management of cardiovascular disease and pregnancy: Developed by the task force on the management of cardiovascular disease and pregnancy of the European Society of Cardiology (ESC). Endorsed by the European Society of Gynecology (ESG). European Heart Journal. 2025;46(43):4462-568.

a. Dengue in Pregnancy and Diagnostic Challenges

Dengue in pregnancy presents with diagnostic challenges due to the physiological changes of pregnancy, as summarised below.

Marker	Pregnancy	Dengue
Haematocrit / Hb	↓ HCT (haemodilution)	↑ HCT (plasma leakage) – masking of haemoconcentration
Platelets	Mild thrombocytopenia	Marked thrombocytopenia – overlap with gestational thrombocytopenia
Heart Rate	↑ baseline HR	Tachycardia – shock may be overdiagnosed/missed
BP / Pulse Pressure	↓ BP, wider pulse pressure	Hypotension, narrow pulse pressure – delayed shock recognition
Fluid Status	Gravid uterus	Ascites/effusion – clinical examination limited
Liver Enzymes	Mild changes	↑ AST/ALT – non-specific overlap
Blood Gas ($PaCO_2$ / pH)	↓ $PaCO_2$ (respiratory alkalosis with compensatory metabolic acidosis)	Metabolic acidosis – severity may be under/overestimated

b. Pregnancy Outcomes in Pregnant Women with Dengue Fever

Evidence indicates that dengue infection significantly elevates risks for both mother and foetus, with pregnant women facing over twice the likelihood of adverse outcomes such as preterm birth, low birth weight, and miscarriage particularly during the third trimester. Compared to non-pregnant dengue patients, pregnant patients also demonstrated substantially higher rates of hospitalisation and a nearly five-fold increased risk of progressing to severe dengue.[128 - 129, level II-2]

c. Maternal and Foetal Outcomes in Gestational Dengue

Evidence highlights that pregnancy significantly complicates the clinical course of dengue, particularly during the critical phase.[128 - 129, level II-2; 130, level III]

Category	Specific Complication
Maternal Complications	Thrombocytopenia
	Shock
	Postpartum Haemorrhage
	Maternal Mortality
Systemic Complications	Acute Kidney Injury (AKI)
	Acute Respiratory Distress Syndrome
	Acute Liver Failure
Foetal & Neonatal Outcomes	Preterm Birth
	Low Birth Weight
	Intrauterine Foetal Demise
	Miscarriage Rate

d. Hospitalisation

All pregnant patients with confirmed dengue infection should be hospitalised to ensure continuous clinical monitoring and timely intervention.[55, 131]

Patient care should be led by a collaborative team of obstetricians and physicians, with early consultation and involvement of the critical care team as clinically indicated

Patient Group	Purpose of Admission	Recommended Ward
Gestation < 28 Weeks	Routine Monitoring	Medical or Obstetric Ward
Gestation ≥ 28 Weeks	Risk of Preterm Labour / Delivery	Obstetric Ward

e. Fluid management in Pregnancy

Fluid management in dengue patient is similar to normal population. However, it should be cautiously given and monitor risk of pulmonary oedema. Refer to **Section 9.3.2(a)** on fluid management

f. Antipyretic

The use of paracetamol is safe in pregnancy, avoid NSAIDs and aspirin due to bleeding risk.[55, 131]

g. Maternal Monitoring

Monitoring frequency in pregnant dengue patients depends on dengue severity. Patients without warning signs are reviewed daily while those with warning signs require 4 - 6 hourly monitoring of vitals and labs, and severe dengue necessitates continuous or 1 - 2 hourly monitoring.[55; 131]

h. Foetal Monitoring

Ultrasound and cardiotocography are essential tools for monitoring of foetal well-being.

i. Labour and Delivery

Labour and delivery should be avoided during the critical phase of dengue infection, and elective procedures such as induction of labour or planned Caesarean section should be postponed until platelet recovery begins, ideally reaching a threshold of >50 x 10^9/L. Vaginal delivery remains the preferred mode of birth, while Caesarean section should only be performed for obstetric indications. To mitigate the high risk of haemorrhage, early group and cross-match of blood must be arranged, and clinicians should employ measures to minimise bleeding, including the avoidance of genital tract trauma, limiting difficult instrumental deliveries, and ensuring active management of the third stage of labour.[55; 131]

Routine prophylactic platelet transfusion is not recommended for patients without active bleeding; instead, transfusion should be individualised based on maternal clinical status and the anticipated mode of delivery. Suggested platelet thresholds for intervention include >30 x 10^9/L for uncomplicated vaginal delivery, >50 x 10^9/L for instrumental vaginal delivery, Caesarean section, and spinal anaesthesia.[55; 131] Due to the risk of spinal haematoma in the presence of severe thrombocytopenia, regional anaesthesia may be contraindicated, necessitating a multidisciplinary discussion regarding the safety of general anaesthesia.

Following delivery, patients require intensive monitoring for at least 24 to 48 hours to detect signs of ongoing plasma leakage, occult bleeding, postpartum haemorrhage, or sudden haemodynamic deterioration. Serial monitoring of the patient's haematocrit and platelet count must be continued into the postpartum period, with prompt blood replacement initiated whenever clinically indicated.[55; 131]

j. Breastfeeding

Breastfeeding remains the recommended standard of care as its benefits significantly outweigh any risk of viral transmission. Dengue is not a contraindication; however, if a mother is too ill to breastfeed, she should be supported in expressing milk until she is well enough to resume.[132; 133, level III]

Recommendation 11

- All pregnant patients with confirmed dengue should be hospitalised and managed by a multidisciplinary team.
- Routine prophylactic platelet transfusions should be avoided in pregnant dengue patients unless indicated for pre-procedure/pre-delivery intervention.
- Vaginal delivery should be prioritised in pregnant dengue patients unless specific obstetric indications necessitate a Caesarean section.
- Intramuscular injections should be avoided in pregnant dengue patients with thrombocytopenia to prevent haematoma formation.
- Mothers diagnosed with dengue fever may continue breastfeeding.

10.2. Dengue in Patients with Heart Failure

Dengue infection in patients with heart failure is associated with a higher risk of shock, bleeding and death. Hence, they may need early hospitalisation for close monitoring.[55]

Treatment with standard fluid regimens also poses a challenge due to increased risk of fluid overload. Evidence on the management of patients with heart failure is derived mainly from Consensus guidelines.

Most dengue guidelines do not provide precise recommendations on fluid regimens in dengue patients with heart failure.

Management strategies for patients with heart failure includes:

- Early admission despite absence of warning signs.[20; 1]
- Modifying the volume of fluid therapy to reduce the risk of fluid overload.[64; 132]
- Early cardiologist consultation may be needed during the critical phase.[134]
- Consider loop diuretics in overloaded haemodynamically stable patients.[53]
- Temporarily discontinue antiplatelets or anticoagulants if platelet count <100,000/μL or if bleeding occurs[53] or for a few days during the critical phase.[134]
- Frequent reassessment of cardiac ischaemia and electrolytes.[53]

10.3. Dengue in Patients with End-Stage Kidney Disease/ Chronic Kidney Disease

Dengue patients with End-Stage Kidney Disease/Chronic Kidney Disease (ESKD/CKD) have higher rates of severe dengue, increased ICU admission and mortality.[135, level III] In a retrospective cohort study, CKD at presentation was an independent predictor of in-hospital mortality (aOR=3.907, 95% CI 1.785 to 8.552). Predictors of mortality were:[136. level II-2]

- altered consciousness (aOR=23.008, 95% CI 2.226 to 237.843)
- pulmonary oedema (aOR=13.525, 95% CI 1.348 to 135.663)
- leukocytosis during hospitalisation (aOR=1.131, 95% CI 1.023 to 1.250)

In another retrospective cohort, severe dengue among adult dengue patients with dialysis-dependent ESKD showed:[137, level II-2]

- 36.7% of patients presented with warning signs and 36.7% developed severe dengue
- 8.2% had in-hospital mortality
- warning signs were significantly associated with plasma leakage (OR=12.36, 95% CI 1.56 to 97.74) and severe haemorrhage (OR=5.1, 95% CI 1.03 to 25.18). These were more likely to be found in older patients >65 years (56% vs 16.7%, p=0.007)

Evidence on the management for patients with CKD Stage 3b and above is derived mainly from guidelines which includes:

- early admission despite absence of warning signs for close monitoring.[1; 20]
- modifying the volume of fluid therapy to reduce the risk of fluid overload[64; 20; 132]

- early consultation with a nephrologist for planning of KRT in oliguric patients with risk of fluid overload[53; 134]

In patients with ESKD, baseline Hb and HCT may differ from the general population due to chronic anaemia or haemodilution from fluid overload. Therefore, haematocrit interpretation should be individualised using the patient's own baseline Hb/HCT and serial trends.

10.4. Dengue in Patients with Liver Cirrhosis

Dengue infection in patients with Liver Cirrhosis is associated with increased risk of severe dengue and mortality. Evidence on the management of patients with liver cirrhosis is derived mainly from Consensus guidelines which include:

- cautious use of paracetamol (acetaminophen) to control fever or pain.[64]
- appropriate management of hepatic failure with fluid and blood transfusion as they are at risk of gastrointestinal (GI) bleeding and severe shock.[53]
- IV Vitamin K may be needed if PT/INR is prolonged.[53]
- Individualised fluid management with close monitoring for those with warning sign and severe dengue.[20; 132]

Recommendation 12

- In dengue patients with heart failure, end-stage kidney disease and liver cirrhosis, fluid management should be individualised with close monitoring

10.5. Dengue in Patients with Diabetes Mellitus

Dengue infection poses a higher risk of adverse outcomes in patients with Diabetes Mellitus (DM). This is due to several factors: metabolic instability, greater risk of dehydration, and the diagnostic challenge of differentiating dengue shock from metabolic complications.

Dengue infection can trigger serious metabolic emergencies like Diabetic Ketoacidosis (DKA) and Hyperosmolar Hyperglycaemic State. These conditions can either mimic the symptoms of Dengue Shock Syndrome (DSS) or occur concurrently.[19; 55; 62]

Hyperglycaemia causes osmotic diuresis, which worsens the hypovolaemic shock. Osmotic diuresis also gives a false impression of good urine output. Hence, dengue patients with diabetes require early and regular metabolic assessment, including monitoring of capillary blood sugar, ketones/acid-base status, and renal function/electrolytes,

to carefully differentiate dengue shock from DKA/HHS, as their symptoms may overlap.[55; 138]

In hospitalised patients, oral hypoglycaemic agents should be discontinued during acute illness as they increase the risk of hypoglycaemia due to poor oral intake, while metformin in particular may lead to lactic acidosis (dehydration/hepatic dysfunction).[55] Insulin-based regimes are preferred in controlling blood sugar.

The CPG DG opines that in dengue patients with DKA/HHS, fluid requirement should follow DKA/HHS regimen whereas in plasma leakage without DKA/HHS, fluid therapy should be guided by clinical assessment.

Recommendation 13

- In hospitalised dengue patients with diabetes:
  - early and serial metabolic assessment should be done to distinguish dengue shock from Diabetic Ketoacidosis (DKA) and Hyperosmolar Hyperglycaemic State (HHS)
  - oral hypoglycaemics should be stopped and consider insulin for glycaemic control.
- In dengue patients with DKA/HHS, fluid requirement should follow DKA/HHS regimen whereas in plasma leakage without DKA/HHS, fluid therapy should be guided by clinical assessment.

10.6. Dengue in Patients with Hypertension

Patients with pre-existing hypertension demonstrate consistently higher baseline blood pressures throughout dengue illness, including during the critical phase. However, similar to normotensive individuals, blood pressure may decline around defervescence due to plasma leakage.[139, level III]

Thus, a critical diagnostic challenge is presented by this situation. For a patient with hypertension, a “normal” blood pressure reading should be interpreted with caution. It may signify relative hypotension and early shock, rather than indicating clinical stability. Therefore, it is essential to ask about the patient’s baseline BP, and closely monitor trends in blood pressure, pulse pressure, CCTVR (colour, capillary refill time, temperature, pulse volume and pulse rate), and urine output.[55]

Dengue assessment and management is further complicated by anti-hypertensives:

- Beta-blockers - can mask tachycardia or worsen bradycardia
- Calcium channel blockers - may cause tachycardia.
- ACEi/ARBs and diuretics - increase the risk of renal injury and hypovolaemia.

Antihypertensives may be withheld during acute dengue, especially in the critical phase, to prevent hypotension. It can be restarted gradually when the patient is in recovery phase with haemodynamic stability.[55]

Recommendation 14

- In dengue patients with hypertension:
  - blood pressure should be interpreted carefully relative to baseline
  - frequent monitoring of trends, pulse pressure, CCTVR and urine output should be done
  - consider withholding antihypertensive medications during the acute phase of dengue, particularly in patients at risk of haemodynamic instability or plasma leakage

10.7. Dengue in Geriatric Population

Managing dengue in older adults (≥60 years) is challenging due to physiological changes, comorbidities, and atypical presentation. Despite some studies suggesting a benign course in selected hospitalised elderly, they face increased risks of complications, delayed diagnosis, and adverse outcomes compared to younger patients. Comorbidities (e.g., hypertension, diabetes, chronic kidney disease, cardiovascular disease) significantly increase the risk of severe dengue, organ dysfunction, and prolonged hospitalisation.[140, level III]

The clinical presentation of dengue in older adults is often atypical. Fever may be absent, low-grade or delayed, and constitutional symptoms like lethargy and poor appetite frequently predominate over classical features of dengue such as myalgia, retro-orbital pain and rash. As early signs can be subtle, relying on classical symptoms may lead to delayed diagnosis, as atypical presentations can mimic conditions like Sepsis, Acute Coronary Syndrome, Stroke/delirium, or decompensated chronic illness.[141, level III] Therefore, a high index of suspicion and vigilant monitoring are vital for safe and timely management.

Leucopenia is an important early laboratory marker that should raise suspicion of dengue, particularly in older adults, with good discriminatory ability for differentiating dengue from other febrile illnesses (AUC=0.91, 95% CI 0.86 to 0.96), and increasing predictive value with age, while thrombocytopenia is a later, less reliable sign, and rising haematocrit can be subtle, especially with coexisting anaemia or chronic disease.[141, level III]

Elderly populations face an increased risk of severe disease due to comorbidities, polypharmacy risks, and reduced physiological reserve like frailty and sarcopenia.

They are also at increased risk of haemorrhagic symptoms, severe dengue, kidney and liver problems, and bacterial coinfections. Signs that may suggest a bacterial coinfection are persistent fever of >5 days, acute kidney injury, leucocytosis and neutrophilia.[89]

Fluid therapy for elderly dengue patients should be individualised and requires a balanced, cautious approach due to the high risk of fluid overload, pulmonary oedema, and heart failure exacerbation.

A lower threshold for hospital admission in the elderly with dengue is especially seen in those with underlying comorbidities, individuals whose home setting is inadequate for safe monitoring, or those presenting with atypical clinical features.

Recommendation 15

- In acutely unwell elderly patients with leucopenia, dengue fever should be considered, even in the absence of classical dengue features.
- Elderly patients with dengue particularly those with underlying comorbidities, or inadequate social support should be hospitalised for monitoring and management to prevent complication.

10.8. Dengue in Obese Patients

Malaysia National Health and Morbidity Survey (NHMS) 2023 reported that 54.4% of adults are overweight or obese (32.6% vs 21.8% respectively).[58] This high burden has direct implications for adverse dengue outcomes.

A recent meta-analysis focusing on hospitalised adult dengue patients revealed that, although there was no difference in overall severe dengue or mortality rates, obese patients with dengue experienced significantly higher incidences of certain complications. These included severe hepatitis (AOR=2.06, 95% CI 1.05 to 4.05), petechiae (AOR=1.35, 95% CI 1.03 to 1.79), dyspnoea (AOR=1.38, 95% CI 1.02 to 1.88), and severe thrombocytopenia ($p<0.001$).[142, level III]

Obese patients with dengue should be considered for admission to ensure early escalation of care is feasible. Clinical assessment may be less reliable due to body habitus; hence the need to rely on haemodynamic trends and objective parameters.

Managing fluid therapy in obese dengue patients demands a careful consideration. The challenge lies in meticulously balancing the risk of under-resuscitation, which can precipitate shock, with the danger of

over-resuscitation, potentially leading to fluid overload and pulmonary oedema.

The use of a weight-based fluid strategy is supported by evidence, particularly for patients with septic shock where fluid administration based on Adjusted Body Weight (AdjBW) was associated with improved mortality (OR=0.29, 95% CI 0.11 to 0.74).[143, level III]

It is critical to manage comorbid conditions associated with obesity focusing on screening for conditions like diabetes, hypertension, OSA and NAFLD, requiring specific care plan.

Recommendation 16
- In obese dengue patients in shock, fluid management should be done with caution by applying adjusted body weight-based fluid dosing with frequent monitoring.

10.9. Dengue in Patients with Haemoglobinopathies

Patients with underlying haemoglobinopathies such as thalassaemia represent a special population in dengue infection, where classical diagnostic and monitoring parameters may be unreliable.

Evidence from paediatric cohorts and case series on dengue thalassemia study, highlights several important notes as the following:[144, level III; 145, level III]

- Dengue may present with persistent fever, progressive anaemia, and thrombocytopenia
- Haemoconcentration may be absent, as anaemia from haemolysis predominates
- HCT is unreliable as a marker of plasma leakage
- Clinical deterioration may be missed or delayed if conventional dengue markers are relied upon

Therefore, heightened clinical vigilance and alternative monitoring strategies are essential to ensure early diagnosis and appropriate management.

Packed cell transfusion is a critical, often life-saving intervention for symptomatic anaemia in this population, where the underlying cause is typically haemolysis rather than plasma leakage or bleeding. Indications for transfusion include symptomatic anaemia, a rapid decline in haemoglobin levels, or clinical signs of tissue hypoxia or shock.

- In dengue patients with haemoglobinopathies (e.g. thalassaemia):
  - maintain a high index of suspicion for dengue when presenting with fever, progressive anaemia, and thrombocytopenia
  - do not rely on haematocrit (HCT) to assess plasma leakage.

- consider early packed red cell transfusion for symptomatic anaemia, rapid Hb decline, or evidence of tissue hypoxia or shock

11. COMPLEMENTARY TREATMENT IN DENGUE

There are plenty of complementary treatments being used in dengue with the objective of preventing severe dengue. However, there is limited evidence to support these practices. Papaya [Carica papaya(CP)] leaf extract has been evaluated as a complementary treatment in dengue infection. Evidence from meta-analyses suggests that CP leaf extract may result in a modest increase in platelet count, (MD=20.27, 95% CI 6.21 to 34.73[146, level I] and MD=35.45, 95% CI 23.74 to 47.15[147, level I]), and a reduction in length of hospital stay (MD=1.90 days, 95% CI 1.62 to 2.18[146, level I] and MD= -1.98 days, 95% CI -2.12 to -1.83[147, level I]). However, no consistent benefit has been demonstrated for other clinically important outcomes, including bleeding complications, disease severity, or mortality.[146 - 147, level I] Quality of primary studies used were mixed. Evidence is limited by heterogeneity of the studies, methodological limitations and insufficient safety data. CP leaf extract should not be used with the sole aim of accelerating hospital discharge, as improvements in platelet count do not reliably reflect overall clinical recovery and have not been shown to correlate with other key clinical parameters.[146 - 147, level I]

- There is lack of evidence to support the use of Carica papaya leaf extract in the management of dengue infection.

12. DENGUE PREVENTION

12.1. Role of Dengue Vaccination in Dengue Prevention

Dengue prevention primarily relies on vector control measures, including elimination of breeding sites, environmental management, larviciding and personal protective measures to reduce transmission.[1; 148, level I] Novel approaches such as Wolbachia-infected mosquito deployment have demonstrated additional benefit.[149, level I] Despite these efforts, dengue remains highly prevalent in endemic settings such as Malaysia, supporting the need for additional preventive strategies including vaccination.

Dengue vaccine development is complicated by the presence of four distinct serotypes of dengue virus and the risk of antibody-dependent enhancement (ADE), whereby prior non-neutralising antibodies may cause increased severity in subsequent dengue infections.[150, level III] The first licensed dengue vaccine, CYD-TDV (Dengvaxia) demonstrated efficacy in reducing dengue infection and hospitalisation, particularly among seropositive individuals.[151, level I] However, subsequent analyses identified an increased risk of severe dengue among seronegative recipients. World Health Organization (WHO) subsequently recommended its use only in individuals with confirmed prior dengue infection.[9, level I; 152]

The live-attenuated tetravalent dengue vaccine TAK-003 (Qdenga), which utilises a DENV-2 backbone demonstrated a favourable safety profile and does not require pre-vaccination serological screening.[5] It is administered as a two-dose schedule given three months apart. In a systematic review and meta-analysis in which approximately 99% of the efficacy data were derived from paediatric and adolescent populations, seroconversion rates exceeded 90% for all serotypes (except DENV-4 at approximately 84%) after the first dose and more than 96% for all serotypes after the second dose. Vaccine efficacy exceeded 80% against DENV-2 and approximately 40% against DENV-1. Protection against DENV-3 and DENV-4 was observed primarily among individuals with prior dengue exposure, while no significant efficacy was demonstrated among seronegative recipients for these serotypes.[153, level I] The primary studies used were of low risk of bias.

An immune-bridging analysis comparing immune responses between children and adolescents (4 - 16 years) living in dengue-endemic regions and adults (18 - 60 years) living in non-endemic regions demonstrated non-inferior neutralising antibody responses for all four serotypes. These data suggest that the vaccine effectiveness observed in paediatric trials may be inferred to the adult population.[154, level II-2]

Real-world effectiveness data further support the protective effects of TAK-003. A case-control study reported vaccine effectiveness of 50.2% after one dose and 61.7% after two doses against symptomatic dengue, with 67.5% protection against dengue-related hospitalisation following a single dose. Protection following a single dose declined after approximately 90 days, highlighting the importance of completing the two-dose vaccination schedule.[155, level II-2]

A position paper by WHO recommended the use of TAK-003 in children aged 6 - 16 years in high-dengue-transmission settings but did not recommend the use of TAK-003 in children aged <6 years because of the lower efficacy of the vaccine in this age group. Persons with comorbidities who live in dengue-endemic countries could be offered vaccination but need to be informed that the vaccine may not protect against DENV3 and DENV4 in seronegative recipients and potential risk of severe dengue cannot be ruled out if exposed to these serotypes.[5]

TAK-003 is currently approved for use in Malaysia for anyone aged ≥4 years. Dengue Prevention Advocacy Malaysia (DPAM) position and recommendations on dengue vaccine include:[156, level III]

- All relevant healthcare professionals (HCPs) should recommend the uptake of the dengue vaccine as per the approved indication.
- HCPs should view each patient visit as an opportunity to recommend dengue vaccination to all individuals aged 4 years and above.
- Dengue vaccine recommendation is especially crucial if the individual falls into one of the following categories which increases their risk of getting dengue or developing severe dengue:
    - Individuals with comorbidities (e.g. diabetes, cardiac disorders etc.).
    - Individuals residing or working in outbreak/hotspot areas.
    - Individuals residing or working in densely populated areas with poor drainage and waste management.
    - Individuals at risk of secondary dengue infection (this statement does not necessitate pre-screening).

The Butantan dengue vaccine (Butantan-DV) is a single-dose live-attenuated tetravalent vaccine based on the NIH TV003 backbone. A phase-2 randomised trial demonstrated that the vaccine was safe and immunogenic in both dengue-naïve and dengue-exposed adults, producing balanced neutralising antibody responses against all four dengue virus serotypes.[157, level I] A subsequent phase-3 trial reported an overall vaccine efficacy of 67.3% against symptomatic dengue and 89% protection against severe dengue or dengue with warning signs. However, no cases of DENV-3 or DENV-4 occurred during follow-up, and vaccine efficacy against these serotypes could not be

assessed.[158, level I] The vaccine was licensed in Brazil in 2025, and regulatory approval outside Brazil has not yet been reported.

Taken together, current evidence suggests that TAK-003 may be considered as an additional preventive measure in high-dengue-transmission settings, particularly among children and adolescents aged 6 - 16 years and adults with comorbidities at increased risk of severe dengue, while vector control and environmental management remain the cornerstone of dengue prevention.

Recommendations 17

- Dengue vaccination may be considered in all adults, particularly those with comorbidities who are at increased risk of severe dengue, unless contraindicated.

13. DISCHARGE CRITERIA

Dengue patients discharge guide are as follows:[19]

Absolute Discharge Criteria

A. Clinical stability

- Afebrile for 24 - 48 hours and clinically improving
- Stable hemodynamic
- Adequate oral intake with improved appetite
- Good urine output (0.5 - 1.5 mL/kg/hr)
- No active bleeding
- No respiratory distress

B. Laboratory stability

- Platelet count: increasing trend (trend matters more than absolute number).
- Haematocrit stable without IV fluids
- Resolving organ dysfunction

Discharge Counselling and Post-Discharge Advice

All discharged patients must receive clear verbal and written instructions, including:

- Adequate oral fluid intake
- Resume old medications as instructed
- Avoid strenuous activity until full recovery
- For admitted patients, attend follow-up review and repeat blood investigations at the nearest primary care clinic as needed

14. IMPLEMENTING THE GUIDELINES

Translating the updated, evidence-based recommendations from the Clinical Practice Guidelines (CPG) for the Management of Dengue in Adults (Fourth Edition) into uniform clinical practice across various healthcare settings is difficult. This challenge is attributed to multiple influencing factors at the system, provider, and patient levels.

14.1. Facilitating and Limiting Factors

Existing facilitating factors for application of the recommendations in the CPG include:

- Nationwide dissemination of the CPG to healthcare providers
- Regular physical and online scientific meetings/trainings/ conferences on dengue management to enhance knowledge and skills of healthcare providers
- State and national level mortality review to identify gaps in dengue management
- Dengue champions being appointed to facilitate training and planning for dengue outbreaks
- Yearly clinical audits in healthcare facilities to assess compliance to guidelines
- National committee for dengue outbreak contingency plan protocol in MoH hospital to prepare for surge in outbreaks
- Regular technical dengue committee meetings at state and national level involving multiple agencies to control dengue outbreak and facilitate collaboration.

Existing barriers for application of the recommendations of the CPG are:

- **Variability in Clinical Expertise and Experience**:
  - Differences in knowledge and skills across care settings
  - Limited familiarity with updated tools and practices
  - Difficulty recognising atypical presentations and warning signs
  - Over-reliance on outdated clinical markers
- **Resource and Infrastructure Constraints**:
  - Limited access to PoCUS and trained staff
  - Limited laboratory test availability (high-titre IgG ELISA)
  - Critical care capacity issues (ICU beds, etc.)
  - High patient volumes during outbreaks causing overcrowding
- **Complexity of Clinical Decision-Making**:
  - Phase-specific management is required
  - Challenges in interpreting clinical/lab trends
  - Differentiating dengue from other illnesses
  - Managing special populations

- **Adherence and Behavioural Barriers**:
  - Inconsistent guideline adherence
  - Lack of awareness of updates
- **System-Level and Coordination Challenges**:
  - Fragmentation between different care levels
  - Inconsistent referral and escalation pathways
- **Patient-Related Factors**:
  - Late presentation to healthcare facilities
  - Poor compliance with follow-up
  - Use of alternative treatments or self-medication
  - Lack of understanding of warning signs

14.2. Potential Resource Implications

For the successful implementation of this Clinical Practice Guideline (CPG), investment is needed across several key areas. Commitment to these domains will help ensure the guideline's effectiveness and sustainability within the healthcare system.

a. Human Resource Development

There must be a strong focus on enhancing the knowledge and skills of healthcare providers. This involves delivering comprehensive training on dengue risk assessment, the effective management of fluids, and the use of PoCUS. Such initiatives will empower staff to provide optimal care for dengue patients.

It is equally important to build competencies at all levels of the healthcare system, ensuring that staff, from primary to tertiary care, are fully equipped to manage dengue cases. Continuous professional development and upskilling will help drive improvements in patient outcomes.

b. Infrastructure and Equipment

Upgrading and expanding healthcare infrastructure is crucial. This includes improving bed capacity to better accommodate patients especially during dengue outbreaks or surges.

Access to essential diagnostic tools, such as high-titre IgG ELISA tests and PoCUS devices, should be prioritised to aid in the timely and accurate diagnosis of dengue and fluid management.

Additionally, intensive care and high-dependency services need to be strengthened to provide advanced support when needed.

c. System Strengthening

Standardising clinical processes is essential for consistent and quality care. The development and use of uniform clinical pathways, clerking

sheets, and monitoring charts will help align practices across different healthcare settings.

The introduction of audit indicators and regular feedback mechanisms will ensure that adherence to guidelines is monitored and areas for improvement are identified.

Reinforcing referral networks and establishing robust systems for surge preparedness will enhance the ability of healthcare facilities to respond efficiently during outbreaks.

Implementation of the dengue CPG is influenced by clinical complexity, resource variability, system constraints, and behavioural factors. Addressing these barriers requires a coordinated, system-wide approach focusing on capacity building, infrastructure strengthening, and continuous quality improvement to ensure safe and effective dengue care across all healthcare settings.

Based on the key recommendations, the following are proposed as clinical audit indicators for the CPG:

$$\text{Case fatality rate of dengue infection (National target < 0.2\%)} = \frac{\text{Number of deaths of dengue infection cases in a period}}{\text{Number of dengue infection cases in the same period}} \times 100\%$$

REFERENCES

1. World Health Organization. Dengue guidelines for diagnosis, treatment, prevention and control. Geneva: World Health Organization; 2009.
2. World Health Organization. Dengue: global situation, surveillance and progress – 2024 update. Wkly Epidemiol Rec [Internet]. 2025. Available from: https://www.who.int/publications/i/item/who-wer10052-665-678
3. Ministry of Health Malaysia. Health facts 2025. Putrajaya: Ministry of Health Malaysia; 2025.
4. Parlimen Malaysia. Penyata rasmi Parlimen Dewan Rakyat Parlimen Kelima Belas Penggal Keempat Mesyuarat Kedua. Kuala Lumpur: Parlimen Malaysia; 2025. Available from: https://parlimen.gov.my/files/hindex/pdf/DR-22072025.pdf
5. World Health Organization. Dengue vaccines: WHO position paper – May 2024. Wkly Epidemiol Rec. 2024;99(21):203-224.
6. Ministry of Health Malaysia. One hour Malaysia clean up: komplikasi denggi [Internet]. Putrajaya: Ministry of Health Malaysia. Available from: https://infosihat.moh.gov.my/images/media_sihat/infografik/jpeg/TRIVIA%208%20Komplikasi%20Denggi-01-01.jpg
7. Yang ZS, Baua AD, Hemdan MS, et al. Dengue virus infection: a systematic review of pathogenesis, diagnosis and management. J Infect Public Health. 2025;18(12):102982.
8. Chem YK, Yenamandra SP, Chong CK, et al. Molecular epidemiology of dengue in Malaysia: 2015-2021. Front Genet. 2024;15:1368843.
9. Sridhar S, Luedtke A, Langevin E, et al. Effect of dengue serostatus on dengue vaccine safety and efficacy. N Engl J Med. 2018;379(4):327-340.
10. Puerta-Guardo H, Glasner DR, Harris E. Dengue virus NS1 disrupts the endothelial glycocalyx, leading to hyperpermeability. PLoS Pathog. 2016;12(7):e1005738.
11. Simmons CP, Farrar JJ, Nguyen VV, et al. Dengue. N Engl J Med. 2012;366(15):1423-1432.
12. McBride A, Duyen HTL, Vuong NL, et al. Endothelial and inflammatory pathophysiology in dengue shock: new insights from a prospective cohort study in Vietnam. PLoS Negl Trop Dis. 2024;18(3):e0012071.
13. Trung DT, Trieu HT, Wills BA. Microvascular fluid exchange: implications of the revised Starling model for resuscitation of dengue shock syndrome. Front Med. 2020;7:601520.
14. Woodcock TE, Woodcock TM. Revised Starling equation and the glycocalyx model of transvascular fluid exchange: an improved paradigm for prescribing intravenous fluid therapy. Br J Anaesth. 2012;108(3):384-394.
15. Levick JR, Michel CC. Microvascular fluid exchange and the revised Starling principle. Cardiovasc Res. 2010;87(2):198-210.
16. Schaefer TJ, Panda PK, Wolford RW. Dengue fever. Treasure Island (FL): StatPearls Publishing; 2026.
17. World Health Organization Regional Office for South-East Asia. Dengue case management for primary health care and home-based care: booklet for health professionals and health-care workers. New Delhi: WHO-SEARO; 2025.
18. Rodrigo C, Sigera C, Fernando D, et al. Plasma leakage in dengue: a systematic review of prospective observational studies. BMC Infect Dis. 2021;21(1):1082.
19. Ministry of Health Malaysia. Clinical practice guidelines: management of dengue infection in adults. 3rd ed. Putrajaya: Ministry of Health Malaysia; 2015.
20. Pan American Health Organization. Algorithms for the clinical management of dengue patients. Washington (DC): PAHO; 2020.
21. Umakanth M, Suganthan N. Unusual manifestations of dengue fever: a review on expanded dengue syndrome. Cureus. 2020;12(9):e10678.

22. Attorney General's Chambers of Malaysia. Prevention and Control of Infectious Diseases (Amendment of First Schedule) Order 2024. Kuala Lumpur: Attorney General's Chambers; 2024.
23. Attorney General's Chambers of Malaysia. Akta Pencegahan dan Pengawalan Penyakit Berjangkit (Pindaan) 2025. Kuala Lumpur: Attorney General's Chambers; 2025.
24. Ministry of Health Malaysia. Pelan strategik pencegahan dan kawalan denggi kebangsaan 2022-2026. Putrajaya: Ministry of Health Malaysia; 2022.
25. Ministry of Health Malaysia. Surat pekeliling Ketua Pengarah Kesihatan Bil. 16/2024: pengukuhan program makmal survelan serotaip virus denggi dan flavivirus. Putrajaya: Ministry of Health Malaysia; 2024.
26. Mata VE, Andrade CAF, Passos SRL, et al. Rapid immunochromatographic tests for the diagnosis of dengue: a systematic review and meta-analysis. Cad Saude Publica. 2020;36(6):e00225618.
27. Pillay K, Keddie SH, Fitchett E, et al. Evaluating the performance of common reference laboratory tests for acute dengue diagnosis: a systematic review and meta-analysis of RT-PCR, NS1 ELISA, and IgM ELISA. Lancet Microbe. 2025;6(7):101088.
28. Ruchusatsawat K, Benjamungkalarak T, Phunikom N, et al. A performance comparison between fluorescent immunoassay and immunochromatography for rapid dengue detection in clinical specimens. Sci Rep. 2022;12(1):17299.
29. Chong ZL, Soe HJ, Ismail AA, et al. Evaluation of the diagnostic accuracy of a new biosensors-based rapid diagnostic test for the point-of-care diagnosis of previous and recent dengue infections in Malaysia. Biosensors. 2021;11(5):161.
30. World Health Organization. Laboratory testing for dengue virus: interim guidance, April 2025. Geneva: World Health Organization; 2025.
31. Tsai JJ, Tsai CY, Lin PC, et al. Comparing the performance of dengue virus IgG and IgG-capture enzyme-linked immunosorbent assays in seroprevalence study. BMC Infect Dis. 2023;23(1):301.
32. Dhanoa A, Hassan SS, Jahan NK, et al. Seroprevalence of dengue among healthy adults in a rural community in Southern Malaysia: a pilot study. Infect Dis Poverty. 2018;7(1):1.
33. Low JG, Oh HM, Leo YS, et al. IgG, IgM, and nonstructural protein 1 response profiles after receipt of tetravalent dengue vaccine TAK-003 in a phase 2 randomized controlled trial. Am J Trop Med Hyg. 2024;111(1):102-106.
34. Pan American Health Organization, World Health Organization. Technical note: detection and differentiation of dengue virus in the context of dengue vaccine administration. Washington (DC): PAHO/WHO; 2024.
35. Yang CH, Lee IK, Chen YC, et al. Prognostic factors in severe dengue patients: a multi-center retrospective cohort study. PLoS Negl Trop Dis. 2025;19(1):e0012846.
36. Talukdar S, Thanachartwet V, Desakorn V, et al. Predictors of plasma leakage among dengue patients in Thailand: a plasma-leak score analysis. PLoS One. 2021;16(7):e0255358.
37. Lee IK, Hsieh CJ, Lee CT, et al. Diabetic patients suffering dengue are at risk for development of dengue shock syndrome/severe dengue: emphasizing the impacts of co-existing comorbidity(ies) and glycemic control on dengue severity. J Microbiol Immunol Infect. 2020;53(1):69-78.
38. Abdul Rahman FK, Wan Puteh SE, Zainuddin MA. E-dengue system insights: exploring the factors influencing dengue-related deaths in an urbanized state in a low-middle income country (LMIC). BMC Public Health. 2024;24(1):3055.
39. Thach TQ, Eisa HG, Hmeda AB, et al. Predictive markers for the early prognosis of dengue severity: a systematic review and meta-analysis. PLoS Negl Trop Dis. 2021;15(10):e0009808.

40. Zhang H, Xie Z, Xie X, et al. A novel predictor of severe dengue: the aspartate aminotransferase/platelet count ratio index (APRI). J Med Virol. 2018;90(5):803-809.
41. Chaudhary S, Manrai K, Dhagat P, et al. Abdominal and chest ultrasonography: a predictor for disease progression in nonsevere dengue. Med J Armed Forces India. 2023;79(4):386-391.
42. Parmar JP, Mohan C, Vora M. Patterns of gall bladder wall thickening in dengue fever: a mirror of the severity of disease. Ultrasound Int Open. 2017;3(2):E76-E81.
43. Jayachandran AK, Nelson V, Anisha K. APRI as a predictor of severe dengue fever. J Family Med Prim Care. 2024;13(2):613-618.
44. Jamil Z, Khalid S, Khan H, et al. Aspartate aminotransferase-to-platelet ratio index (APRI) as a novel score in early detection of complicated dengue fever. J Multidiscip Healthc. 2024;17:2321-2330.
45. Vasikasin V, Rojdumrongrattana T, Chuerboonchai W, et al. Effect of standard dose paracetamol versus placebo as antipyretic therapy on liver injury in adult dengue infection: a multicentre randomised controlled trial. Lancet Glob Health. 2019;7(5):e664-e670.
46. Chia PY, Htun HL, Leo YS, et al. Safety of temporary interruption of antiplatelet therapy in dengue fever with thrombocytopenia. J Infect. 2021;82(2):270-275.
47. Chia PY, Htun HL, Ling WP, et al. Hyperlipidemia, statin use and dengue severity. Sci Rep. 2018;8(1):17147.
48. Htun HL, Yeo TW, Tam CC, et al. Metformin use and severe dengue in diabetic adults. Sci Rep. 2018;8(1):3344.
49. Minh Nguyen N, Thi Hoai Tam D, Quang Chanh H, et al. Safety and tolerability of metformin in overweight and obese patients with dengue: an open-label clinical trial (MeDO). PLoS Negl Trop Dis. 2025;19(7):e0013281.
50. Meco M, Agosteo E, Zulli P, et al. Beyond glycemia: pharmacology-driven ketogenesis and euglycemic DKA with SGLT2 inhibitors-a practical review for acute care. J Pers Med. 2026;16(3):142.
51. Ministry of Health Malaysia. Penggunaan protokol Malaysian Triage Scale 2022 ke arah penyeragaman kaedah saringan pesakit di kaunter triage Jabatan Kecemasan dan Trauma, hospital-hospital Kementerian Kesihatan Malaysia. Putrajaya: Ministry of Health Malaysia; 2023.
52. Ministry of Health and Family Welfare India. National guidelines for clinical management of dengue fever 2023. New Delhi: National Center for Vector Borne Diseases Control; 2023.
53. World Health Organization Timor-Leste. National guidelines for clinical management of dengue 2022. Dili: Ministry of Health Timor-Leste; 2022.
54. World Health Organization Nepal. Short guidelines on dengue case management. Kathmandu: World Health Organization Nepal; 2019.
55. Pan American Health Organization. Dengue guidelines for patient care for the region of America. 2nd ed. Washington (DC): PAHO; 2016.
56. Sigera PC, Weeratunga P, Deepika Fernando S, et al. Rational use of ultrasonography with triaging of patients to detect dengue plasma leakage in resource limited settings: a prospective cohort study. Trop Med Int Health. 2021;26(8):993-1001.
57. Azmi MY Jr, Junidah R, Siti Mariam A, et al. Body mass index (BMI) of adults: findings of the Malaysian Adult Nutrition Survey (MANS). Malays J Nutr. 2009;15(2):97-119.
58. Institute for Public Health Malaysia. National Health and Morbidity Survey (NHMS) 2023: non-communicable diseases and healthcare demand - key findings. Setia Alam: National Institutes of Health; 2024.

59. Ministry of Health Sri Lanka. Guidelines on management of dengue fever & dengue haemorrhagic fever in adults. Colombo: Ministry of Health Sri Lanka; 2024.
60. Directorate General of Health Services Bangladesh. Pocket guideline for dengue clinical case management 2022 (revised). Dhaka: DGHS Bangladesh; 2022.
61. Bhalla A, Singh H, Suri V, et al. ISCCM position statement: management of severe dengue in intensive care unit. Indian J Crit Care Med. 2024;28(Suppl 2):S42-S58.
62. World Health Organization Regional Office for South-East Asia. Comprehensive guidelines for prevention and control of dengue and dengue haemorrhagic fever. Revised and expanded ed. New Delhi: WHO SEARO; 2011.
63. Madanayake P, Jayawardena A, Wijekoon SL, et al. Fluid requirement in adult dengue haemorrhagic fever patients during the critical phase of the illness: an observational study. BMC Infect Dis. 2021;21(1):286.
64. World Health Organization. WHO guidelines for clinical management of arboviral diseases: dengue, chikungunya, Zika and yellow fever. Geneva: World Health Organization; 2025.
65. Pan American Health Organization. Guidelines for the clinical diagnosis and treatment of dengue, chikungunya, and Zika. Washington (DC): PAHO; 2022.
66. Antequera Martín AM, Barea Mendoza JA, Muriel A, et al. Buffered solutions versus 0.9% saline for resuscitation in critically ill adults and children. Cochrane Database Syst Rev. 2019;7(7):CD012247.
67. Prescott HC, Antonelli M, Alhazzani W, et al. Surviving Sepsis Campaign: international guidelines for management of sepsis and septic shock 2026. Crit Care Med. 2026;54(4):725-812.
68. KDIGO Acute Kidney Injury Work Group. KDIGO 2026 clinical practice guideline for acute kidney injury (AKI) and acute kidney disease (AKD). Kidney Int. 2026.
69. Chaves RCF, Barbas CSV, Queiroz VNF, et al. Assessment of fluid responsiveness using pulse pressure variation, stroke volume variation, plethysmographic variability index, central venous pressure, and inferior vena cava variation in patients undergoing mechanical ventilation: a systematic review and meta-analysis. Crit Care. 2024;28(1):289.
70. Toscani L, Aya HD, Antonakaki D, et al. What is the impact of the fluid challenge technique on diagnosis of fluid responsiveness? A systematic review and meta-analysis. Crit Care. 2017;21(1):207.
71. Messina A, Calabrò L, Pugliese L, et al. Fluid challenge in critically ill patients receiving haemodynamic monitoring: a systematic review and comparison of two decades. Crit Care. 2022;26(1):186.
72. Monnet X, Marik P, Teboul JL. Passive leg raising for predicting fluid responsiveness: a systematic review and meta-analysis. Intensive Care Med. 2016;42(12):1935-1947.
73. Mallat J, Siuba MT, Abou-Arab O, et al. Changes in pulse pressure variation induced by passive leg raising test to predict preload responsiveness in mechanically ventilated patients with low tidal volume in ICU: a systematic review and meta-analysis. Crit Care. 2025;29(1):18.
74. Wang X, Liu S, Gao J, et al. Does tidal volume challenge improve the feasibility of pulse pressure variation in patients mechanically ventilated at low tidal volumes? A systematic review and meta-analysis. Crit Care. 2023;27(1):45.
75. Mulder MP, Potters JW, van Loon LM, et al. Context-specific clinical applicability of the end-expiratory occlusion test to predict fluid responsiveness in mechanically ventilated patients: a systematic review and meta-analysis. Eur J Anaesthesiol. 2025;42(8):737-746.

76. Wang J, Zhou D, Gao Y, et al. Effect of VTILVOT variation rate on the assessment of fluid responsiveness in septic shock patients. Medicine (Baltimore). 2020;99(47):e22702.
77. Wenwen Y, Ping X, Yue D, et al. Accuracy of indices of inferior vena cava in predicting fluid responsiveness in patients with shock: a systematic review and meta-analysis. Intensive Crit Care Nurs. 2025;89:104015.
78. Cardozo Júnior LCM, Lemos GSD, Besen B. Fluid responsiveness assessment using inferior vena cava collapsibility among spontaneously breathing patients: systematic review and meta-analysis. Med Intensiva. 2023;47(2):90-98.
79. Si X, Xu H, Liu Z, et al. Does respiratory variation in inferior vena cava diameter predict fluid responsiveness in mechanically ventilated patients? A systematic review and meta-analysis. Anesth Analg. 2018;127(5):1157-1164.
80. Klompmaker P, Mousa A, Allard DJ, et al. Ultrasound assessment of fluid tolerance in critically ill patients: epidemiology of systemic and pulmonary congestion. Ultrasound Med Biol. 2026;52(3):596-601.
81. Jung B, Jabaudon M, De Jong A, et al. Sodium bicarbonate for severe metabolic acidemia and acute kidney injury: the BICARICU-2 randomized clinical trial. JAMA. 2025;334(22):2000-2010.
82. Malavige GN, Ogg GS. Molecular mechanisms in the pathogenesis of dengue infections. Trends Mol Med. 2024;30(5):484-498.
83. Rajapakse S, de Silva NL, Weeratunga P, et al. Prophylactic and therapeutic interventions for bleeding in dengue: a systematic review. Trans R Soc Trop Med Hyg. 2017;111(10):433-439.
84. Baksaas-Aasen K, Gall LS, Stensballe J, et al. Viscoelastic haemostatic assay augmented protocols for major trauma haemorrhage (ITACTIC): a randomized, controlled trial. Intensive Care Med. 2021;47(1):49-59.
85. Wikkelsø A, Wetterslev J, Møller AM, et al. Thromboelastography (TEG) or thromboelastometry (ROTEM) to monitor haemostatic treatment versus usual care in adults or children with bleeding. Cochrane Database Syst Rev. 2016;(8):CD007871.
86. Kumar R, Ng LXL, Wong YJ, et al. Rotational thromboelastometry reduces the need for preemptive transfusion in cirrhosis: a randomized controlled trial (NCT:05698134). J Clin Exp Hepatol. 2025;15(1):102409.
87. Ministry of Health Malaysia. Consensus statement on patient blood management. Putrajaya: Ministry of Health Malaysia; 2024.
88. Ministry of Health Sri Lanka. Guidelines on management of dengue fever & dengue haemorrhagic fever in children and adolescents. Colombo: Ministry of Health Sri Lanka; 2023.
89. National Centre for Infectious Diseases (NCID). Singapore national dengue clinical guideline. Singapore: NCID; 2026.
90. Chakraborty S, Alam S, Sayem M, et al. Investigation of the efficacy and safety of eltrombopag to correct thrombocytopenia in moderate to severe dengue patients - a phase II randomized controlled clinical trial. EClinicalMedicine. 2020;29-30:100624.
91. Saddique J, Ghaffar M, Shabbir MR, et al. Platelet transfusion and tranexamic acid in the treatment of bleeding in dengue fever. J Pharm Res Int. 2022;34(39B):58-65.
92. Sam JE, Gee TS, Nasser AW. Deadly intracranial bleed in patients with dengue fever: a series of nine patients and review of literature. J Neurosci Rural Pract. 2016;7(3):423-434.
93. Meena S, Dulara S, Gautam S, et al. Peri-operative considerations in intracranial hemorrhage in dengue virus infected patients: a prospective mini case series. Nepal J Neurosci. 2016;13:43-46.

94. Leowattana W, Leowattana T. Dengue hemorrhagic fever and the liver. World J Hepatol. 2021;13(12):1968-1976.
95. Wang XJ, Wei HX, Jiang SC, et al. Evaluation of aminotransferase abnormality in dengue patients: a meta-analysis. Acta Trop. 2016;156:130-136.
96. Campana V, Inizan C, Pommier JD, et al. Liver involvement in dengue: a systematic review. Rev Med Virol. 2024;34(4):e2564.
97. Teerasarntipan T, Chaiteerakij R, Komolmit P, et al. Acute liver failure and death predictors in patients with dengue-induced severe hepatitis. World J Gastroenterol. 2020;26(33):4983-4995.
98. Chia PY, Thein TL, Ong SWX, et al. Severe dengue and liver involvement: an overview and review of the literature. Expert Rev Anti Infect Ther. 2020;18(3):181-189.
99. Wendon J, Cordoba J, Dhawan A, et al. EASL clinical practical guidelines on the management of acute (fulminant) liver failure. J Hepatol. 2017;66(5):1047-1081.
100. Teerasarntipan T, Thanapirom K, Chaiteerakij R, et al. Validation of prognostic scores for predicting acute liver failure and in-hospital death in patients with dengue-induced severe hepatitis. World J Gastroenterol. 2024;30(45):4781-4790.
101. Dissanayake D, Gunaratne W, Kumarihamy K, et al. Use of intravenous N-acetylcysteine in acute severe hepatitis due to severe dengue infection: a case series. BMC Infect Dis. 2021;21(1):978.
102. Giri S, Anirvan P, Vaidya A, et al. Dengue-related acute liver failure-a scoping review. Indian J Gastroenterol. 2024;43(2):407-424.
103. Rahim A, Hameed A, Ishaq U, et al. Cardiovascular sequelae of dengue fever: a systematic review. Expert Rev Cardiovasc Ther. 2022;20(6):465-479.
104. Robinson J, Hartling L, Vandermeer B, et al. Intravenous immunoglobulin for presumed viral myocarditis in children and adults. Cochrane Database Syst Rev. 2020;8(8):CD004370.
105. See KC. Dengue-associated hemophagocytic lymphohistiocytosis: a narrative review of its identification and treatment. Pathogens. 2024;13(4):287.
106. Bhushan B, Sardana V, Maheshwari D, et al. Immune-mediated neurological manifestations of dengue virus- a study of clinico-investigational variability, predictors of neuraxial involvement, and outcome with the role of immunomodulation. Neurol India. 2018;66(6):1634-1643.
107. Fong SL, Wong KT, Tan CT. Dengue virus infection and neurological manifestations: an update. Brain. 2024;147(3):830-838.
108. Agarwal N, Yadav A, Yadav R. Acute respiratory distress syndrome associated with dengue fever in children - a case series. Int J Pharm Clin Res. 2024;16(1):434-437.
109. Seow D, Khor YH, Khung SW, et al. High-flow nasal oxygen therapy compared with conventional oxygen therapy in hospitalised patients with respiratory illness: a systematic review and meta-analysis. BMJ Open Respir Res. 2024;11(1):e002100.
110. Roca O, Caralt B, Messika J, et al. An index combining respiratory rate and oxygenation to predict outcome of nasal high-flow therapy. Am J Respir Crit Care Med. 2019;199(11):1368-1376.
111. Duan J, Han X, Bai L, et al. Assessment of heart rate, acidosis, consciousness, oxygenation, and respiratory rate to predict noninvasive ventilation failure in hypoxemic patients. Intensive Care Med. 2017;43(2):192-199.
112. Grasselli G, Calfee CS, Camporota L, et al. ESICM guidelines on acute respiratory distress syndrome: definition, phenotyping and respiratory support strategies. Intensive Care Med. 2023;49(7):727-759.

113. Qadir N, Sahetya S, Munshi L, et al. An update on management of adult patients with acute respiratory distress syndrome: an official American Thoracic Society clinical practice guideline. Am J Respir Crit Care Med. 2024;209(1):24-36.
114. Jia L, Wang P, Li C, et al. The efficacy and safety of vasopressors for septic shock patients: a systemic review and network meta-analysis. Shock. 2023;60(6):746-752.
115. Kalimouttou A, Kennedy JN, Feng J, et al. Optimal vasopressin initiation in septic shock: the OVISS reinforcement learning study. JAMA. 2025;333(19):1688-1698.
116. Mathew R, Di Santo P, Jung RG, et al. Milrinone as compared with dobutamine in the treatment of cardiogenic shock. N Engl J Med. 2021;385(6):516-525.
117. Bignardi PR, Pinto GR, Boscarioli MLN, et al. Acute kidney injury associated with dengue virus infection: a review. J Bras Nefrol. 2022;44(2):232-237.
118. Oliveira JF, Burdmann EA. Dengue-associated acute kidney injury. Clin Kidney J. 2015;8(6):681-685.
119. Li JH, Cai JH, Wang MJ, et al. Early strategy vs. late initiation of renal replacement therapy in adult patients with acute kidney injury: an updated systematic review and meta-analysis of randomized controlled trials. Eur Rev Med Pharmacol Sci. 2023;27(13):6046-6057.
120. Deshwal R, Sharma A. Serum calcium levels as a marker of dengue severity: a clinical observational study. J Assoc Physicians India. 2025;73(12):48-50.
121. Yanda S, Karimulla H. Study of serum electrolytes and to correlate with severity in patients with dengue. Eur J Cardiovasc Med. 2022;12(1):107-111.
122. MR Rajalekshmy, M Vadivelan. Electrolyte abnormalities in patients with dengue infection admitted to a tertiary care teaching hospital in Southern India. JIACM. 2019;20(1):47-51.
123. Fatima N, Ashfaq M, Hussain W, et al. Electrolyte imbalance and its association with the severity of dengue fever. Prof Med J. 2025;32(10):1272-1279.
124. Zhang Z, Xu X, Ni H, et al. Predictive value of ionized calcium in critically ill patients: an analysis of a large clinical database MIMIC II. PLoS One. 2014;9(4):e95204.
125. Kander T, Frigyesi A, Kjeldsen-Kragh J, et al. Bleeding complications after central line insertions: relevance of pre-procedure coagulation tests and institutional transfusion policy. Acta Anaesthesiol Scand. 2013;57(5):573-579.
126. Coz Yataco A, Soghier I, Hébert PC, et al. Transfusion of fresh frozen plasma and platelets in critically ill adults: an American College of Chest Physicians clinical practice guideline. Chest. 2025;168(3):661-676.
127. Parienti JJ, Mongardon N, Mégarbane B, et al. Intravascular complications of central venous catheterization by insertion site. N Engl J Med. 2015;373(13):1220-1229.
128. Martin BM, Evans AA, de Carvalho DS, et al. Clinical outcomes of dengue virus infection in pregnant and non-pregnant women of reproductive age: a retrospective cohort study from 2016 to 2019 in Paraná, Brazil. BMC Infect Dis. 2022;22(1):5.
129. Tougma SA, Zoungrana/Yaméogo WN, Dahourou DL, et al. Dengue virus infection and pregnancy outcomes during the 2017 outbreak in Ouagadougou, Burkina Faso: a retrospective cohort study. PLoS One. 2020;15(9):e0238431.
130. Brar R, Sikka P, Suri V, et al. Maternal and fetal outcomes of dengue fever in pregnancy: a large prospective and descriptive observational study. Arch Gynecol Obstet. 2021;304(1):91-100.
131. Ministry of Health Sri Lanka. Clinical management of dengue infection in pregnancy. Colombo: Ministry of Health Sri Lanka; 2019.

132. Centers for Disease Control and Prevention. Dengue clinical management pocket guide. Atlanta: Centers for Disease Control and Prevention; 2025.
133. Iqtadar S. Dengue guidelines in pregnancy. J Pak Soc Intern Med. 2024;5(1):452-455.
134. Ministry of Health Sri Lanka. National guidelines - guidelines on management of dengue fever & dengue haemorrhagic fever in adults. Colombo: Ministry of Health Sri Lanka; 2012.
135. Vidanapathirana M. Dengue haemorrhagic fever in chronic kidney disease and heart failure: challenges in fluid management. Trop Med Health. 2024;52(1):33.
136. Lee IK, Lee NY, Huang WC, et al. In-hospital mortality predictors among hospitalized adults and those with chronic kidney disease with dengue. J Microbiol Immunol Infect. 2023;56(5):996-1006.
137. Chen HJ, Tang HJ, Lu CL, et al. Warning signs and severe dengue in end stage renal disease dialysis patients. J Microbiol Immunol Infect. 2020;53(6):979-985.
138. World Health Organization. Handbook for clinical management of dengue. Geneva: World Health Organization; 2012.
139. Yeung W, Lye DCB, Thein TL, et al. Blood pressure trend in hospitalized adult dengue patients. PLoS One. 2020;15(7):e0235166.
140. Jeng MJ, Lee NY, Lee IK, et al. Prognosis and mortality risk in elderly patients with dengue virus infection: excess fatality and the urgent need for revising current WHO criteria for elderly patients. Travel Med Infect Dis. 2025;65:102855.
141. Low JG, Ong A, Tan LK, et al. The early clinical features of dengue in adults: challenges for early clinical diagnosis. PLoS Negl Trop Dis. 2011;5(5):e1191.
142. Chiu YY, Lin CY, Yu LS, et al. The association of obesity and dengue severity in hospitalized adult patients. J Microbiol Immunol Infect. 2023;56(2):267-273.
143. Taylor SP, Karvetski CH, Templin MA, et al. Initial fluid resuscitation following adjusted body weight dosing is associated with improved mortality in obese patients with suspected septic shock. J Crit Care. 2018;43:7-12.
144. Natesirinilkul R, Tantiworawit A, Charoenkwan P. Clinical course of dengue in patients with thalassaemia. Paediatr Int Child Health. 2013;33(1):32-36.
145. Pongtanakul B, Narkbunnam N, Veerakul G, et al. Dengue hemorrhagic fever in patients with thalassemia. J Med Assoc Thai. 2005;88 Suppl 8:S80-S85.
146. Charan J, Saxena D, Goyal JP, et al. Efficacy and safety of Carica papaya leaf extract in the dengue: a systematic review and meta-analysis. Int J Appl Basic Med Res. 2016;6(4):249-254.
147. Rajapakse S, de Silva NL, Weeratunga P, et al. Carica papaya extract in dengue: a systematic review and meta-analysis. BMC Complement Altern Med. 2019;19(1):265.
148. Bowman LR, Donegan S, McCall PJ. Is dengue vector control deficient in effectiveness or evidence?: a systematic review and meta-analysis. PLoS Negl Trop Dis. 2016;10(3):e0004551.
149. Fox T, Sguassero Y, Chaplin M, et al. Wolbachia-carrying Aedes mosquitoes for preventing dengue infection. Cochrane Database Syst Rev. 2024;4(4):CD015636.
150. Wilder-Smith A, Ooi EE, Horstick O, et al. Dengue. Lancet. 2019;393(10169):350-363.
151. Hadinegoro SR, Arredondo-García JL, Capeding MR, et al. Efficacy and long-term safety of a dengue vaccine in regions of endemic disease. N Engl J Med. 2015;373(13):1195-1206.
152. World Health Organization. Dengue vaccine: WHO position paper – September 2018. Wkly Epidemiol Rec. 2018;93(36):457-476.
153. Flacco ME, Bianconi A, Cioni G, et al. Immunogenicity, safety and efficacy of the dengue vaccine TAK-003: a meta-analysis. Vaccines (Basel). 2024;12(7):722.

154. LeFevre I, Bravo L, Folschweiller N, et al. Bridging the immunogenicity of a tetravalent dengue vaccine (TAK-003) from children and adolescents to adults. npj Vaccines. 2023;8(1):75.
155. Ranzani OT, Lazar Neto F, Mareto LK, et al. Effectiveness of the TAK-003 dengue vaccine in adolescents during the 2024 outbreak in São Paulo, Brazil: a test-negative, case-control study. Lancet Infect Dis. 2026;26(1):91-100.
156. Dengue Prevention Advocacy Malaysia. Position paper on dengue vaccination: a recommendation by Dengue Prevention Advocacy Malaysia. Kuala Lumpur: Malaysian Paediatric Association; 2025.
157. Kallas EG, Precioso AR, Palacios R, et al. Safety and immunogenicity of the tetravalent, live-attenuated dengue vaccine Butantan-DV in adults in Brazil: a two-step, double-blind, randomised placebo-controlled phase 2 trial. Lancet Infect Dis. 2020;20(7):839-850.
158. Nogueira ML, Cintra MAT, Moreira JA, et al. Efficacy and safety of Butantan-DV in participants aged 2-59 years through an extended follow-up: results from a double-blind, randomised, placebo-controlled, phase 3, multicentre trial in Brazil. Lancet Infect Dis. 2024;24(11):1234-1244.

Appendix 1

CLINICAL QUESTION

Diagnosis

1. What are the accuracy of clinical findings and laboratory tests to differentiate dengue from other acute febrile illnesses?
2. What is the accuracy of laboratory tests/investigations in the diagnosis of dengue?
   - Rapid Combo Test
   - Immunoassay test
   - Molecular test
3. What is the accuracy of laboratory tests/investigations in the diagnosis of dengue in dengue-vaccinated persons?
4. What are the significant predictors of severe dengue or hospitalisations?
   - Clinical signs
   - Laboratory markers
   - Imaging/ultrasound findings

Assessment/Prognosis

5. What are the effective and safe triaging system of patients with dengue in primary care/emergency department?
6. What is the accuracy of dengue severity scoring tools in dengue?
7. What is the effective and safe monitoring of dengue outpatients?
8. What is the effective and safe monitoring for hospitalised dengue patients?

Treatment

9. Are the following treatments effective and safe in patients with dengue at primary care?
   - Oral fluid therapy
   - Non-steroidal anti-inflammatory drugs (NSAIDs)
   - Antibiotics
   - Steroids
   - Concurrent medication
10. Are following treatments effective and safe in inpatients with dengue?
    a. Fluid
    b. Antibiotics
    c. Management of bleeding:
    d. Management of hepatitis
    e. Treatment of Expanded Dengue Syndrome
    f. Management of acute kidney injury (AKI)
    g. Intensive care management
11. What are the effective and safe treatment (management) of patients with dengue in the following special population?
    a. Pregnancy
    b. Heart Failure
    c. End-stage Renal Failure (ESRF)

d. Liver Cirrhosis
e. Uncontrolled hypertension
f. Diabetic ketoacidosis (DKA)/hyperglycaemic hyperosmolar syndrome (HHS)
g. Geriatric
h. Obesity

12. What are the effective and safe complementary and alternative therapies for dengue patients?

Prevention

13. How effective and safe is dengue vaccination in reducing dengue infection and hospitalisation?

Appendix 2

EXAMPLE OF SEARCH STRATEGY

Clinical Question: What are the effective and safe complementary and alternative therapy of dengue patients?

Database: Ovid MEDLINE(R) ALL <1946 to February 20, 2025>
Search Strategy:

1 DENGUE/ (16258)
2 Dengue.tw. (30597)
3 ((classical or fever) adj1 dengue*).tw. (6894)
4 (classical dengue adj2 fever*).tw. (57)
5 SEVERE DENGUE/ (1853)
6 (dengue h?emorrhagic adj2 fever).tw. (3021)
7 ((h?emorrhagic or severe) adj1 dengue*).tw. (4499)
8 dengue shock syndrome.tw. (1239)
9 1 or 2 or 3 or 4 or 5 or 6 or 7 or 8 (31528)
10 COMPLEMENTARY THERAPIES/ (18647)
11 ((alternative or complementary) adj1 (medicine or therap*)).tw. (43028)
12 DIETARY SUPPLEMENTS/ (81675)
13 dietary.tw. (338993)
14 ((dietary or food or herbal) adj1 supplement*).tw. (36681)
15 n?utraceutical*.tw. (14177)
16 nutriceutical*.tw. (179)
17 10 or 11 or 12 or 13 or 14 or 15 or 16 (456457)
18 9 and 17 (62)
19 limit 18 to (english language and humans and yr=”2015 -Current” and “all adult (19 plus years)”) (3)

Appendix 3

SAMPLE COLLECTION FOR DENGUE TESTS

Test	Sample Type	Volume	Container	Storage and Transport Instructions	Referral laboratory	Turnaround time
Rapid Diagnostic Tests for Dengue (as POCT)	Blood	1 ml	Plain tube or EDTA (refer to manufacturer insert)	As per manufacturer insert	NA	NA
Rapid Diagnostic Tests for Dengue	Blood	1 ml	Plain tube or EDTA (refer to manufacturer insert)	As per laboratory handbook	As per laboratory handbook	As per laboratory handbook
Dengue Serology IgM/IgG (ELISA)	Blood	3–5 ml	Plain tube without anticoagulants.	Allow to clot at ambient temperature.	As per laboratory handbook	As per laboratory handbook
Dengue RT-PCR (for surveillance)	Serum	2-4 ml	Plain tube with serum separator	After sample is collected, immediately send the sample to the laboratory. If there is any delay, keep the sample at 2-8°C up to 48 hours. If exceeding 48 hours, the sample should be stored at -70°C.	MKAK (use MKAK-BPU-U01/Rev2018 form)	14 days
Dengue RT-PCR (for diagnosis)	Serum/plasma	1-3 ml	Plain tube/ EDTA	With ice	IMR (use Sample management information system- SMIS)	1-5 days
	Cerebrospinal fluid (CSF)	1- 3 ml	Sterile leakproof container	Keep sample in a freezer at -70°C; transport on ice.		
	Tissue (post-mortem)	1.5 cm cubes from various parts of affected organs	Sterile container containing Viral Transport Medium (VTM) or normal saline (NS) to keep tissue moist	Keep at 4°C or in a cold box with ice.		

Source:

1. National Public Health Laboratory (NPHL). The National Public Health Laboratory Test Handbook. NPHL; Sg Buloh: 2022
2. The Specimen Management Information System (SMIS). http://smis.imr.gov.my/html/smpl/login.php

Appendix 4

DENGUE CLERKING SHEET IN PRIMARY CARE

PRIMARY HEALTH CARE:_____

Name:_____ I/C:_____
Age:_____ Latest Phone No.:_____
Occupation:_____ Allergy History:_____
LMP (for females only):_____

Date:_____

Chief complaint:

Fever onset (date):_____

History of presenting illness:

Weight/ height /BMI:_____

Medication: □NSAID □Steroid □Anticoagulant □Aspirin

Co-morbidity:□Previous dengue illness □ Obesity (BMI >27kg/m^2) □Smoking □ Pregnancy

□ Other comorbid:_____

SUSPECTED DENGUE

- Anyone with a recent history of fever in your house/workplace: □Yes □ No
- Are you living or travelling to a dengue endemic area/recent fogging? □ Yes □ No
- Any travelling history? □Yes □ No
- **Fever, and 2 of the following criteria:**
  - □ Aches and pains
  - □ Rash
  - □ Intense abdominal pain
  - □ Persistent vomiting (>three times per day)
  - □ Lethargy, restlessness, irritability or confusion
  - □ Bleeding manifestation (mucosa, menses, PR bleed)

***Proceed with lab investigation if there is a fever with two other signs or symptoms in SUSPECTED DENGUE.**

<table>
<tr><td rowspan="4">FBC</td><td>WBC:</td><td rowspan="4">Dengue Combo</td><td>NS1:</td></tr>
<tr><td>Hb:</td><td>IgM:</td></tr>
<tr><td>Platelet:</td><td rowspan="2">IgG:</td></tr>
<tr><td>HCT:</td></tr>
<tr><td>OTHER TEST</td><td colspan="3">BFMP, RBS, Measles, etc.):</td></tr>
</table>

FOLLOW-UP VISIT					
HISTORY	1st Visit	2nd Visit	3rd Visit	4th Visit	5th Visit
Date/ time:					
• How many days of fever/ illness?					
• Any concurrent: (Yes/ No) If yes, how many days?					
• Aches and pains (pain score)					
• Rash					
Daily Function					
• Oral fluid intake: Poor/ Adequate					
• Urination:___ times per day. (normal/ reduced)					
• Can you perform daily activities? (Yes/ No)					
• Any lethargy? (Yes/No)					
Warning Symptoms (Yes/No)					
• Worsening clinical symptoms when fever subsides					
• Intense abdominal pain? If Yes, duration & pain score					
• Persistent vomiting. If Yes, episodes per day					
• Lethargy, restlessness, irritability or confusion.					
• 3rd space fluid accumulation					
• Postural hypotension.					
• Liver enlargement >2 cm.					
• Reduced or absent urine output for 4-6 hours					
• Bleeding manifestation (mucosa, menses, PR bleed) • If Yes, site and duration?					
• Lab: Raised HCT (≥20% from baseline) with rapid drop in platelets. (Normal: Male:. 60y.o: HCT ≤ 46, Male> 60y.o: HCT < 42, Female: HCT < 40)					
General Condition & Vital Signs					
• Conscious Level: Alert/ Confuse/ Unconscious					
• Hydration: Good/ Dehydrated					
• Temperature (°C)					
• Blood Pressure (mmHg)					
• Pulse pressure {SBP-DBP) (normal/ narrow)					
• Pulse Rate (beats/min) - (normal/ tachycardia/ bradycardia)					
• SPO_2:					
• Blood glucose:					
CCTVR					
• Colour: Pink/ Pallor					
• Capillary Refill Time (CRT) *abnormal >3 second					
• Temperature of Extremities: Warm/ Cold					
• Pulse Volume: Good/ Poor					
• Pulse Rate: Normal/Tachycardia/Bradycadia					
SystemicExamination					
• CNS:					
• CVS:					
• Lung:					
• Abdomen:					

Investigations:					
• WBC					
• Hb					
• Platelet					
• HCT					
• AST					
Sign Of Shock (based on General Condition, Vital Signs & CCTVR)					
• Not in shock/ Compensated/ Decompensated shock					
Sign Of Severe Dengue					
• Severe plasma leakage [Shock (DSS) / Fluid accumulation with respiratory distress]					
• Severe bleeding manifestation					
• Severe organ impairment (CNS impaired consciousness / Impaired cardiac function/ AST or ALT> 1000)					
• Expanded dengue syndrome					
Diagnosis					
• Dengue Day/ Other Diagnosis:					
• Dengue Phase: (Febrile/ Critical/ Recovery)					
• Warning Signs:(Yes / No)					
• In Shock: No/ Compensated/ Decompensated Shock)					
• Severe Dengue: (Yes/ No)					
• Comorbidity / Special Population: (Yes/ No)					
Management					
• Notification: (Yes/ No) Date:					
• Dengue monitoring card given: (Yes/ No) Date:					
If sent home: • Advice to drink 200 ml per hour when awake. • Avoid taking any NSAID/ steroids/ intramuscular injection • TCA stat if warning sign develops • Symptomatic medications like PCM, cough syrups, etc, with doses and durations. • Other management:					
• Date to repeat FBC					
Referral (if indicated): _____ • Refer ED Hospital. Indication: _____ • Start on IV Drip Normal Saline_____ml /kg/ hour = _____ ml/ hour • Time started drip: _____ • Total fluid given: iv: _____ oral: _____ ml • Name of the receiving doctor: _____ • The dengue monitoring record is attached. (Yes/ No)					
• Name of doctor, signature and official stamp					

Appendix 5

A. DENGUE MONITORING RECORD

Patient's Name:
Address:

I/C No. / Passport:
Date of onset of fever:
Date of Dengue Rapid Combo Test:
Result of Dengue Rapid Combo Test:
Date of AST:
Result of AST:

Date	Day of fever	Temp (°C)	BP (mmHg)	Pulse Pressure (mmHg)	PR (min)	Hb (g/dL)	Hct (%)	WCC ($x10^3$/µl)	Platelet ($x10^3$/µl)	Freq of Urine Output (min 4-6 times/d)	Attending Doctor	Next Appointment

B. HOME CARE ADVICE

HOME CARE ADVICE IN DENGUE

What to do

- Drink adequate fluids. Aim 200 ml of fluid every hour (one small cup) while awake. Examples of fluids include fruit juice, oral rehydration salts (ORS), coconut water, or soup.
- Take paracetamol for fever or pain when necessary, but not to exceed 4g per day.
- Get adequate rest.
- Monitor the temperature and dengue warning signs.
- Seek immediate medical care if warning signs develop.
- Attend daily follow-up appointments as advised.
- Look for mosquito breeding places in and around the home and prevent mosquito bites using mosquito repellent.

What to avoid

- DO NOT take steroids, aspirin, ibuprofen, diclofenac, or any other NSAIDs. If you are already taking these medications, please consult your doctors.
- DO NOT consume traditional or herbal remedies.
- DO NOT take intramuscular injections.
- DO NOTdo massage, cupping or gua sha.

NASIHAT PEMANTAUAN DEMAM DENGGI DI RUMAH

Apa yang perlu dilakukan

- Minum air secukupnya sebanyak 200 ml setiap jam (satu cawan kecil) sewaktu jaga. Contoh minuman termasuk jus buah, larutan garam rehidrasi (ORS), air kelapa, atau sup.
- Ambil *paracetamol* untuk demam atau sakit badan jika perlu, tetapi tidak melebihi 4g sehari.
- Dapatkan rehat yang mencukupi.
- Pantau suhu badan dan tanda amaran demam denggi.
- Dapatkan rawatan segera jika tanda amaran muncul.
- Hadiri temujanji susulan.
- Periksa tempat pembiakan nyamuk di rumah serta cegah gigitan nyamuk menggunakan semburan penghalau nyamuk.

Apa yang perlu dielakkan

- JANGAN ambil steroids, aspirin, ibuprofen, diclofenac, atau ubat anti-radang bukan steroid (NSAIDs) lain. Sila rujuk doktor jika anda sedang mengambil ubat-ubatan ini.
- JANGAN ambil ubat-ubat tradisional atau herba.
- JANGAN ambil suntikan intramuskular.
- JANGAN berurut, bekam atau gua sha.

DENGUE WARNING SIGNS

As the **fever starts to subside**, be cautious as the condition may become dangerous and worsen in some dengue patients. **Go to the hospital immediately if any of the following occur**:

- General condition worsening or symptoms becoming more severe
- Severe abdominal pain
- Persistent vomiting
- Extreme fatigue
- Difficulty breathing or rapid breathing
- Fainting spells
- Bleeding from the gums or nose, or blood in vomit, urine, or stool
- Reduced urine output less than 4 times per day

TANDA-TANDA AMARAN DEMAM DENGGI

Apabila **demam mula kebah**, anda perlu berhati-hati kerana keadaan boleh menjadi berbahaya dan semakin teruk bagi sesetengah pesakit demam denggi. **Segera ke hospital jika anda mempunyai salah satu tanda-tanda berikut**:

- Keadaan badan semakin lemah atau simptom semakin teruk
- Sakit perut yang teruk
- Muntah berterusan
- Keletihan melampau
- Susah bernafas atau pernafasan laju
- Pitam
- Pendarahan pada gusi atau hidung, atau darah dalam muntah, air kencing atau najis
- Kekerapan kencing kurang dari 4 kali sehari

Appendix 6

CAPILLARY REFILL TIME ASSESSMENT

CRT standardisation protocol

To ensure reliability and reduce variability in CRT measurement, standardised procedures are essential for all patients.

Procedure for measuring CRT:

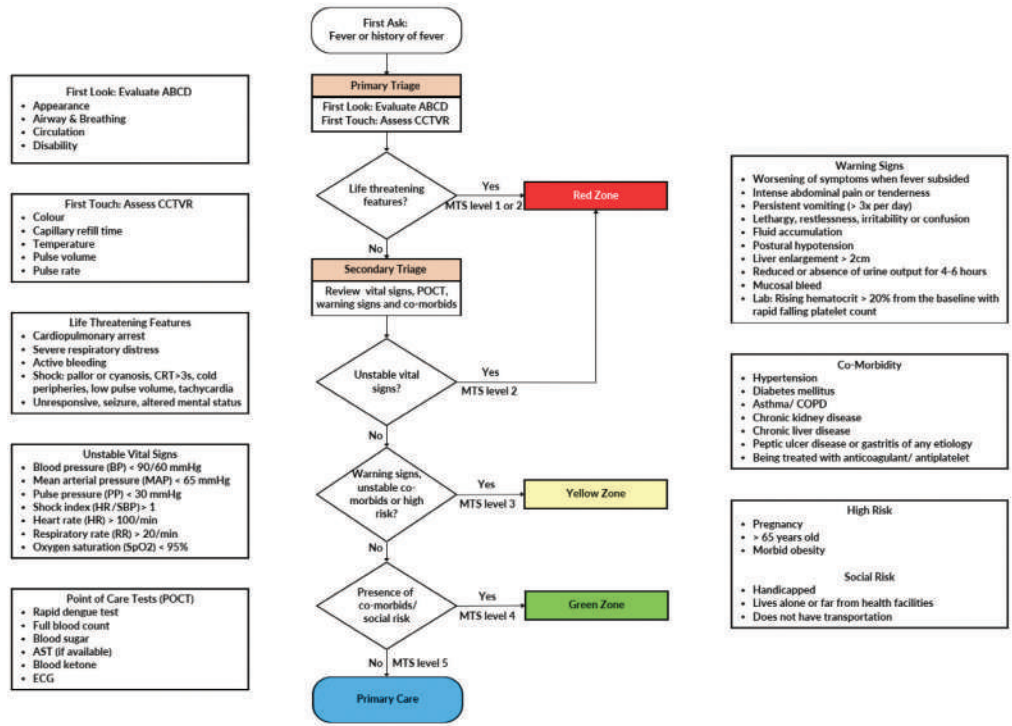
1. **Preparation**:
   - Ensure the room is warm.
   - Select a measurement site free from skin abnormalities.
   - Position the patient's hand at the level of the thorax/heart.
2. **Application of Pressure**:
   - Apply firm pressure to the ventral surface of the distal phalanx of the index finger for 10 seconds.
3. **Measurement**:
   - Remove the pressure.
   - Measure the time taken for the skin colour to return to its pre-existing state.
4. **Interpretation:**
   - A CRT exceeding 3 seconds is considered abnormal (WHO 2025).



Notes: 1 – Place the patient's hand at the level of their heart (mid chest). 2 – Apply firm pressure to the ventral surface of the distal phalanx of one finger for 10 seconds. This can be done using your thumb, or if available a glass microscope slide so that blanching of the skin underneath can be confirmed. 3 – Release the pressure and measure the time taken for the skin colour to return to the same as before.

Source: World Health Organization. WHO guidelines for clinical management of arboviral diseases: dengue, chikungunya, Zika and yellow fever Geneva: WHO; 2025.

SYSTEMATIC TRIAGE OF DENGUE IN ADULTS

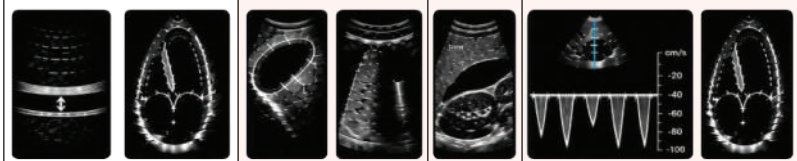


POCUS GUIDED DENGUE MANAGEMENT

POCUS GUIDED DENGUE MANAGEMENT

A PHASE GUIDED APPROACH

1 PHASE	**FEBRILE PHASE** (Day 2 - 7)	**CRITICAL PHASE** (Day 3 - 7)	**RECOVERY PHASE** (Day 7 onwards)
2 POTENTIAL COMPLICATIONS	DEHYDRATION	LEAKING | BLEEDING | SHOCK | ORGAN DYSFUNCTION	REABSORPTION | FLUID OVERLOAD
3 POCUS FINDINGS	SMALL / FLAT IVC (<1.5 cm); HYPERDYNAMIC HEART WITH SMALL LV, RV	GBWT >3 mm; PLEURAL EFFUSION; ASCITES; LOW LVOT VTI; REDUCED CARDIAC FUNCTION	DILATED IVC (POOR VARIABILITY); B-LINES (FLUID OVERLOAD)
4 POCUS GUIDED MANAGEMENT	• Assess volume • Guide fluids • Screen for early leakage	• Detect leakage • Assess hemodynamics • Differentiate type of shock • Guide fluids / blood / vasoactives • Serial reassessment	• Confirm resolution • Detect fluid overload • De-escalate fluids



IMPORTANT: SIGNS CAN COEXIST AND OVERLAP ACROSS DIFFERENT PHASES, CLINICAL JUDGEMENT AND SERIAL POCUS ARE ESSENTIAL.

POCUS USE:
- ✓ When clinically indicated
- ✓ By trained operators
- ✓ Where resources are available

IMPORTANT:
POCUS COMPLEMENTS CLINICAL ASSESSMENT. ALWAYS INTEGRATE WITH CLINICAL JUDGMENT AND PATIENT CONTEXT

POCUS CAN HELP TO:
- ✓ Detect hypovolemia early
- ✓ Identify plasma leakage and complications
- ✓ Guide fluid therapy and monitor response
- ✓ Support safe de-escalation during recovery

Note: PoCUS - Point-of-care Ultrasound, IVC - Inferior Vena Cava, LVOT VTI - Left Ventricular Outflow Tract Velocity Time Integral, GBWT - Gallbladder Wall Thickening, B-lines - Lung comet-tail artifacts.

REFERENCES:
1. World Health Organization. Dengue Guidlines For Diagnosis, Treatment, Prevention And Control. Geneva: WHO; 2009.
2. World Health Organization. WHO guidlines for clinical management of arboviral diseases: dengue, chikungunya and yellow fever. Geneva: WHO; 2025.

Appendix 9

INPATIENT DENGUE ASSESSMENT AND CLERKING SHEET

Date of clerking :		**Time :**	
Patient demographic data		**Risk factors /Special populations**	
Name		☐ Obesity	☐ Diabetes mellitus
Age		☐ Elderly (>65)	☐ HPT
Gender	☐ Male ☐ Female	☐ Pregnancy	☐ Chronic liver disease
IC		☐ Chronic kidney disease	☐ Chronic heart failure
RN		☐ Hemolytic diseases	☐ Asthma/COPD
Admission via	☐ ETD ☐ OPD ☐ Others.....	☐ On anticoagulation/ antiplatelet/IM injection	☐ Peptic ulcer disease
Previous dengue vaccination	☐ Yes ☐ No Date of vaccination:		☐ Previous dengue infection ☐ Others.....

Current medications	**Dengue diagnostic test**		**Date**	
Medication list and date of last taken medication	NS1	IgM	IgG	
	Baseline FBC (if known)		Date :	
	Hb	WCC	HCT	PLT
	FBC on admission			
	Hb	WCC	HCT	PLT
	Renal profile			
	Na	K	Urea	Creat
	Liver function tests			
	AST	ALT	Alb	T bil
	Others			
	VBG	pH	HCO3	Lactate
	CK	Trop	INR	APTT

History of presenting illness	
Date of fever onset	
Current day of illness	
Date and time of defervescence	

Warning signs (For patients in critical phase)

- ☐ Lack of clinical improvement or worsening symptoms when fever subsides
- ☐ Intense abdominal pain or tenderness
- ☐ Lethargy/restlessness/irritability/confusion
- ☐ Vomiting (≥3 times per day/unable to tolerate fluid)
- ☐ Postural hypotension
- ☐ Liver enlargement >2cm
- ☐ Reduced or absent urine output for 4-6 hours
- ☐ Bleeding manifestations (mucosa, menses, PR)
- ☐ Raised HCT with rapid drop in platelets
  normal HCT values:
  Male ≤60 years-46
  Male > 60 years-42
  Female(non-pregnant)-40
- ☐ Third-space fluid accumulation (pleural effusion, ascites, gallbladder wall thickening on POCUS)

Illness severity assessment

a) Level of daily activities during illness
- ☐ Normal as usual
- ☐ Reduced (not attending school/work but ambulant)
- ☐ Bedbound/ needs assistance

b) Fluid intake (past 24-48 hours)
- ☐ Normal
- ☐ Reduced
- ☐ Poor/ unable to tolerate orally

c) Frequency of urination (past 24 hours)
- ☐ ≥4 times
- ☐ <4 times

d) Ongoing persistent diarrhoea
- ☐ Yes
- ☐ No

e) Abnormal laboratory findings:

Clinical examination				
Weight:	Height:	BMI (kg/m²)	**Vital signs**	
Adjusted body weight (ADW): (if BMI≥27.5) (If pregnancy to use pre-pregnancy / booking weight)			BP:	PR:
			Pulse Pressure (SBP - DBP):	Baseline BP (if any):
Mental status	☐ alert	☐ lethargic	RR:	SpO_2:
Tongue	☐ moist	☐ dry	Temp:	Glucose:
Bleeding (if any)	☐ petechiae	☐ bruises	Last menstrual period (LMP) (For female):	

Peripheral perfusion		
C- Colour	☐ Pink	☐ Pallor
C- Capillary refill time (CRT)	☐ ≤3 sec	☐ >3 sec
T- Temperature	☐ Warm	☐ Cold peripheries
V- Volume (pulse)	☐ Good	☐ Low
R- Rate (PR)	☐ Normal	☐ Tachycardia or Bradycardia

CLINICAL REMARKS

1. **Signs of reduced perfusion: tachycardia, cold peripheries, CRT> 3sec, PP 20-30 mmHg, low pulse volume**
2. **Compensated shock: systolic pressure maintained but has signs of reduced perfusion**
3. **Decompensated shock: hypotension with signs of reduced perfusion**

Targeted bedside examination	
Respiratory system (pleural effusion)	Cardiovascular system
Abdominal system (tenderness/hepatosplenomegaly/ascites)	Other findings including POCUS if relevant

Diagnosis				
Days of illness		Severe dengue	☐ Yes	☐ No
Phase: ☐ Febrile ☐ Critical ☐ Recovery Classification of dengue ☐ DNWS ☐ DWWS (only in critical phase) ☐ SD		☐ Severe plasma leakage ☐ Severe bleeding ☐ Severe organ impairment o Hepatitis o Encephalitis o Myocarditis o Others		
		In shock	☐ Yes	☐ No
		(if yes)	☐ Compensated	☐ Decompensated

Cumulative I/O since presentation to healthcare	
Total intake (oral plus IV fluid/blood products)	
Total output (urine plus GI losses)	
Balance	

Dengue notification done			
☐ Yes	☐ No	Date and time	

Management

1) Fluid management (type, rate, amount)

2) Blood investigations

3) Time for next blood taking, bedside review and revision of drip regime

4) Monitoring frequency:

   a) Vital signs: _____hrly

   b) Urine output:_____hrly

   c) Glucose: _____hrly (if applicable)

Notify doctor if

- SBP <_____ mm Hg or DBP <_____mmHg
- Heart Rate <_____ or >_____ per/min
- Respiratory rate >25 breaths per/min
- Pulse Pressure <30 mmHg
- SpO_2 <95%
- Urine output < 0.5ml/kg/hour
- New complaint / warning signs

5) Review old medication and need to continue/discontinue

6) Other plans including pass over, escalation of care:

Clerking completed by

Signature and stamp	Date and time :

DENGUE MONITORING CHART

| DENGUE MONITORING CHART | | |

Appendix 11

FLUID QUOTA FOR DENGUE IN CRITICAL PHASE WITHOUT SHOCK

Fluid Quota
For the entire critical phase, the recommended total fluid volume is **Maintenance + 5% deficit, where a 5% deficit is equivalent to 50ml/kg**. This fluid quota includes both oral and intravenous fluid intake.

Important Considerations:

- Not all patients will require the full fluid quota Maintenance + 5% deficit; many may need less as the rate, peak, and duration of plasma leakage vary between individuals.
- **5% deficit represents the expected plasma volume loss during critical phase**
- **Crystalloid is the intravenous fluid of choice for maintenance therapy**.

The fluid quota calculation:
The total fluid allowance (combining oral and IV routes) should be calculated as the maintenance requirement plus 5% deficit.

This entire amount is to be administered over 48 hours.

Example for a 60kg person	
Calculation	**Total fluid**
Maintenance fluid for 1 day (A) (40 ml + 20 ml + 40 ml=100 ml/hr)	2400ml
5% deficit (B) (50 ml/kg)=50 ml x 60 kg	3000ml
Total fluid to be given over 48 hrs (A + B)	5400ml

Sources:
1. Bhalla A, Singh H, Suri V, et al. ISCCM Position Statement: Management of Severe Dengue in Intensive Care Unit. Indian journal of critical care medicine : peer-reviewed, official publication of Indian Society of Critical Care Medicine. 2024;28(Suppl 2):S42-s58.
2. National Center for Vector Borne Diseases Control India. National Guidelines for Clinical Management of Dengue Fever 2023. New Delhi: National Center for Vector Borne Diseases Control; 2023.
3. National Malaria Elimination and Aedes Transmitted Disease Control Program DGHS, Bangladesh. Pocket Guideline for Dengue Clinical Case Management 2022 (Revised). Dhaka: DGHS Bangladesh; 2022.

Appendix 12

PASSIVE LEG RAISE MANOEUVRES

The PLR test is used to predict if an increase in circulating volume will lead to a rise in cardiac output and, consequently, improved end-organ perfusion (i.e., *preload responsiveness*).

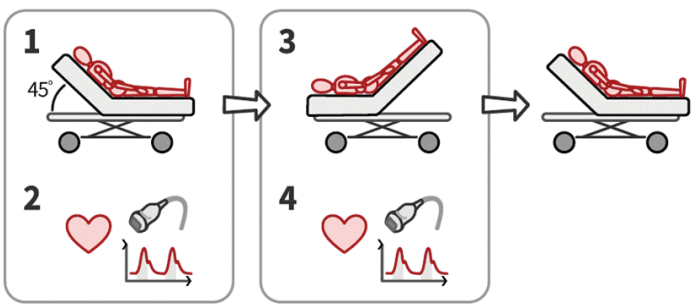
Measuring Cardiac Output Response

Possible measures for assessing the cardiac output response to PLR or a fluid bolus include:

- Echocardiography (LVOT VTI/ stroke volume)
- Pulse contour CO monitoring
- Thermodilution systems (PAC/ transpulmonary)

Note: Blood pressure is an indirect and less reliable indicator of preload responsiveness.

PLR Test Protocol



Adapted from: WHO guidelines for clinical management of arboviral diseases: dengue, chikungunya, Zika and yellow fever. Geneva: World Health Organization; 2025.

1. **Baseline Positioning**: Position the patient semi-recumbent (at 45°).
   Notes: Ensure you are using a bed in which the patient can be rapidly repositioned without discomfort (electric bed).
2. **Baseline Measurement**: Record the cardiac output at baseline (an average of multiple measurements may be used).
3. **Maneuver**: Quickly lower the patient flat and lift both legs to at least 45° from the horizontal. (The legs should not be manually raised by hand)
4. **Repeat Measurement**: Record the cardiac output again as soon as possible after the maneuver (the maximal effect occurs between 30 and 90 seconds).
5. **Repositioning**: Reposition the patient to the semi-recumbent position

Cautions and Contraindications for PLR test

Known Limitations:

- **Severe Hypovolaemia**: Due to small volumes of venous blood in the legs available for recruitment.
- **Discomfort or Adrenergic Stimulation**: If provoked by the test.

Contraindications:

- **Traumatic Brain Injuries**: Due to the risk of increasing intracranial pressure.
- **Deep Vein Thrombosis of the Lower Limbs**: Due to the risk of dislodging a clot.
- **Lower Extremity Fractures or Leg Amputations**: Due to effects on the volume of blood recruited.
- **Venous Compression Stockings**.

Interpreting the Results

Results	Interpretation	Action
Positive Test (Increase in CO ≥10%)	Preload responsiveness & lower risk of fluid overload	May proceed with fluid bolus ensuring continued monitoring and reassessment.
Negative Test	Further fluid administration is unlikely to improve the patient's physiology and increases the risk of fluid overload.	Consider other causes of shock. Alternative therapies, such as vasoactive drugs, might be more suitable.
The PLR test can be repeated after any fluid administration.		

Source: World Health Organization. WHO guidelines for clinical management of arboviral diseases: dengue, chikungunya, Zika and yellow fever Geneva: WHO; 2025.

Appendix 13

ROX INDEX AND HACOR SCORE

ROX Index

$$\text{ROX index} = \frac{SpO_2/FiO_2}{\text{Respiratory rate}}$$

A ROX <2.85, <3.47, and <3.85 at 2, 6, and 12 hours of HFNC initiation, respectively, are predictors of HFNC failure.

SpO_2=peripheral oxygen saturation, FiO_2= fraction of inspired oxygen, HFNC=high-flow nasal cannula

Source: Roca O, Caralt B, Messika J, et al. An Index Combining Respiratory Rate and Oxygenation to Predict Outcome of Nasal High-Flow Therapy. Am J Respir Crit Care Med. 2019 Jun 1;199(11):1368-1376.

HACOR score

Variable	Category	Points
Heart rate (beats per minute)	≤120 ≥121	0 1
Acidosis (pH)	≥7.35 7.30–7.34 7.25–7.29 <7.25	0 2 3 4
Consciousness (GCS)	15 13–14 11–12 ≤10	0 2 5 10
Oxygenation (PaO_2/FiO_2, mmHg)	≥201 176–200 151–175 126–150 101–125 ≤100	0 2 3 4 5 6
Respiratory rate (breaths per minute)	≤30 31–35 36–40 41–45 ≥46	0 1 2 3 4
Score of >5 at 1 hour predicts NIV failure		

GCS=Glasgow Coma Scale, HACOR=heart rate, acidosis, consciousness, oxygenation, and respiratory rate, NIV=non-invasive ventilation

Source: Duan J, Han X, Bai L, et al. Assessment of heart rate, acidosis,consciousness, oxygenation, and respiratory rate to predict noninvasive ventilation failure in hypoxemic patients. Intensive Care Med 2017; 43:192–199

LIST OF ABBREVIATIONS

ABW	adjusted body weight
ADE	antibody-dependent enhancement
AE	adverse event/effect
AGREE II	Appraisal of Guidelines, Research and Evaluation II
ALT	alanine transaminase
APRI	Aspartate Aminotransferase-to-Platelet Ratio Index
aPTT	activated partial thromboplastin time
AST	aspartate aminotransferase
AUC	area under the curve
BP	blood pressure
BMI	body-mass index
CCTVR	Colour, CRT, Temperature, Pulse Volume, Pulse Rate
CFR	case fatality rate
CI	confidence interval
CK	creatine kinase
CO	cardiac output
CPG	clinical practice guidelines
CRT	capillary refill time
CQ	clinical question
CXR	chest X-ray
DG	Development Group
DHF	dengue haemorrhagic fever
DIC	disseminated intravascular coagulation
DNWS	dengue without warning signs
DSS	dengue shock syndrome
DWWS	dengue with warning signs
ECG	electrocardiogram
EDS	Expanded Dengue Syndrome
EEOT	end-expiratory occlusion test
e.g.	example
EGL	endothelial glycocalyx layer
ELISA	enzyme-linked immunosorbent assays
ER	external reviewer
ESKD	end-stage kidney disease
FBC	full blood count
FIA	fluorescent immunoassay
FMS	Family Medicine Specialist
GCS	Glasgow Coma Score
GI	gastrointestinal
GRADE	Grading of Recommendations, Assessment, Development and Evaluations
Hb	haemoglobin
HCT	haematocrit
HDU	high-dependency unit
HHS	hyperosmolar hyperglycaemic state
HR	heart rate
IC	immunochromatography
ICD-11	International Statistical Classification of Diseases, 11th Revision
ICT	immunochromatographic tests
ICU	intensive care unit

IgG	immunoglobulin G
IgM	immunoglobulin M
IV	intravenous
IVC	inferior vena cava
L	litre
LFT	liver function test
LVOT	left ventricular outflow tract
MACCE	major adverse cardiac and cerebrovascular events
MD	mean difference
mg	milligram
ml/kg	millilitre per kilogram
ml/kg/h	millilitre per kilogram per hour
mmHg	millimetre mercury
mmol/L	millimole per liter
MoH	Ministry of Health
MTS	Malaysian Triage Scale
NIV	non-invasive ventilation
NPV	negative predictive value
NS	non-significant
NS1	non-structural protein 1
NSAIDs	Non-steroidal anti-inflammatory drugs
OR	odds ratio
ORS	oral rehydration salts
p	p-value
PAHO	Pan American Health Organization
PCR	polymerase chain reaction
PLR	passive leg raise
PoCUS	point-of-care ultrasound
PPV	positive predictive value
PPV	pulse pressure variation
PR	pulse rate
PT	prothrombin time
RC	Review Committee
RCT	randomised controlled trial
RDT	rapid diagnostic test
RNA	ribonucleic acid
RP	renal profile
RR	respiratory rate
RT-PCR	reverse transcriptase–polymerase chain reaction
SD	severe dengue
SEARO	South East Asia Regional Office
SGLT2i	Sodium-Glucose Transport 2 Inhibitors
SUCRA	Surface Under the Cumulative Ranking
SIGN	Scottish Intercollegiate Guidelines Network
SMD	standardised mean difference
Sn	sensitivity
Sp	specificity
SpO_2	oxygen saturation
SV	stroke volume
SVV	stroke volume variation
T	temperature

TVC	tidal volume challenge
U/L	unit/liter
VTI	velocity–time integral
WCC	white cell count
WHO	World Health Organization

ACKNOWLEDGEMENT

The CPG DG members would like to express their gratitude and appreciation to the following for their contributions:

- Panel of external reviewers who reviewed the draft
- Technical Advisory Committee of CPG for their valuable input and feedback
- Health Technology Assessment and Clinical Practice Guidelines Council for approving the CPG
- Puan Subhiyah Ariffin on retrieval of evidence
- All those who have contributed directly or indirectly to the development of the CPG

DISCLOSURE STATEMENT

The panel members of both DG and RC had completed disclosure forms. None held shares in pharmaceutical firms or acted as consultants to such firms. Details are available upon request from the CPG Secretariat.

SOURCE OF FUNDING

The development of the CPG on Management of Dengue in Adults (Fourth Edition) was supported in its entirety by the MoH Malaysia.

MALAYSIAN HEALTH TECHNOLOGY ASSESSMENT SECTION
Medical Development Division
Ministry of Health Malaysia
Level 4, Block E1, Precinct 1
62590 Putrajaya, Malaysia